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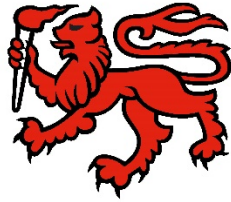
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**Enhancing TAM 3 for DRS technologies and related
stakeholder management practices: a case study of
international cricket**

By

Muhammad Siddique Fawad Qureshi

Masters by Research in Computer Science – University of Wollongong

Submitted in fulfilment of the requirements for the Degree of
Doctor of Philosophy

June 2022

University of Tasmania

Declaration of Originality

This thesis contains no material which has been accepted for a degree or diploma by the University or any other institution, except by way of background information and duly acknowledged in the thesis, and to the best of my knowledge and belief, no material previously published or written by another person except where due acknowledgement is made in the text of the thesis, nor does the thesis contain any material that infringes copyright.

Muhammad Qureshi 17/12/2021

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Statement of Ethical Conduct

The research associated with this thesis abides by the international and Australian codes on human and animal experimentation, the guidelines by the Australian Government's Office of the Gene Technology Regulator and the rulings of the Safety, Ethics and Institutional Biosafety Committees of the University. Ethics Approval No. H0018397.

Muhammad Qureshi 17/12/2021

Abstract

The Technology Acceptance Model (TAM) has become one of the most widely used information systems frameworks relating to new technology acceptance. TAM 3 has recently come under increased criticism, however, for its over-emphasis on the individual user and its lack of explicit concern of multiple stakeholder management practices necessary for the acceptance of complex technologies in professional contexts. The multiple stakeholder management issues have manifest most notably with the introduction of Decision Review System (DRS) technologies in the professional sports context. DRS include technologies are both retrospective in nature (i.e., they capture objective information that may not have been observable or processed by the referee/umpire accurately) as well as predictive (i.e., they are tasked with *inferring* new information from an incomplete data set – e.g., predicting the path of an object in motion). DRS technologies also deviate from those that had been used in professional sports since the mid-1980s, in that they are not designed to provide support for the referee/umpire to make their decision, but rather to empower players to challenge the referee/umpire's decision if they perceive that an error had been made. The introduction of these technological (and process) changes have been met with significant levels of stakeholder dissatisfaction and stakeholder rejection (in the form of outright boycotts). Given the criticisms of TAM 3, and the intersecting stakeholder management issues inherent to the implementation of DRS technologies in international sporting competitions, the research question addressed in this thesis is: *To what extent does TAM 3 need to be modified to account for DRS technologies and related stakeholder management practices?* In order to address this research question, the research was designed as a two-stage process; firstly, a detailed history of the ICC's implementation of DRS technologies was constructed from secondary data. Secondly, critical incidents from the historical account of the ICC's implementation of DRS technologies were used to form the base interview questions asked of 32 professional stakeholders in Stage Two.

This study posits that four elements of multiple stakeholder engagement (i.e., stakeholder 'identification', 'analysis', 'prioritisation' and 'feedback') along with three elements of multiple stakeholder education (i.e., 'specialised education', 'general education', and 'transparency') enabled the ICC to overcome DRS technology acceptance issues experienced by its multiple salient stakeholder groups. In terms of multiple stakeholder engagement practices, this thesis found support for a direct relationship with TAM 3's 'Job Relevance', 'Result Demonstrability' and 'Experience' variables. In terms of multiple stakeholder education, this thesis found support for a direct relationship with TAM 3's 'Perception of External Control' variable, and an indirect relationship (via the concept of 'transparency') with TAM 3's 'Result Demonstrability' variable. Based on analysis of the case data, a modified TAM 3 that is inclusive of the stakeholder management practices related to DRS technology acceptance is presented, as well as a range of future research opportunities in this regard. The findings of this thesis provide the basis for three key areas of future research. Firstly, the modified TAM 3 for DRS technologies and related stakeholder management practices presented in this thesis would be well-served by further validation and/or modification by empirical studies in an array of other professional sports settings and organisational contexts. Secondly, there is also the opportunity to explore the modified TAM 3 for DRS technologies and related stakeholder management practices across various technologies. Lastly, the research work of this study will benefit significantly through examination based on mixed research methods to explore the statistical significance and strength of relationships between elements of the modified TAM 3 for DRS technologies and related stakeholder management practices.

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List of Tables

Table 1.1 Exploration of TAM in various domains	2
Table 3.1 Sampling techniques and their suitability for this research	39
Table 3.2: Examples of stakeholders with multiple roles	41
Table 3.3: Number of interviews conducted with participant stakeholders.....	43
Table 3.4: Number of interviews conducted with non-participant stakeholders	43
Table 3.5: Codes and Coding Rules for the characteristics of DRS technologies	50
Table 3.6: Codes and Coding Rules for extant TAM 3 variables	51
Table 3.7: Coding and sub-category development.....	52
Table 3.8: Generic category development	53
Table 3.9: Main category development.....	54
Table 4.1: Elements to be considered in DRS technology implementation.....	60

List of Figures

Figure 2.1: The Original TAM adapted by Venkatesh and Davis (1996)	11
Figure 2.2: The extended version of Original TAM including determinants of PU named TAM 2 proposed by Venkatesh and Davis (2000)	12
Figure 2.3: The extended version of TAM 2 including determinants of PEOU named TAM 3 proposed by Venkatesh and Bala (2008)	14
Figure 4.1: T Signal used by the captain requesting a review of the decision made by the on-field umpire	61
Figure 4.2: Path calculation through Hawk-Eye technology. The image on the left shows an actual impact of the ball, whereas the image on the right provides a calculated path produced by the Hawk-Eye technology used in the first DRS technology trial	62
Figure 4.3: Sound Trakker Technology used in the DRS technology trial	63
Figure 4.4: The controversial Leg Before Wicket	64
Figure 4.5: The controversial Caught Behind the Wicket	65
Figure 4.6: Hot Spot detects contact between bat and ball and displays the point of contact using a bright spot	66
Figure 4.7: Indian captain MS Dhoni unsuccessfully reviews an LBW appeal against England's Ian Bell during the 2011 World Cup	73
Figure 4.8: The Real-Time Snickometer detects contact between bat and ball	75
Figure 4.9: The Australians stand in disbelief after running out of reviews when the on-field umpire gave a Not Out decision to an obvious catch	79
Figure 4.10: English batsman James Taylor talks to the on-field umpire after his partner James Anderson was given Run Out	81
Figure 4.11: Under the new 'Umpire's Call' rules, this decision will change from 'Not Out' to 'Out'	84
Figure 4.12: Under the new 'Umpire's Call' rules, this decision will remain 'Not Out'	84
Figure 4.13: The Real-Time Snickometer detects contact before and after the ball goes past the bat	87
Figure 4.14: The image on the left shows an actual impact of the ball, whereas the image on the right provides a calculated path produced by the Hawk-Eye technology	88
Figure 4.15: The Third Umpire declared a review taken by Indian captain Virat Kohli as 'null and void' since the replay was broadcast on the stadium screen prior to the Player Review being requested	90
Figure 4.16: The image on the left shows Indian captain having a conversation with the on-field umpires after his referral was rejected by the Third Umpire; the image on the right provides calculated path produced by the Hawk-Eye technology	91

Figure 4.17: There was no detection of contact between the bat and ball by Hot Spot technology in Tim Paine's Caught Behind decision	92
Figure 4.18: The Real-Time Snickometer detected contact between the ball and the bat by Hot Spot for Tim Paine's Caught Behind decision	93
Figure 6.1: The elements of multiple stakeholder engagement as an external variable of TAM 3.....	137
Figure 6.2: The external variables of TAM 3 with elements of multiple stakeholder engagement	137
Figure 6.3: The external variables of TAM 3 with elements of multiple stakeholder education	140
Figure 6.4: The proposed elements of multiple stakeholder engagement and multiple stakeholder education for TAM 3	140
Figure 6.5: The elements of multiple stakeholder engagement and multiple stakeholder education	141
Figure 6.6: A modified TAM 3 for DRS technologies and related stakeholder management practices	142

List of Acronyms

ATU	Attitude Towards Use
CAATT	Computer Assisted Audit Tools and Techniques
CBW	Caught Behind the Wicket
DRS	Decision Review System
FIFA	International Federation of Association Football
ICC	International Cricket Council
LBW	Leg Before Wicket
PDA	Personal Digital Assistant
PEOU	Perceived Ease of Use
PU	Perceived Usefulness
SSHERC	Social Sciences Human Research Ethics Committee
TAM	Technology Acceptance Model

List of Definitions

Crease Line	A series of line markings in the game of cricket that define the areas of fair play and designate the required positions of the batters and fielders at various points of play.
Dead Ball	A particular state of play in which the players cannot enact any of the performative aspects of the game of cricket.
Edge	Descriptive of situations where the ball comes into contact with the edge of the bat and not the face of the bat.
Hail Mary	A sporting manoeuvre attempted against heavy odds.
Howler	An obvious error made by a referee or umpire.
Leg Before Wicket	A mode of dismissal in the game of cricket where (in the opinion of the umpire), part of the batter's body (and not their bat) prevented a fairly delivered ball from otherwise going on to strike the batter's wicket.
Line Decision	A decision relating to whether a batter is within their ground at the time that the wicket is broken fairly by the fielding side in a game of cricket.
Not Out	The status bestowed upon a batter that has not been dismissed under the Laws of Cricket.
Off Stump	The stump of the wicket (see below) that is furthest away from the batter's stance.
Out	When a batter's innings is brought to an end by a form of dismissal under the Laws of Cricket.
Run Out	When the batters are attempting to run between the wickets, and the fielding team succeed in fairly breaking the wicket before a batter has made their ground behind the crease line at that wicket.
Shambles	In sporting parlance, a 'shambles' represents a state of chaos or disorder.
Stumping	A method of dismissing a batter which involves the wicket-keeper fairly breaking the wicket while the batter is out of their ground.
Test Series	A series of five-day, two-innings per side cricket matches played between representative national teams.
Third Umpire	An off-field umpire used in all international cricket matches that makes the final decision in appeals referred to them by the two on-field umpires or captains of the competing teams.
Wicket	A series of three (typically) wooden stumps pitched at each end of a 20-yard playing surface in the game of cricket. The stumps are connected by two (typically) wooden bails across their tops; each wicket is 9 inches in width.

List of the ICC's DRS Technologies

Hawk-Eye	Hawk-Eye technology is a computer vision system used in numerous sports such as cricket, tennis, Gaelic football, badminton, hurling, rugby union, association football and volleyball, to visually track the trajectory of the ball and display a profile of its statistically most likely path as a moving image. In cricket, it uses six specially placed cameras around the ground to track the path of the ball, from when it was released from the bowler's hand right up until when it becomes dead. The images captured by the camera are then turned into a 3D image by a special computer to show how the ball will travel on an imaginary cricket pitch.
Hot Spot	Hot Spot technology is an infrared imaging system used in cricket to determine whether the ball has struck the batsman, bat or pad. The technology relies on infrared cameras that detect the heat signature of ball impact.
Snickometer	Snickometer technology makes use of both the audio and visual evidence to detect any edges. A very sensitive microphone is placed at the stumps and this microphone is connected to an oscilloscope which tracks the sound waves. In case of any edges, the microphone records even the slightest of the sounds. The recorded trace is then played along with the slow-motion video which makes it easy to determine if there was any contact.
Sound Trakker	Sound Trakker technology uses sound from the stump microphones with the replays at average speed and in slow motion. The live sound is picked up using the stump microphone, gets filtered and relayed using an attached oscilloscope that traces the relevant sound waves. Cameras record the visual and replay it in slow motion with the sound attached. If the ball has touched the bat, the sound graph gets ticketed for impact - a single, sharp spike on the graph confirming the ball is hitting the bat. The Third Umpire then correlates the sound with the ball's slow-motion replays, passing the bat to make their decision.
Ultra-Edge	Ultra-Edge technology provides both audio and heat-based technologies by syncing up vision from ultra-motion cameras and audio from the stump microphones to deliver the most accurate and cost-effective system for detecting whether a batsman has edged the ball.

List of the Cricketing Roles

Administrator	An official that assists with performing duties in administrating and developing cricket in an appointed region.
Coach	An official in charge of practical training, monitoring, and engaging a group of players.
Commentator	A narrator who describes the events of a game usually during a live broadcast on radio or television.
Educator	A person responsible for developing high-quality material for theoretical training of umpires in providing both the technical and practical knowledge of the game.
Journalist	A person able to study the game and then present it in their own words in a manner that strikes a connection with the reader.
Match Referee	An official appointed to oversee international cricket matches, ensuring that every participant of the game correctly follows all the rules and regulations of the game.
Player	An athlete trained to compete in the sports performing different roles such as a bowler, a batter, a fielder or a wicket-keeper.
Scorer	An official appointed to record all runs scored, all wickets taken, and the number of overs bowled.
Technologist	An expert employed to implement the technical equipment used for different technological aspects of the game.
Umpire	An official authorised to make decisions on the playing field in accordance with the Laws of Cricket.

Table of Contents

Declaration of Originality.....	i
Authority of Access.....	ii
Statement of Ethical Conduct.....	iii
Abstract.....	iv
Acknowledgments.....	vi
List of Tables.....	vii
List of Figures.....	viii
List of Acronyms.....	x
List of Definitions.....	xi
List of the ICC's DRS Technologies.....	xii
List of the Cricketing Roles.....	xiii

Chapter One: Rationale and Introduction to the Thesis

1.1 Objective of the chapter.....	1
1.2 Rationale for the thesis.....	1
1.3 Broad research opportunity.....	5
1.4 Structure of the thesis.....	5

Chapter Two: Literature Review

2.1 Objectives of the chapter.....	7
2.2 Characteristics of DRS technology.....	7
2.2.1 Undermining the role of the original decision-maker.....	7
2.2.2 No support for the original decision.....	8
2.2.3 Changes the balance of stakeholder power in the decision-making process.....	9
2.3 The development of TAM.....	10
2.3.1 TAM: The original construct.....	10
2.3.2 TAM 2.....	11
2.3.3 TAM 3.....	13
2.3.4 Perceived Usefulness (PU).....	15
2.3.5 Perceived Ease of Use (PEOU).....	16
2.3.6 Behavioural Intent.....	17
2.3.7 Subjective Norm.....	18
2.3.8 Image.....	20
2.3.9 Job Relevance.....	21
2.3.10 Output Quality.....	22
2.3.11 Result Demonstrability.....	23
2.3.12 Computer Self-efficacy.....	24
2.3.13 Perceptions of External Control.....	25
2.3.14 Computer Anxiety.....	26

2.3.15 Computer Playfulness.....	27
2.3.16 Perceived Enjoyment.....	28
2.3.17 Objective Usability	29
2.4 Summary of TAM 3's development.....	30
2.5 Stakeholder management practices.....	30
2.6 Specific research question	32
2.7 Chapter summary	33

Chapter Three: Research Design and Methodology

3.1 Objective of the chapter.....	34
3.2 Research paradigm	34
3.2.1 Philosophical assumptions.....	35
3.3 Research design	36
3.3.1 Case study protocol	37
3.3.2 Unit of analysis and observation	38
3.3.3 Sampling and interviewees.....	39
3.3.3.1 Conduct of interviews.....	42
3.3.4 Data collection and procedures.....	44
3.3.4.1 Adhering ethical guidelines	44
3.3.4.2 Interview protocol.....	45
3.3.5 Administration of the data collection process	46
3.3.6 Data organisation and storage.....	46
3.4 Methodological and analytical decisions	47
3.4.1 Methodological choice	47
3.4.2 Analytical decisions.....	48
3.4.2.1 Familiarisation with the data	49
3.4.2.2 Development of codes.....	50
3.5 Ensuring the quality of research.....	54
3.5.1 Credibility.....	54
3.5.1.1 The legitimacy of research and process	55
3.5.1.2 Bias control	55
3.5.2 Transferability	56
3.5.3 Dependability.....	57
3.5.4 Confirmability	57
3.6 Intra-coder reliability	58
3.7 Chapter summary	58

Chapter Four: International Cricket Council's Decision Review System: An Historical Overview

4.1 Objectives of the chapter.....	59
4.1.1 March 2008: The ICC confirms the DRS technology trial during an international cricket competition	59
4.1.1.1 Hawk-Eye	61

4.1.1.2 Sound Trakker	62
4.1.1.3 Slow-Motion, Super Slow-Motion or Ultra-Motion cameras	63
4.1.2 July 2008: Trial of DRS technology in international Cricket competitions	63
4.1.2.1 Controversy #1: Leg Before Wicket.....	64
4.1.2.2 Controversy #2: Caught Behind the Wicket	64
4.1.3 March 2009: The ICC extends the scope of DRS technology.....	66
4.1.4 November 2009: The BCCI and other cricketing boards oppose using DRS technology due to cost	68
4.1.5 Controversy #3: The ICC investigate referral complaint made by the ECB	69
4.1.6 October 2010: DRS technologies available for the 2011 World Cup	70
4.1.7 Controversy #4: The 2011 World Cup	71
4.1.8 February 2013: Introduction of the Real-Time Snickometer	74
4.1.9 July 2013: Take DRS technology off players and leave it to the umpires	76
4.1.10 July 2013: The DRS technology is not the solution, it's part of the problem.....	77
4.1.11 September 2013: ICC announces DRS technology changes.....	78
4.1.12 November 2013: Australian Captain Michael Clarke claims DRS technology 'distorts the game'	78
4.1.13 January 2015: Indian cricketing legend Sachin Tendulkar wants standardised DRS technology	80
4.1.14 Controversy #5: The ICC admits Third Umpire mistake in 2015 World Cup	81
4.1.15 November 2015: Serious flaws in DRS technologies indicate a need for a review	82
4.1.16 July 2016: Umpire's Call review for the DRS technology	83
4.1.17 October 2016: When will we see a uniform application of the DRS technology everywhere?.....	85
4.1.18 June 2017: Teams to retain review for Umpire's Call verdict	86
4.1.19 March 2019: Indian captain Virat Kohli says, "DRS technology is not consistent at all."	86
4.1.20 August 2020: Third Umpire to make the original decision for front foot No Balls	89
4.1.21 Controversy #6: Virat Kohli complains after failed DRS technology protocol in a T20 game against Australia in 2020	90
4.1.22 Controversy #7: Tim Paine queries DRS technology verdict with umpires.....	92
4.1.23 January 2021: Ex-ICC Elite Umpire calls for end to Umpire's Call	94
4.2 Chapter summary	94

Chapter Five: Research Findings

5.1 Objectives of the chapter	96
5.2 Characteristics of DRS technology	96
5.2.1 Undermining the role of the decision-maker	96
5.2.2 DRS technology undermining the role of the original decision-maker.....	97
5.2.3 DRS technologies provide no support to the original decision	100
5.2.4 DRS technologies not providing support for the original decision	101
5.2.5 Changes the balance of stakeholder power in the decision-making process.....	103
5.2.6 DRS technologies changing the balance of stakeholder power in the decision-making process	104

5.3 Extant TAM 3 and its role in DRS technology acceptance in the ICC context	106
5.3.1 TAM 3's 'Job Relevance'	106
5.3.2 TAM 3's 'Result Demonstrability'	108
5.3.3 TAM 3's 'Experience'	110
5.3.4 TAM 3's 'Perceptions of External Control'	112
5.4 Additional Factors in DRS technology acceptance in the ICC context	113
5.4.1 Multiple stakeholder engagement	113
5.4.2 Multiple stakeholder education	117
5.5 Chapter summary	121

Chapter Six: Discussion and Conclusion

6.1 Objectives of the chapter	123
6.2 Characteristics of DRS technology in the professional sports context	123
6.2.1 Undermining the role of the original decision-maker	123
6.2.2 No support for the original decision	124
6.2.3 Changes the balance of stakeholder power in the decision-making process	125
6.3 Extant TAM 3 variables related to multiple stakeholder acceptance of DRS technologies	126
6.3.1 Job Relevance	126
6.3.2 Result Demonstrability	128
6.3.3 Experience	130
6.3.4 Perceptions of External Control	131
6.4 Enhancing TAM 3 with stakeholder management practices	133
6.4.1 Multiple stakeholder engagement	133
6.4.2 Multiple stakeholder education	138
6.5 A proposed modified TAM 3	141
6.6 Limitations of the research	142
6.7 Future research opportunities	143
6.8 Chapter summary	144
References	145
Appendix A - Case Study Protocol	186
Appendix B – The ICC's Process for DRS Technology Use	190
Appendix C – Ethics Approval	202
Appendix D – Information Sheet	205
Appendix E – Consent Form	208
Appendix F – Interview Questions	210
Appendix G – Interview Invitation to Cricket Australia	211
Appendix H – The Preamble to the Laws of Cricket: The Spirit of Cricket	212

Chapter One

Rationale and Introduction to the Thesis

1.1 Objective of the chapter

The objective of this chapter is to introduce this thesis. The chapter commences with a discussion of the rationale for this thesis. Next, the broad research opportunity is noted and briefly discussed. The chapter concludes with a discussion of the structure of the thesis.

1.2 Rationale for the thesis

The Technology Acceptance Model (TAM) was first posited by Davis (1989) to explore the factors inherent to the effective implementation and acceptance of new technologies in professional settings. TAM is an intent-based model based on the tenets of the Theory of Reasoned Action (see Ajzen & Fishbein 1980) and Theory of Planned Behaviour (see Ajzen 1985) to explain and predict users' acceptance of information technology. The original TAM formulation linked two specific factors (i.e., the perceived usefulness (PU) and ease of use (PEOU) of a given technology) with a user's ultimate acceptance and usage of the given technology in a professional setting (Davis 1989). TAM has been empirically examined in a range of professional settings, that when taken together, have established an array of external variables that moderate or mediate the relationship between an individual user's perceptions of a given technology and their eventual acceptance of it (see Table 1.1 over page).

Domain	Example Research Publications
e-Commerce	Leonard & Jones (2014) Brohi <i>et al.</i> (2017) Nissen <i>et al.</i> (2019) Solangi <i>et al.</i> (2019) Hassen <i>et al.</i> (2020)
Education	Weng, Cheong & Cheong (2010) Saleh & Drew (2014) Azawei, Parslow & Lundqvist (2017) Ates (2018) Zardari <i>et al.</i> (2021)
Finance	Martin & Matlay (2001) Calantone, Griffith & Yalcinkaya (2006) Ahmad <i>et al.</i> (2013) Lee (2016) Singh, Gaur & Ramakrishnan (2017)
Government	Shareef (2011) Fletcher, Sarkani & Mazzuchi (2014) Ahram <i>et al.</i> (2017) Ebrahimzadeh (2017) Salem (2019)
Health Care	Holden & Karsh (2010) Lakshmi & Rajaram (2012) Ahlan & Ahmad (2014) Tadesse <i>et al.</i> (2019) Usmanova <i>et al.</i> (2020)
Information Technology	Hong, Thong & Tam (2006) Li, Qi & Shu (2008) Chandio <i>et al.</i> (2017) Arain <i>et al.</i> (2018) Muchran & Ahmar (2019)
Professional Sports	Hur & Claussen (2007) Ibrahim (2014) Pool, Amani & Lashaki (2014) Ha, Kang & Ha (2015) Bae <i>et al.</i> (2017)

Table 1.1: Exploration of TAM in various domains

The results of these examinations culminated in the development of TAM through two subsequent iterations (TAM 2 in 2000 and TAM 3 in 2008) to become one of the most widely used theoretical frameworks in the information systems discipline (Holden & Karsh, 2010; Koul & Eydgahi 2018; Lee, Kozar & Larsen 2003; Li 2010; Usmanova *et al.* 2020). TAM 2 focused on identifying determinants of PU and moderating variables, whereas TAM 3 centred on specifying additional external variables related to PEOU (see Chapter Two). The theoretical and empirical developments have benefitted researchers and practitioners alike; it has enabled

researchers to explore the relative strengths in the relationships between variables in a range of different professional settings (and form the basis for the exploration of new relationships - see Alharbi & Drew 2014; Mousa *et al.* 2020; Szajina 1996; Taherdoost 2019) whilst enabling practitioners the ability to invest the resources and training necessary for effective and efficient implementation of new technologies (see Bach, Zoroja & Celjo 2017; Matikiti, Mpinganjira & Roberts 2018; Praveena & Thomas 2014). Koul and Eydgahi (2018) acknowledged TAM 3 as having developed into one of the most influential and widely used information systems frameworks and stated that it continues to serve as a ‘hotspot’ for contemporary technology acceptance research.

Recently, however, TAM 3 has been criticised for its emphasis on the individual user and the concomitant lack of explicit concern of multiple stakeholder management practices necessary for the acceptance of more complex technologies in professional contexts. For example, it has been noted that TAM 3 fails to explicitly engage with multiple stakeholders simultaneously in the technology implementation process; Raza and Standing (2010), Straub (2017), Pardo and Coronado (2016) note that TAM 3 fails to identify, analyse, prioritise and gain feedback from multiple stakeholders to the technology acceptance process. Similarly, Mingers and Walsham (2010), Farre *et al.* (2019), Freiesleben *et al.* (2021) state that TAM 3 requires more inclusive methods to ensure that salient stakeholders are included within the ‘boundaries of the model’. Govindaraju, Chandra and Siregar (2012), Pardo and Coronado (2016), Ajibade (2018) go further to suggest that it is entirely impracticable for TAM 3 to emphasise feedback from the individual user of a given technology and not include those stakeholders that are indirectly impacted by the implementation of new technologies. Another criticism has focused on the paucity of explicit stakeholder education and training inherent to TAM 3; Zahid *et al.* (2013), Almaiah, Al-Khasawneh and Althunibat (2020), Ratjeana *et al.* (2020), for example, state that TAM 3 does not consider factors such as education and training as important external variables

influencing multiple users' acceptance. Similarly, Hess, McNab and Basoglu (2014), Bario-Garcia (2015), Ratjeana *et al.* (2020) state that TAM 3 only considers training (or retraining) specific/direct users of a given technology and ignores the training implications of other stakeholder groups impacted by its implementation.

The multiple stakeholder management issues identified above have manifested most notably with the implementation of Decision Review System (DRS) technologies in the professional sports context (Arastey 2018; Dobell 2011; Shivakumar 2018; Sportskeeda 2021; Steen 2011; Wilson 2013). DRSs include technologies that are both retrospective in nature (i.e., they capture objective information that may not have been observable or processed by the referee/umpire accurately) as well as predictive (i.e., they are tasked with *inferring* new information from an incomplete data set – e.g., predicting the likely path for an object in motion) (Chalkley, MacMahon & Ball 2013; ICC 2021; Shenoy 2021; Steen 2011). DRS technologies also deviate from those that had been used in professional sports since the mid-1980s in that they are not designed to provide support for the referee/umpire to make their decision but to empower players to challenge the referee/umpire's decision if they perceive that an error had been made (Monga 2011; Waingankar 2017). The introduction of these technological (and process) changes have been met with significant levels of stakeholder dissatisfaction and stakeholder rejection (in the form of outright boycotts). In terms of stakeholder dissatisfaction, professional players and coaches alike expressed ongoing frustration with the lack of consistency and reliability with the technology, as well as a lack of consultation (and by extension, comprehension) about the technology's capabilities and the manner in which it is deployed by technicians and match officials (see Brettig 2012; Jones 2018; Mennon 2013; Ryall 2012; The International News 2021; Weaver 2009). Similarly, professional referee/umpiring stakeholders expressed their dissatisfaction with the implementation of DRS technologies in terms of its inherent contradiction to the spirit of

sportsmanship (i.e., the ability to challenge the referee/umpire's decision effectively undermines their role in the sport – see Cherny 2021; ESPNcricinfo 2009a; Nair 2021; Saltau 2009; Smith 2021), and the lack of adequate training in its use (The Indian Express 2021; The International News 2019). In terms of stakeholder rejection, the Board of Cricket Control in India (BCCI) and the International Federation of Association Football (FIFA) both rejected the introduction of DRS technologies, given that there were serious misgivings about their accuracy and reliability in performing the tasks as it charged with (Bradley 2018; Selvey 2015; Steen 2011; The Telegraph 2019); the BCCI and FIFA went so far as to refuse to allow the use of DRS technologies in any competitions in which they were involved (Jacobs 2010; Ryall, 2012; The Economist 2013; Wilson 2012).

1.3 Broad research opportunity

Given the criticisms of TAM 3 noted above and the intersecting stakeholder management issues inherent to the implementation of DRS technologies in international sporting competitions, the broad research opportunity for this thesis is to examine the extent to which TAM 3 needs to be modified to incorporate the unique characteristics of DRS technologies and related stakeholder management practices.

1.4 Structure of the thesis

This thesis comprises six chapters. Following this introductory chapter, Chapter Two provides an overview of extant literature pertaining to the unique characteristics of DRS technologies and the extant factors of TAM 3. The literature review will culminate in the presentation of the specific research question to be analysed in this thesis. Chapter Three outlines the case method used to gather the requisite data needed to answer the specific research question presented in Chapter Two. The ICC's implementation of DRS technologies in their international cricket

competitions (and the role of stakeholder management practices) provides the focus of this study. The research was informed via qualitative analysis of secondary data relating to the history of the ICC's implementation of DRS technology and a series of semi-structured interviews undertaken with 32 professional stakeholders from a range of stakeholder groups. These key individuals were closely associated with critical incidents inherent to the ICC's implementation of DRS technologies (i.e., from 2005 to 2020). Chapter Four presents an historical account of critical incidents occurring in the ICC's implementation of DRS technologies in their international cricket competitions and stakeholder management practices that were evident over time; the historical account is based on both primary and secondary data sources. It provides the grounding from which interview questions for key interviewees were designed and the basis upon which the specific research question was analysed. Chapter Five presents an analysis of the case data in terms of TAM 3 presented in Chapter Two. Drawing on the case data and using the computer software package QSR NVivo (version 12), the common themes emerging from the interviews are identified and triangulated between respondents. Chapter Six provides a discussion of the key findings of the thesis and theoretical implications for TAM 3. Based on analysis of the case data, a modified TAM 3 that is inclusive of the stakeholder management practices related to DRS technology acceptance is proposed. Chapter Six concludes with a consideration of the practical implications of the proposed model, followed by the acknowledgement of the limitations of this study and a brief discussion of future research opportunities in this area.

Chapter Two

Literature Review

2.1 Objectives of the chapter

The objectives of this chapter are four-fold. Firstly, it will present a review of the DRS technology literature to identify and define its characteristics (i.e., how the stakeholder management requirements of DRS technology differentiate it from other decision-related technologies). Secondly, the chapter discusses the development of the extant TAM theory since its inception and all the critical factors included in different versions of TAM for technology acceptance. Thirdly, it will provide an overview of stakeholder management practices emanating from the Stakeholder Theory of Corporate Social Responsibility (CSR). Lastly, the chapter presents the specific research question to be addressed in this thesis.

2.2 Characteristics of DRS technology

A review of the DRS technology literature indicates three characteristics that differentiate it from other decision-related technologies: (i) DRS technologies undermine the role of the decision-maker, (ii) DRS technologies provide no support for the original decision, and (iii) DRS technologies change the balance of stakeholder power in the decision-making process. Each of these characteristics are discussed in the sections below.

2.2.1 Undermining the role of the original decision-maker

The first key characteristic relates to the DRS technologies' capacity to undermine, even if inadvertently, the role of the original legitimate decision-maker in the decision-making process (see Baldwin 2014; Collins & Evans 2008; Steen 2011). The factors that relate to undermining the role of the original decision-maker include undue scrutiny, stress, and anxiety. In terms of

undue scrutiny, for example, Collins and Evans (2008) examined the change in the relationship between the spectators and the on-field umpires that emerged due to the decision review technologies as the on-field umpires reported finding themselves under increased undue scrutiny on their decision-making. The study found that the sport's governing bodies are attempting to introduce more technology that dilutes the on-field umpires' role in the game and affects the relationship between the umpires and spectator stakeholders. In terms of stress and anxiety, Baldwin's (2014) study on the role of video replay technology in the elite rugby league competition indicated DRS technologies as the sources of stress and anxiety in a cross-section of semi-elite rugby league referees. The research highlighted that when an emphasis is placed on the use of DRS technology to overturn or confirm decisions (and the relentless reviewing of the on-field decisions), the authority of the on-field referees is undermined, resulting in stress and anxiety. The study also indicated that a lack of training of the referees on using DRS technology compounds the stress and anxiety factors reported by the referees. Based on the literature reviewed, this study associates any stress, anxiety, or scrutiny factors to the characteristic of 'undermining the role of the original decision-maker'.

2.2.2 No support for the original decision

The second key characteristic relates to the DRS technologies' tendency to provide no support for making the original decision in the decision-making process (see Leveaux 2009; Ryall 2012). This factor highlights that the use of DRS technologies is only triggered after an original decision has been made and communicated to the playing group. For example, Leveaux (2009), in his study on using DRS technology to support referee's decision found that different stakeholders (e.g., commentators, coaches) are provided with access to DRS technology information and often use the technology to criticise umpiring decisions (even when they are not overturned). Ryall (2012) explored the arguments against Goal-Line Technology relating

to FIFA's resistance to implementing DRS technology. The research highlighted one of FIFA's concerns (i.e., referees' authority will be undermined as they will tend to abdicate their responsibilities) after it was demonstrated that football referees were referring to the DRS technologies to manage the match where no technological support was required. The research implied that FIFA's main concern related to the referees abdicating their responsibilities (i) because the DRS technologies could be used at any point to confirm or overturn an on-field decision, and (ii) the reliance on DRS technologies resulted in inferior quality of referee preparation and performance. Therefore, based on the literature, this study considers any aspect of the DRS technology used after the original decision has been communicated as evidence of 'no support to the original decision'.

2.2.3 Changes the balance of stakeholder power in the decision-making process

The third key characteristic relates to the DRS technologies' tendency to change the balance of stakeholder power in the decision-making process (see Borooah 2016; Nlandu 2012; Steen 2011). The changing of the balance of stakeholder power in the decision-making process relates to the empowerment of traditionally passive stakeholders in the decision-making process. Steen (2011), for example, discussed the 'paradigm change' brought about by the implementation of DRS technologies in professional sporting contexts. Before introducing DRS technologies, Steen (2011) noted that on-field umpires were considered the final arbiters of contested sport; however, since its implementation, professional sportspeople have been empowered to shed their passive role in the decision-making process instead become active participants. This was noted in the reported change in behaviour of both stakeholder groups (i.e., the umpires and the professional sportspeople) in that (i) the umpires were hesitant to make strong judgements in contentious situations instead of increasing their reliance on the technology to make the de facto original decision, and (ii) professional sportspeople were far more open in their dissent

towards the decision-maker on their field of play. Collins (2012) expanded on the range of stakeholders that were 'empowered' to openly criticise the performance of the original on-field decision-maker for including coaches, expert commentators, peers and the viewing public; this increased the pressure perceived by the original decision-maker in that their performance was not judged on real-time decision-making, but rather on close inspection of information that could only be gathered and reviewed multiple times at different speeds and angles. Nlandu (2012) went as far as to claim that DRS technologies served only to increase the dissatisfaction of previously passive actors in the decision-making process and actively 'scapegoat' the original decision-maker should they have made an error. Based on the literature reviewed, this study considers any empowerment of traditionally passive actors in the decision-making process as a 'change the balance of stakeholder power in the decision-making process.

2.3 The development of TAM

As noted in Chapter One, the theoretical development of TAM has progressed in three distinct stages known as 'TAM', 'TAM 2', and 'TAM 3'; each will be described in the following sections:

2.3.1 TAM: The original construct

The original construct of TAM proposed by Davis (1989) was based on the Theory of Reasoned Action (see Ajzen & Fishbein 1980) and Theory of Planned Behaviour (see Ajzen 1985). Firstly, TAM was developed to explain the acceptance and use of new technology. Secondly, it was used to identify the factors which lead to users' acceptance or rejection of technology by integrating technological aspects with organisational behaviour concepts (Davis 1989). In its simplest form, TAM proposes a relationship between four constructs: perceived usefulness (PU), perceived ease of use (PEOU), attitude towards use (ATU) and behavioural intent to use new technology. Davis (1989) applied a regression analysis to determine the relationships that

existed in technology acceptance; he suggested that PU and PEOU constructs are the two antecedent factors related to an individual's behavioural intent to use technology. The model theorises that the effect of external variables (such as system characteristics, user training, user participation in design, and the nature of the implementation process) on behavioural intent will be mediated by PU and PEOU, as presented in Figure 2.1:

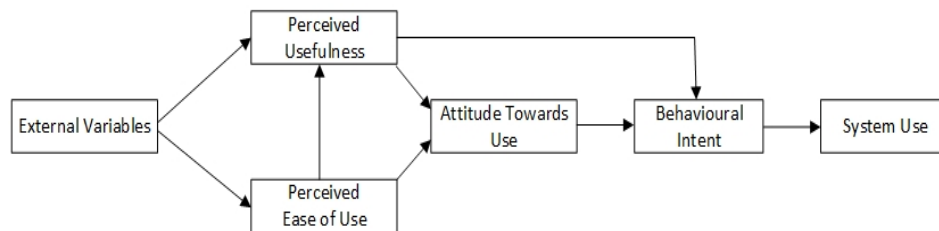


Figure 2.1: The Original TAM adapted by Venkatesh and Davis (1996)

2.3.2 TAM 2

Venkatesh and Davis (2000) proposed TAM 2 as an extension of TAM by specifying the external variables and moderators for PU. The external variables were identified as determinants of PU: subjective norm, image, job relevance, output quality, result demonstrability, and PEOU and two moderator variables are called experience and voluntariness. The three constructs of PU, PEOU and behavioural intent were kept from the Original TAM, but ATU was excluded because of the following reasons:

- The relationship between PU and behavioural intent is more significant than the relationship between ATU and behavioural intent, and
- ATU is not able to fully mediate the relationship between PEOU and behavioural intent

TAM 2 reflects two theoretical processes in social influence and cognitive instrumental in explaining the various determinants of PU and behavioural intent. The two external variables of subjective norm and image represent the social influence processes. On the other hand, the cognitive instrumental processes include job relevance, output quality, result demonstrability

and PEOU (NB: these will be discussed in more detail later in the chapter). TAM 2 maintained the concept of PEOU from the original TAM as a direct determinant of PU, as shown in Figure 2.2:

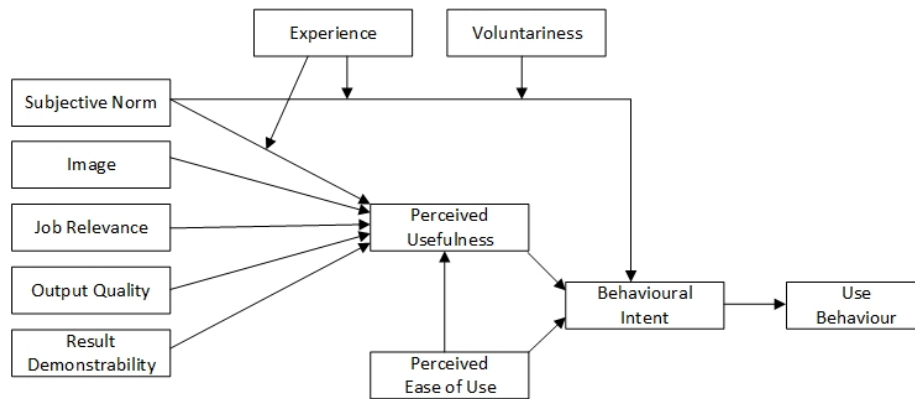


Figure 2.2: The extended version of Original TAM including determinants of PU named TAM 2 proposed by Venkatesh and Davis (2000)

TAM 2 suggests that three social influence mechanisms in compliance, internalisation and identification play a role in understanding social influence processes:

- *Compliance* represents a situation in which an individual performs a behaviour to attain certain rewards or avoid punishment (Miniard & Cohen 1979)
- *Identification* refers to an individual's belief that performing a behaviour will elevate their social status within a referent group because important referents believe the behaviour should be performed (Venkatesh & Davis 2000)
- *Internalisation* is defined as incorporating a referents belief into one's belief structure (Warshaw 1980)

TAM 2 indicates that subjective norms and image positively influence PU through processes of internalisation and identification, respectively (Venkatesh & Davis 2000). It further posits that the effect of subjective norm on both PU and behavioural intent will attenuate over time as users gain more experience with a system (Venkatesh & Davis 2000). The four constructs in job relevance, output quality, result demonstrability, and PEOU capture the influence of

cognitive instrumental processes on PU. TAM 2 indicated that individuals' assessment of the match between important work goals and the consequences of performing job tasks using a system serves as a basis for forming perceptions regarding the systems' usefulness (Lu, *et al.* 2003; Venkatesh & Davis 2000) and posits that PEOU and result demonstrability will positively influence PU. Job relevance and output quality will have a moderating effect on PU, such that the higher the output quality, the stronger the effect job relevance will have on PU (Venkatesh & Davis 2000).

2.3.3 TAM 3

Venkatesh and Bala (2008) proposed TAM 3 as an extension of TAM 2 by specifying additional external variables related to PEOU and were categorised into the following two groups:

- *Anchors*: were considered as general beliefs about computers and computer usage included (see Wansink, Kent & Hoch 1998)
- *Adjustments*: were considered as beliefs that are shaped based on direct experience with the target system included (see Wansink, Kent & Hoch 1998)

The external variables of computer self-efficacy, perception of external control, computer anxiety and computer playfulness are categorised as anchors; the remaining two external variables of perceived enjoyment and objective usability were named adjustment variables, as shown in Figure 2.3.

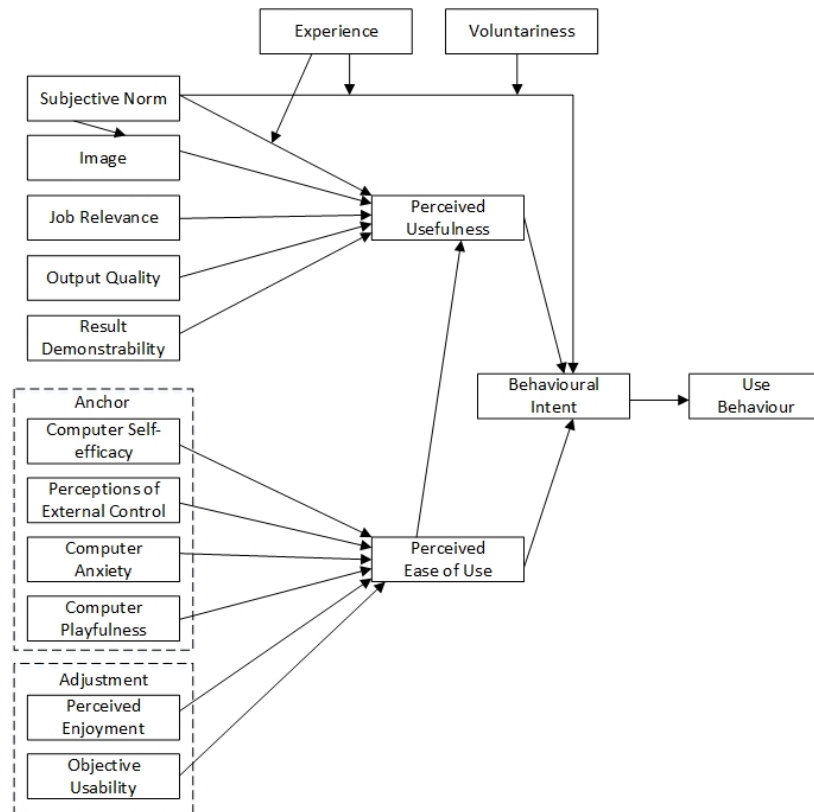


Figure 2.3: The extended version of TAM 2 including determinants of PEOU named TAM 3 proposed by Venkatesh and Bala (2008)

TAM 3 was developed based on the anchoring and adjustment framing of human decision-making (Ang, Ramayah & Amin 2015; Venkatesh & Bala 2008; Wong, Russo & McDowell 2013). Combining the second and third model determinants added more richness and understanding to the extended model (Venkatesh & Bala 2008). The model theorises that the newly added variables are believed to play a role in determining PEOU after individuals gain experience with the new system. The role of two anchors in computer self-efficacy and perceptions of external control will continue to become strong. However, the effects of the other two anchors in computer playfulness and computer anxiety were theorised to diminish over time. Over the last three decades, there has been substantial empirical support for TAM's efficacy (see Adams *et al.* 1992; Agarwal & Karahanna 2000; Dishaw & Strong 1999; Gefen, Karahanna, & Straub 2003; Karahanna, Agarwal & Angst 2006; Srite & Karahanna 2006;

Venkatesh & Morris 2000; Venkatesh *et al.* 2003). TAM 3's usefulness has been verified across distinct types of technologies, users, and cultures in information systems. For example, TAM was tested on word processing tools (Davis, Bagozzi & Warshaw 1989; Davis 1989), the World Wide Web (Agarwal & Karahanna 2000), computer banking systems (Adamson & Shine 2003; Ahmad *et al.* 2013; Brown *et al.* 2002; Calantone, Griffith & Yalcinkaya 2006), email (Davis 1989; Gefen & Straub 1997; Malhotra & Galleta 1999; Szanja 1996), graphics software (Venkatesh & Davis 1996), spreadsheet software (Mathieson 1991), online shopping (Gefen, Karahanna, & Straub 2003; Datta 2011; Leonard & Jones 2014; Nissen *et al.* 2019), ERP systems (Amoako-Gyampah & Salam 2004), groupware (Lou, Luo & Strong 2000), online gaming (Hsu & Lu 2004), sports websites (Ba *et al.* 2017; Ha, Kang & Ha 2015; Hur & Claussen 2017; Ibrahim 2014; Pool, Amani & Lashaki 2014) and telemedicine (Ahlan & Ahmad 2014; Chau & Hu 2002a; Holden & Karsh 2010; Lakshmi & Rajaram 2012; Tadesse *et al.* 2019; Usmanova *et al.* 2020). The individual elements and external variables mentioned above will be discussed in detail in the following sections.

2.3.4 Perceived Usefulness (PU)

The original TAM construct of PU has been maintained in all three versions without any significant modifications (Davis 1989). The PU is defined as the degree to which an individual perceives that using an IT system will enhance their job performance (i.e., the extent to which users of a system are optimistic that their productivity and effectiveness in their work could be improved using the system – see Hong, Thong & Tam 2006; Mou, Shin & Cohen 2017; Rouibah & Abbas 2006). Therefore, users' behavioural intent is determined by the perception of usefulness they have on the system (Arain *et al.* 2019; Sharma, Oman & Yadav 2011). In relation to this, several studies have found that PU has a significant and strong influence on an individual's behavioural intent to use a new technology or system (Guritno & Siringoringo

2013; Suki & Suki 2011). For instance, the study on e-procurement reveals that PU significantly influences the behavioural intent to use the e-procurement system in the public sector institutions (Sambasivan, Wemyss & Rose 2010). Similarly, it was found in another study that behavioural intent to use electronic payment among bank customers was influenced by its PU in banking transactions (Tella & Olasina 2014; Singh, Gaur & Ramakrishnan 2017). Similar studies have revealed that PU is a factor that influences behavioural intention in most of the organisations that have adopted the Accounting Information System (AIS) (Diatmika, Irianto & Baridwan 2016; Lee 2016; Martin & Matlay 2001). Alok and Mocherla (2016), in their empirical study on Enterprise Resource Planning (ERP) systems in the manufacturing industry, also found that when employees view the ERP system as useful, it translates to a higher intention to use it. The definition of PU in this study, therefore, will encompass any users' perceptions of how technology will positively contribute to the performance of a task.

2.3.5 Perceived Ease of Use (PEOU)

The original TAM construct of PEOU has been maintained in all three versions without any significant modifications (Davis 1989). The PEOU is defined as the degree to which an individual believes that using an IT system is intuitive in nature (i.e., PEOU is tied to an individual's assessment of the effort involved in the process of using the system - see Davis 1989). Therefore, users' behavioural intent is determined to increase when the system is not difficult to use (Davis 1989). TAM posits that PU will be influenced by PEOU because, other things being equal, the easier a technology is to use, the more valuable it can be. PEOU is a key driver of user acceptance and usage as it provides leverage points to create favourable perceptions and foster user acceptance and actual use (Bamigbola & Adetimirin 2020; Venkatesh 2000). With an increase in direct experience with the target system, individuals adjust their system-specific PEOU to reflect their interaction with the system.

Studies have found the significant influence of PEOU on behavioural intent is largely based on the perceived simplicity of the system (Mohd *et al.* 2020; Yeom *et al.* 2013). For instance, the study conducted on using the Computer Based Assessment system among undergraduate student's shows that PEOU significantly influences behavioural intention to use the system (Terzis 2011). In addition, research on the use of 3G mobile service found that PEOU influences behavioural intent amongst smartphone users (Suki 2011). In a similar study, PEOU directly affected behavioural intent to use electronic air-ticketing due to the ease of using the system amongst different age groups (Gurito & Siringoringo 2013; Kim, Kim & Shin 2009; Li, Qi & Shu 2018). The finding from Zhang, Zhao and Tan (2008) also asserted that one's PEOU influenced students' acceptance of web-based learning. According to Phua, Wong and Abu (2011), website designers play an important role in developing suitable and easy to use education-oriented websites for students and teachers as PEOU is important in influencing the user's behavioural intent whether to accept or reject the websites. Ong and Lai (2006) confirmed that if the level of PEOU towards online learning is high, then the acceptance and usage of online education will also increase. When the users perceive that technology is easy to use, they will assume that the system is simple and will be satisfied with using it (Ramayah, Jantan & Noraini 2003; Sun & Zhang 2008; Taylor & Todd 1995). The definition of PEOU in this study, therefore, will encompass any users' perceptions of how intuitive a given technology is to engage with and use in the performance of a task.

2.3.6 Behavioural Intent

The original TAM construct of behavioural intent has been maintained in all three versions without significant modifications (Davis 1989). Behavioural intent is defined as the degree to which a person has formulated consciously plans to perform or not perform some specified behaviour (i.e., behavioural intent is used to measure the strength of an individual's intention

to perform a specific behaviour for using technology – see Davis 1989). The effect of PEOU on behavioural intent will be moderated by experience as individuals will place less importance on PEOU while forming their behavioural intent in using the technology. For instance, the study by Shihab, Meilatinova and Hidayanto (2017) helps understanding the determinants for the acceptance of Computer Assisted Audit Tools and Techniques (CAATT) found empirical evidence that behavioural intent directly influenced CAATT adoption. The results confirmed that the auditors tended to adopt CAATT due to the excellent quality outcomes and lesser technological complexities. In addition, Mohamad, Radzi and Hanafiah (2021) examined how mobile technology acceptance influences customers' intention to book hotel rooms via smartphone. The results confirmed that PU and PEOU both significantly influenced consumers' behavioural intent towards the use of mobile hotel booking as they consider the online environment useful and easily accessible, which boosts their intention to book online. Hsu, Chen and Lin (2017) investigated how knowledge about a green IT product (e-books) influences consumers' buying intention. The findings confirmed that both PU and PEOU had a strong effect on consumers' behavioural intent as consumers with higher knowledge about the green e-book knowledge intend to purchase e-books if they are confident about their ability (e.g., if it is not from their budget) and/or if they have positive attitudes toward e-books (e.g., the attitude that e-books are good for the environment). The definition of behavioural intent in this study, therefore, will encompass the factors that show an individual's intention of using any technology.

2.3.7 Subjective Norm

The TAM external variable of subjective norm was introduced in TAM 2 and defined as an individual's perception that those who are important to him or her would approve or disapprove the performance of a given behaviour (i.e., subjective norm refers to the perceived social

pressure on the user's decision to use or not to use the technology – see Venkatesh & Davis (2000). Therefore, the higher the perceived expectations from the associates, the stronger the subjective norm, and the stronger the subjective norm, the stronger the intention to perform that behaviour (Hong *et al.* 2018; Wang *et al.* 2003). TAM 2 posited that subjective norms have a direct effect on PU and PEU, which in turn impacts behavioural intent and then actual usage. Marcinkiewicz and Regstad (1996) supported this, who studied the influence of subjective norm on computer use and found that subjective norm was most predictive for computer usage. The study by Lu, Yao and Yu (2005) also asserted that subjective norm plays a significant role in shaping individuals' perceptions of PU and PEOU. Schepers and Wetzels (2007) confirmed that subjective norm to be particularly important in influencing users' attitude towards the use of technology. The work on the effects of subjective norm on learner's e-learning acceptance by Abdullah and Ward (2016) found that subjective norm was related to peers, teachers, and educational institution policies that influence students' tendency to use an e-learning system. Schepers and Wetzels (2007) in the meta-analysis of the TAM demonstrated that the subjective norm had stronger effects on students for technology acceptance since they were more technology-ready, sensitive to trends, and easily influenced by peers' opinions than non-students or older users. Studies by Dye (2000), Lee and Connor (2003) and Fletcher, Sarkani and Mazzuchi (2014) confirmed that subjective norm can be created by word of mouth when considering technology acceptance in a consumer market since a positive or negative 'buzz' can make or break a product in such a situation. Herrero and Rodríguez (2008) and Pavlou and Fygenson (2006) presented an e-commerce acceptance model to identify subjective norm that can predict a buyer's attitude and then purchase intention. The definition of Subjective Norm in this study, therefore, will encompass the social influence aspect, that is, the perception that relevant others think that one should or should not use the technology.

2.3.8 Image

The external variable of image was introduced in TAM 2 and is defined as the degree to which an innovation is perceived to enhance ones' status within a social system (i.e., image reflects individuals' considering use of technology to improve their standing amongst the peer group – see Venkatesh & Davis 2000). Therefore, if peers at work are favourable towards new technology, employees will more likely form favourable perceptions towards the system through social influence processes (Venkatesh & Davis 2000). TAM 2 posits that image and subjective norm will positively influence PU through processes of internalisation and identification, respectively. Also, peer support may influence subjective norm and image associated with using a technology (Venkatesh & Davis 2000). For instance, the study conducted by Laukkanen *et al.* (2007) examined mature consumers in the mobile banking context. Using an Internet survey, the authors found that image was especially salient for aging consumers to adopt the technology. Lu, Yao and Yu (2005) and Schepers and Wetzels (2007) found social influences from social networks and the sense of image play important roles in shaping individuals' perceptions of PU. A study by Chau and Hu (2002b) on technological acceptance by healthcare professionals concluded to have significant positive effects of image on PU on physicians when making the technology acceptance decision, as they value their peer's assessment on the use of a particular technology. In addition, Wu *et al.* (2011) found that users' image for using Web 2.0 websites positively influences PU. Users' perception is influenced by the degree of interaction and sharing by Web 2.0. The perception then determines whether the user will continue to use Web 2.0 websites, and it then further increases the time and frequency of using the websites. The definition of image in this study, therefore, will encompass any users' perceptions about the desire to maintain certain behaviour that can persistently enhance the quality standing amongst others.

2.3.9 Job Relevance

The TAM external variable of job relevance was introduced in TAM 2 and is defined as the degree to which a person believes that the technology is relevant to his or her job (i.e., job relevance is a function of the importance within one's job of the set of tasks the system can support – see Venketash & Davis 2000). Therefore, job relevance is regarded as a cognitive judgment that directly affects PU, distinct from social influence processes. TAM 2 posits that job relevance has a positive effect on PU. According to Speier, Vessey and Valacich, (2003), if a system can accurately provide users relevant information in a timely manner and in an understandable format and help them make better decisions, users will more likely perceive greater job relevance of the system. For instance, Amir and Ismael (2016) used TAM as a predictor to explain the agricultural experts' behaviour in acceptance of technology. They found job relevance affecting the perception of employees on the actual application of technology in doing the responsibilities of the job. Their study suggests that demonstrating the capabilities of technology in relevance to doing the job duties better breaks down the resistance to technology acceptance by the employees. The study by Liang, Xue and Byrd (2003) on Personal Digital Assistance (PDAs) usage in healthcare professionals found job relevance to have strong predicting power for PU in the healthcare settings. They related job relevance to the healthcare professional's cognitive thinking as the enhancement in PU will be achieved if the healthcare professionals know exactly their job goals and the functionalities of the PDA's. Therefore, healthcare organisations can provide their employee's adequate information access to establish the mental linkage between job goals and PDA functions. The study conducted by Alharbi and Drew (2014) helped to understand the academics behavioural intent to use Learning Management Systems (LMS), correlated job relevance strongly with PU and found a significant positive relationship between the two variables. The results showed that when academics believe that LMS use for teaching is relevant to their job duties, it affects their

behavioural intent and develops a positive attitude towards utilising the system. The definition of job relevance in this study, therefore, will encompass any user's perceptions of how applicable a given technology is to their job.

2.3.10 Output Quality

The external variable of output quality was introduced in TAM 2 and is defined as the degree to which an individual believes that the system performs his or her job tasks well (i.e., output quality is the rate at which an individual believes that a system is performing its task – see Vankatesh & Davis 2000). Therefore, high-quality output systems are more likely to be utilised (Vankatesh & Bala 2008). TAM 2 posits that output quality has a positive effect on PU and is a factor that affects acceptance. For instance, the study conducted by Jauk *et al.* (2021) also confirmed output quality highly relevant for the application of complex machine learning models in healthcare. In addition to this, Aqleh *et al.* (2019) studied factors that affected the acceptance of Electronic Medical Records (EMR) among physicians in Bahrain. They found that acceptance of EMRs is positively affected by output quality as it provides faster and easy access to medical information, helping healthcare professionals with their decision making. A study by Ismail, Celebi and Nadiri. (2019) on identifying the factors that lead to students' trust and satisfaction in using Student Information System (SIS) in Cyprus university found output quality to positively affect the PU of the SIS. The research indicates a significant relationship between the degree to which students' trust and satisfaction over the system can perform the required tasks and the student's perceived usefulness of SIS over time. The definition of experience in this study, therefore, will encompass any users' perceptions of how well the system performs the given tasks.

2.3.11 Result Demonstrability

The external variable of result demonstrability was introduced in TAM 2 and is defined as the degree to which an individual accepts the tangibility, observability, and communicability of the results of using a system (i.e., result demonstrability implies that individuals can learn about the usage of innovation with less effort and explain the usage and benefit to others clearly – see Venkatesh & Davis 2000). Therefore, when tangible results of the technology are directly apparent, it is easier to communicate or show the benefits of the innovation to others. The potential adopters can comprehend the advantages of using new technology for their job well (Yi, Fiedler & Park 2006). TAM 2 posits that result demonstrability directly and positively affects PU (Li 2010; Venkatesh & Bala 2008). For instance, the study by Yi *et al.* (2006) and Imtiaz *et al.* (2019) demonstrated that individual perception of result demonstrability was a significant determinant of PU, indicating that when technology makes observable relevant results, that are easily recognisable by individuals, PU is affected. In addition to this, Ebnehoseini *et al.* (2020) investigated the factors affecting the acceptance of Electronic Health Record (EHR) technology in Iranian hospitals. They found result demonstrability had a direct effect on the behavioural intent of the healthcare workers in using the system and was one of the key factors applicable to assess the acceptance of EHR in hospitals. The work by Nam, Bahn and Lee (2013) on exploring the acceptance of Assistive Technology (AT) by special education teachers found result demonstrability to affect teachers PU significantly. The study outcomes showed that acceptance of AT since tangible results of the technology was directly apparent to teachers when they experienced and understood the advantages of using a new AT system. Shareef *et al.* (2011), Ahram *et al.* (2017) and Kooli *et al.* (2019) found result demonstrability to have a significant effect on PU, which in turn influences Business-to-Business marketers' intention to use social media sites. Business-to-Business firms encourage their employees to use social media sites, and because of their use, the performance of the

marketers is highlighted and praised. The definition of job demonstrability in this study, therefore, will encompass users' perceptions of all the quantifiable outcomes or benefits from using the technology.

2.3.12 Computer Self-efficacy

The construct of computer self-efficacy or perceptions of internal control was introduced in TAM 3 and is defined as the degree to which a person believes their ability to accomplish a certain task using technology (i.e., computer self-efficacy reflects one's beliefs regarding the capability of performing specific tasks successfully – Alfadda & Mahdi 2021). Therefore, the easier the system is to use, the greater will be the user's computer self-efficacy regarding their capacity to use the system comfortably. TAM 3 posits computer self-efficacy to be a crucial factor affecting individuals' intention to use technology (Kabir, Saidin & Ahmi 2017). For instance, the study conducted by Celik and Yesilyurt (2013) and Pellas (2014) stated that students who have higher computer self-efficacy are more likely to use e-learning and computer-supported education. In the context of integrated pest management practices, perceived computer self-efficacy shows how individuals consider their knowledge, skill, and capability for the advantageous application of Integrated Pest Management (IPM) (Rohollah, Leila & Mohammad 2020). According to Weng, Cheong and Cheong (2009), students with high self-efficacy perceived failure experiences as challenges rather than threats due to more substantial self-efficacy expectations. They found that students' attitudes and self-efficacy perceptions concerning computer-supported education are the primary factors in success in computer-supported education practices. Said *et al.* (2019) also found computer self-efficacy to positively affect the PU of e-learning systems. The results of their study suggest that when the students have sufficient computer skills and expose a positive tendency to interact spontaneously with the e-learning system, their use of the system would increase. The

definition of computer self-efficacy in this study, therefore, will encompass any users' perceptions about their ability to successfully execute a behaviour required to produce the desired outcome.

2.3.13 Perceptions of External Control

The external variable of perceptions of external control or facilitating conditions was introduced in TAM 3 and is defined as the degree to which an individual perceives that an organisational and technical infrastructure exists to support the use of the system (i.e., perceptions of external control is the state at which all the necessary facilities, tools, equipment and assistance are provided to an individual to support the use of a system - see Venkatesh & Bala 2008). For instance, a study conducted on the use of technology revealed that perceptions of control have significant effects on behavioural intent to use web-based training systems (Alrawashdeh, Muhairat & Alqatawnah 2012; Chintalapati & Daruri 2017; Salloum *et al.* 2019). Similarly, other related studies showed that perceptions of external control influence students' intention to use an interactive whiteboard and e-learning system in a learning environment (Teo 2010; Teo & Khine 2009; Wong & McDowall 2013). It was found that there is a significant relationship between perceptions of external control and behavioural intent to use wireless connectivity on mobile phone technology (Lu, Yueh & Lin 2015). Venkatesh *et al.* (2003) found that the influence of perceptions of control on usage is moderated by age and experience of the individual. Many studies have suggested that perceptions of external control affect the use, either directly or through behavioural intent (Taylor & Todd 1995; Xie *et al.* 2021). Groves and Zemel (2000) found that, among the perceptions of external control provided to teachers, technical support was ranked highly on the list of factors that affected teachers' perception of technology acceptance. Teo and Milutinovic (2015) used perceptions of external control in their version of the TAM 3 to assess factors that affect technology use in

mathematics teaching amongst pre-service teachers. Fathema, Shannon and Ross (2015) integrated perceptions of external control as one factor influencing the implementation of a Learning Management System. The study shows that user is likely to develop a more approving attitude towards the technology when perceptions of external control (i.e., training, tutorials, support, etc.) are present. The definition of perceptions of external control in this study, therefore, will encompass any users' perceptions about the environmental facilitations that influence system utilisation.

2.3.14 Computer Anxiety

The external variable of computer anxiety was introduced in TAM 3 and is defined as a persons' apprehension, or even fear, when they are faced with the possibility of using computers (i.e., computer anxiety relates to users' negative reaction towards using a computer with which they are not too familiar – see Donmez & Kir 2019; Venkatesh & Bala 2008). TAM 3 posits that with increasing experience, general computer anxiety is expected to continue to affect system-specific PEOU (Venkatesh *et al.* 2003). As user experience with the specific system increases, knowledge and anxiety-related adjustment are expected to be achieved. Computer anxiety can be controlled by providing effective computer training to employees to reduce their levels of anxiety when using a computer in their daily office tasks (Achim & Kassim 2015; Taylor & Todd 1995). Several studies have found computer anxiety a strong predictor in the acceptance of technology (Busch 1995; Inan & Lowther 2010; Wu *et al.* 2008). For instance, the study by Agbatogun (2010) asserted that computer anxiety has both direct and indirect effects on learners in becoming knowledgeable users of a computer. Durndell and Lightbody (1994) found that anxiety fallouts from little computer experience; the more exposure, the less anxiety someone will experience. Saade and Otrakji (2007) confirmed that with the continuous development of richer and more integrated interfaces, anxieties about learning to use the new

interface and executing tasks effectively becomes of primary importance in an LMS environment. Rosen and Weil (1995) suggested that the quality of support, friendliness, and enthusiasm of the student using the LMS profoundly influences the prevalence of computer anxiety. According to Konerding (1998), Hong *et al.* (2002) and Thatcher *et al.* (2007) an increase in system training can lower computer anxiety, thus improving performance and willingness to learn. The definition of computer anxiety in this study, therefore, will encompass any users' anxious or emotional responses when it comes to while using technology.

2.3.15 Computer Playfulness

The external variable of computer playfulness was introduced in TAM 3 and is defined as the degree of cognitive spontaneity in an individual's interaction with using any new system (i.e., computer playfulness is as a motivational characteristic in using a computer; that is, feeling good or having fun in performing computer-related activities - Venkatesh & Bala 2008). Therefore, an individual with a higher degree of playfulness or pleasure in computer-related activities will positively perceive its ease of use, which will translate to continual usage. TAM 3 posits that Computer playfulness is an anchoring factor determining PEOU for technology use (Venkatesh & Bala 2008). A study conducted by Lin, Wu and Tsai (2005) indicated that computer playfulness is a vital consideration of World Wide Web usage. In a similar study on web usage, Moon and Kim (2001) found the dimensions of computer playfulness were the extent to which the individual (i) perceives that their attention is focused on the interaction with the Web; (ii) is curious during the interaction; and (iii) finds the interaction intrinsically enjoyable or interesting. Yuan *et al.* (2017) explored the relationships among employees' usage intention pertaining to mobile information devices. The results of the study show that PEOU for the members of the organisation was affected by computer playfulness. A user-friendly and captive design style based on intuitive and interesting principles helped the organisation

members feel at ease when using the system and naturally enhanced utilisation. In addition, Padilla-Meléndez, Guila-Obra and Garrido-Morena (2012) examined perceived playfulness in the context of a Blended Learning Setting (BLS) with existing gender differences. They found playfulness and PU to be key drivers for the acceptance and use of BLS depending on the user's gender, where playfulness directly impacted PU, affecting the student's attitude towards technology by making learning via BLS 'useful fun.' The definition of computer playfulness in this study, therefore, will encompass any users' perceptions of how much pleasure and satisfaction they get while using any technology.

2.3.16 Perceived Enjoyment

The external variable of perceived enjoyment was introduced in TAM 3 and is defined as the extent to which the activity of using a specific system is perceived to be enjoyable, aside from any performance consequences resulting from system use (i.e., perceived enjoyment is referred to the activity of using a specific system is perceived to be enjoyable in its own right, aside from any performance consequences resulting from system use - see Venkatesh & Bala 2008). Therefore, when there is an increase in direct experience with a specific system, the function of system-specific perceived enjoyment is expected to increase (Venkatesh 2000). TAM 3 posits that perceived enjoyment is an anchoring factor determining PEOU for technology use (Venkatesh & Bala 2008). For instance, a comparative study on factors that influence students to use web-based learning information systems conducted by Saade, Nebebe and Tan (2007) revealed that perceived enjoyment impacted students' PEOU of an online learning environment. In a similar context, Ongena, Wijngaert and Huizer (2013) examined the antecedents of user acceptance of an online audio-visual heritage archive using TAM 3. They found that perceived enjoyment was a better predictor of users' PEOU of visual heritage archives. Teo and Noyes (2011) found that perceived enjoyment was a significant predictor of

PU in explaining pre-service teachers' intention to use technology. They indicated that teacher educators should focus on aspects of their curriculum in ways that promote enjoyment for the pre-service teachers helping them to acquire relevant skills for their future jobs as teachers and at the same time experience enjoyment from these activities. The definition of perceived enjoyment in this study, therefore, will encompass any users' perceptions of how an individual perceives the process of using technology-related activity is enjoyable to them.

2.3.17 Objective Usability

The external variable of objective usability was introduced in TAM 3 and is defined as a comparison of systems based on the actual level (rather than perceptions) of effort required to complete specific tasks (i.e., objective usability refers to how functional and utilisable a system is and the influence it has on user's perception of a system's ease of use - see Venkatesh & Bala 2008). TAM 3 posits that objective usability measured in the early stages of user interaction appears to serve as a very useful predictor of long-term perceptions of PEOU of a target system. Objective usability can also be evaluated through user observation using technology (Muchran & Ahmar 2019; Raaij & Schepers 2008; Zhang, Maron & Charles 2013). Objective usability on PEOU is shown to grow as experience with technology increases (Venkatesh 2000; Venkatesh & Bala 2008). Wahab *et al.* (2011) stated five factors that must be considered in defining objective usability: ease of learning, task efficiency, ease of remembering, understandability, and subjective satisfaction. Muchingami (2018) study found objective usability as a major component that drives customers' behavioural intent to use mobile banking. The research results indicate that commercial banks exhibited to their customers the benefits they were bound to have by subscribing and using mobile banking over traditional banking methods for them to switch to the modern banking way. Ajala and Adetimirin (2018) studied the adjustment factors for using online databases by postgraduate students in Nigeria. They

discovered that objective usability was one of the main factors influencing postgraduate students' use of online databases. Objective usability of an online database refers to how easy such a database is to use, independent of the user's experience. It is rated by how easy it is for a user to use such an online database even if they have no prior experience in its use (Ebrahimzadeh 2017; Pani & Mahapatra 2013; Salem 2019). The definition of objective usability in this study, therefore, will encompass observations on how the user interaction with technology is influenced by their direct experience with it.

2.4 Summary of TAM 3's development

A review of the extant literature indicates how TAM 3 has developed over time (and in a wide array of professional contexts, see Table 1.1) to include concepts critical to the acceptance of new technologies and the drivers/barriers of their ultimate acceptance/rejection of it. It has also provided an outline of the theoretical criticisms directed at TAM 3's development noted in Chapter One – i.e., that the development of TAM 3 has not given due consideration to the range of multiple stakeholder management practices that may be critical to the acceptance of new technologies in dynamic and complex contexts (Ajibade 2018; Almaiah, Al-Khasawneh & Althunibat 2020; Dube, Van & Zuva 2020; Freiesleben *et al.* 2021; Li 2020; Okafor, Nico & Azman 2016; Raza & Standing 2010; Strub 2010; Zhang *et al.* 2020). The following section provides an overview of the most common stakeholder management practices represented in the Stakeholder Management literature.

2.5 Stakeholder management practices

Stakeholder management practices emerged from Freeman's (1984) Stakeholder Theory of Corporate Social Responsibility and the myriad studies that followed detailing the range and complexities of effective stakeholder relationship management. Whilst subsequent research

into Stakeholder Theory has provided finer-grained insights into the nature of stakeholder roles (see Agle, Mitchell & Wood 1997; Bhattacharya & Sen 2006; Kujala, Lehtimäki & Myllykangas 2016), stakeholder power and influence (see Frooman 1999; Henriques & Sadorsky 1999; Olsen 2017), stakeholder communications (see Rubinelli & Von Groote 2017; Verley 2007) and business ethics (see Bonnafous-Boucher & Rendtorff 2016; Jahn & Brühl. 2016; Moriarty 2016), the critical stakeholder management practices common therein can be categorised into three main components: stakeholder identification, stakeholder analysis, and stakeholder engagement (Lumpkin & Bacq 2019). Stakeholder identification involves management processes relating to the mapping of all salient stakeholder groups that are directly or indirectly affected by organisations mission and enacted policies – not just those with a direct financial interest (Elsbach & Cable 2019; Laplante, Laplante & Voas 2016). Stakeholder identification requires managers to undertake an array of external analyses (often referred to as ‘boundary spanning activities’) to develop a detailed understanding of which groups represent Primary Stakeholders (i.e., direct beneficiaries of any organisational strategy or policy – e.g., ‘target markets’), Secondary Stakeholders (i.e., those that are involved with or responsible for Primary Stakeholders – e.g., parents, spouses, carers etc.), and Key Stakeholders (i.e., government agencies or officials that are able to devise, enact and/or enforce legislation), so that an integrated and strategically relevant set of investments, communications and other interactions can be implemented effectively (see Mitchell *et al.* 2021; Perrault 2017; Raha, Hajdini & Windsperger 2021).

Once the array of salient stakeholder groups has been identified, stakeholder analysis is undertaken (typically as a marketing research exercise) to gather data and generate meaningful information about each stakeholder group (in terms of salient characteristics *inter alia*: their relative power, influence, legitimacy, urgency, communication requirements, preferences etc.) as they relate to the organisation’s strategy or policy (Almughairy 2021; Anastasi 2018). Once

these relationships to the organisation's strategy or policies are established, managers are then better equipped to implement more effective stakeholder engagement practices (Varvasovszky & Brugha 2000). Stakeholder engagement practices include any organisational process that seek to involve, collaborate, communicate and/or respond to stakeholder groups in the implementation of an organisational strategy or policy (Lindgreen 2019; Masefield *et al.* 2021 Pedrini & Ferri 2019). Stakeholder engagement involves meaningful consultation with stakeholder groups to (i) increase their required knowledge and understanding of the organisation's strategies and policies, and (ii) gather feedback to understand the nature of, and ultimately reduce, stakeholder concerns and conflicts (Elsbach & Cable 2019; Laplante, Laplante & Voas 2016; Raha, Hajdini & Windsperger 2021; Turner & Zolin 2012). Taken together, the stakeholder management practices noted above present a sound basis for addressing the theoretical criticisms currently directed at TAM 3 (see Ajibade 2018; Freiesleben *et al.* 2021; Ratjeana *et al.* 2020).

2.6 Specific research question

Given the theoretical criticisms and demonstrable stakeholder management issues evident in professional sports' acceptance of DRS technologies noted in Chapter One, there is an opportunity to explore how TAM 3 (as one of the predominant technology acceptance models in use) can be modified to improve its predictive and explanatory capabilities in these regards. This thesis, therefore, adopts a TAM 3 lens through which to investigate the intersection between the characteristics of DRS technologies and the related stakeholder management practices necessary to optimise their acceptance. This research, therefore, will seek to address the following specific research question: *To what extent does TAM 3 need to be modified to account for DRS technologies and related stakeholder management practices?*

2.7 Chapter summary

This chapter firstly identified the DRS technologies characteristics and their related stakeholder management issues by reviewing the DRS technology literature. Secondly, the chapter provided an overview of the evolution of extant TAM, TAM 2 and TAM 3, and definitions of all the external variables included within each subsequent iteration of TAM. Next, the chapter introduced the concept of stakeholder management practices and the roles that stakeholder identification, stakeholder analysis, and stakeholder engagement play in the process. Lastly, it presented the specific research question to be addressed in this thesis. The method used to address the specific research question forms the basis of the chapter that follows.

Chapter Three

Research Design and Methodology

3.1 Objective of the chapter

The objective of this chapter is to describe the research methodology adopted in this thesis. Consistent with the research aim, this thesis undertakes a quasi-longitudinal qualitative approach, accommodating theory building based upon existing knowledge and theoretical relationships yet remains sensitive to any patterns or phenomena that may emerge. This chapter explains the rationale and process of a single case study inquiry and details the methodological principles and procedures engaged in investigating the research question posed at the end of Chapter Two.

3.2 Research paradigm

The research strategy and design were guided by the research's underlying philosophical assumptions, commonly known as paradigms (Creswell & Creswell 2017; Neuman 2013). To manage the research process and execute it effectively and efficiently, the researcher had to clearly define the philosophical assumptions related to addressing the research question (Creswell & Creswell 2017). The philosophical assumptions of this research were embedded in the quasi-longitudinal constructivist paradigm of research. Mills, Bonner and Francis (2006) and Lincoln, Lynham and Guba. (2011) define constructivism as a research approach that emphasises individual's participation and considers reality as social construction by human actors. Following Blaikie (2009) and Lincoln, Lynham and Guba (2011), this research assumed that the construction of reality is possible through interaction with the social actors.

This research aimed to explore how multiple stakeholder management practices associated with the acceptance of DRS technologies in a professional sporting context could inform the modification of TAM 3 in these regards. For this, the views of multiple stakeholders in an exemplar sport were considered central to elicit the factors inherent to multiple stakeholder acceptance and effective use of DRS technologies (Steen 2011). Moreover, the necessity for multiple stakeholder views required a constructivist paradigm of research because it enhanced the ability of the researcher to gather multiple perspectives on the phenomenon (Blaikie & Priest 2017).

3.2.1 Philosophical assumptions

The research paradigm guides the whole research process with the help of the underlying assumptions (Creswell & Creswell 2017) and makes readers aware of the researcher's ontological, epistemological, and methodological choices. The researchers then can validate the appropriateness of the decisions regarding research philosophy and its suitability with the intended outcomes. The ontological belief of this research assumed that the interviewees in this research have specific views and the construction of reality is relative to each participant's context. The relativist ontological belief is associated with the constructivist paradigm of research and assumes that the world consists of multiple individual realities and is influenced by each individual's context (Blaikie 2009; Mills, Bonner & Francis 2006; Lee 2012; Lincoln, Lynham & Guba 2011). This research aimed to explore the reality interpreted by different actors over their time and involvement in implementing the ICC's DRS technology. The role of multiple stakeholders in experiencing the ICC's implementation of DRS technologies is considered central in this research. It was also assumed that various stakeholders would report multiple observations as a constructed reality according to their contextual influences (Creswell & Creswell 2017).

Epistemology clarifies the relationship between the inquirer and the knowable (Lee 2012). Lee (2012) and Mills, Bonner and Francis (2006) discuss that the constructivist paradigm of research contains the subjectivist epistemology where the knowable can be co-constructed by the inquirer and the interviewee (Lincoln, Lynham & Guba 2011). In this research, multiple stakeholders in sports could improve the understanding of the inquirer about DRS technologies by constructing the reality based on their particular context. The research paradigm and underlying philosophical assumptions highly influence the methodology (Mills, Bonner & Francis 2006). Methodological assumptions include the research logic and methods used to investigate the matter (Creswell 2013). The research strategy, often referred to as logic, determines how to answer the research question by elaborating a starting point, a series of steps and an endpoint (Blaikie 2009). This research used an inductive approach to proceed among the commonly used research strategies (inductive or deductive). Inductive logic assisted the researcher in investigating the phenomenon through the interviewees' knowledge of the DRS technology acceptance from their stakeholder perspective (Blaikie 2009). In this research, the role of multiple stakeholders in the ICC's implementation of DRS technologies as interviewees was significant because they could comment on their individual and collective experiences (both positive and negative). Moreover, the inductive logic appeared to be a suitable research strategy for this research which led to the choice of the research design discussed in the next section.

3.3 Research design

The research gap identified in Chapter Two indicates that extant TAM 3 requires modification to account for the unique characteristics of DRS technologies and related stakeholder management practices. This study, therefore, seeks to modify TAM 3 for DRS technologies and related stakeholder management practices in the professional sports context; hence

providing the answer to the following research question: *To what extent does TAM 3 need to be modified to account for DRS technologies and related stakeholder management practices?*

In order to answer to the specific research question, this study was designed in two stages: firstly, a detailed history of the ICC's implementation of DRS technologies was constructed from secondary data (and presented in Chapter Four). Secondly, the critical incidents arising from the historical account of the ICC's implementation of DRS technologies were used to form the interview questions asked of the professional stakeholders in Stage Two. According to Bromley (1990, p.302), case study research is defined as a "systematic inquiry into an event or a set of related events which aims to describe and explain the phenomenon of interest".

Hakim (1987, p.61) notes that case studies:

... take as their subject one or more elected examples of a social entity – such as communities, social groups, organisations, events, life histories, work teams, roles or relationships – that are studied using a variety of data collection techniques....Using a variety of data collection techniques and methods allows a more rounded, holistic study than with any other design.

Similarly, Yin (2018) suggests that case studies are not a specific research technique as such but rather a way of organising social data to preserve the unitary character of the phenomenon being studied. Being amenable to accumulating data from various sources holistically, case studies have been widely applied to research into government policy creation (for example, see Boddy, 2000; Gordon & McCann, 2000; Peters & Hood, 2000).

3.3.1 Case study protocol

The procedures employed by this study while conducting case research are documented in the case study protocol (see Appendix A). The protocol instrument details the guidelines and rules that were followed while implementing the single case study, thus representing a constant reminder of the rationale for the study and enabling other researchers to understand the

method used (Yin 2018). Yin (2018) suggested that the case study protocol is a valuable tool in the case study method. The protocol provides a checklist that increases the reliability of the case study research and guides the investigator when undertaking data collection from a single case study. The case study protocol was devised before the commencement of the research action, given the importance of using this tool to identify the crucial factors essential in the planning and execution of this research. Thus, the design of this research, the nature of the data gathering process, and the primary form of the thesis were determined by the case study protocol. The following sub-section discusses the various elements of research design adopted in this research.

3.3.2 Unit of analysis and observation

Yin (2018) suggests that the unit of analysis and unit of observation can be distinguished based on their levels. The unit of analysis is what is to be analysed, while the unit of observation is what is to be studied to analyse the unit of analysis (Lavrakas 2008). This research aimed to explore the extent to which TAM 3 needed to be modified to incorporate DRS technologies and the related stakeholder management practices necessary in the professional sports context. Professional sports have adopted the DRS technologies to a greater extent (Collins 2012). These new technologies are primarily used to adjudicate decisions rather than support the generation of the 'original' decision. Examples of these technologies include Hawk-Eye used in Tennis, Goal Line in Football, Television Match Official in Rugby, Decision Challenge System in Baseball and Decision Review System in Cricket (Hawk-Eye 2021). However, compared to other professional sports, the ICC's DRS technologies are used for the highest proportion of decisions across a given match (Borooah 2016) and makes it a useful exemplar case for this research. Therefore, the unit of analysis in this study is the ICC's implementation of its DRS technologies, and the unit of observation was the stakeholder management practices

evident in professional sports context (NB: the ICC's current processes for DRS technology use is included in Appendix B).

Sampling technique	Aim	Suitable for this research:	Rationale
Convenience	Statistical generalisability	No	The aim of this research is not statistical generalisability.
Snowball	Statistical generalisability	No	The aim of this research is not statistical generalisability.
Quota	Ensuring the representation of the population from each criterion defined by the investigator	Yes	This research aims to find representatives from different categories of cricket stakeholders as a professional sport and investigate the phenomenon based on their opinion.
Judgmental	Finding the best results through careful selection of the participants	No	Though this research aims to find the most suitable interviewees, each category of the stakeholders should be represented in the sample, which is not possible through this technique because it does not divide the population into different types of stakeholders.
Theoretical	Theory development and new themes observation	No	This research aims to offer future theory development propositions, not theory development, at this stage.

Table 3.1: Sampling techniques and their suitability for this research - adapted from Blaikie (2009)

3.3.3 Sampling and interviewees

Sample selection is an essential step in the research because it helps reach the aims and goals of the study (Blaikie & Priest 2019). The method literature emphasises the importance of sample selection and the type of sampling procedure to be followed to draw conclusions aligned with the aims and objectives of the research (Blaikie 2007; Blaikie & Priest 2019; Cassell, Cunliffe & Grandy 2017; Creswell & Creswell 2017). There are five different non-probability sampling techniques: convenience, snowball, quota, judgmental and theoretical sampling (Blaikie 2009) explained in Table 3.1 and their suitability for this research. Creswell (2013) and Creswell and Creswell (2017) suggest using the non-probability sampling technique for the qualitative studies because it allows the researcher to select and reach the interviewees that can best contribute to the research findings. In this thesis, a non-probability quota sampling

assisted in exploring the phenomenon by selecting professional interviewees directly involved or affected by the ICC's implementation of its DRS technologies (Creswell & Creswell 2017).

Sharma (2017) further argues that quota sampling improves the representation of participant groups. This thesis sought to interview multiple stakeholders to the ICC's international cricket competition as its key interviewees; multiple stakeholders were broadly divided into 'participant' and 'non-participant' stakeholder groups. Participant stakeholders include all those that are directly involved in the performance and management of the game of international cricket (e.g., players, umpires, coaches, match referees and scorers). Non-participant stakeholders include all other groups that are indirectly affected and/or are able to influence the game of international cricket (e.g., administrators, managers, educators, journalists, and media). Creswell and Creswell (2017) discuss the investigator's role in selecting the participant by asserting that the investigator determines the criteria for the selection of the interviewees who can best inform the various aspects of the phenomenon under discussion. After the review of the literature following criteria was developed to select the interviewees for this study:

- i. Participants may possess multiple perspectives; and
- ii. At least one of their designations must be in a professional capacity

A rigorous approach was then followed to recruit the interviewees required to represent the salient stakeholder groups. Firstly, the leading cricketing authority in Australia (i.e., Cricket Australia) was contacted by email explaining the project and requesting the stakeholders' direct contact or sharing the project details with interested interviewees. The information sheet, consent form and ethics documents for the potential participants were also attached with the initial email to the cricketing body (see Appendices C, D, E, F and G).

Primary Role	Level	Other Roles
Umpire	International	Player/Coach
Referee	National	Team Manager
Educator	National	Umpire/Referee
Administrator	National	Match Referee
Scorer	International	Umpire/Administrator
Coach	National	Player
Player	State	Volunteer
Cricket Writer	International	Player
Player	State	Coach Junior Cricket
Technologist	National	Player/Coach/Umpire
Umpire	International	Player/Match Referee
Umpire	National	Player
Commentator	International	Player/Administrator
Technologist	National	Player
Player	International	Coach Premier Cricket

Table 3.2: *Examples of stakeholders with multiple roles*

A list of stakeholders was obtained from Cricket Australia, and the stakeholders were then contacted through email. The time and medium of the interview were decided as per the convenience of the interviewees. The majority of stakeholders recruited for this study performed a variety of cricket roles. As such, they provided multiple perspectives that increased the number of views collected via the interviews. The interviews covered each role of stakeholders by asking the same set of questions for each role separately. However, multiple interviews were conducted to cover each role individually for the same stakeholder in some cases. Table 3.2 provides an overview of the multi-perspective stakeholders with their primary role and at the level performed. The experience levels were categorised as ICC (International), Cricket Australia (National), and State Cricket. It also highlights any other role(s) that the stakeholder had performed.

3.3.3.1 Conduct of interviews

Whiting (2008) discusses the types of interviews for the research as structured, semi-structured, and unstructured. Qualitative studies based on inductive research logic can benefit most from semi-structured or unstructured interviews (Jansen 2010; Whiting 2008). For this study, semi-structured interviews were conducted because the researcher has no prior frame, unlike the unstructured interview requirement. Qualitative interviews are preferably undertaken face-to-face because of their strong ability to build rapport with the interviewees and gather the most relevant rich data (Farooq & De Villiers 2017). However, in this study, mixed of face-to-face and telephone interviews were conducted. The main reason to use a mixed technique was the physical location of the interviewees in many cases. There were many targeted interviewees who were located worldwide; contacting them involved cost and time-related issues. However, for the local interviewees, interviews were planned to be face-to-face. Still, only four interviews were conducted face to face as due to pandemic (i.e., COVID-19) and social distancing protocols, the investigator had to switch the interview technique from face-to-face to online ZoomTM meetings. Farooq and De Villiers (2017) state that video-assisted interviews can serve a similar purpose for rapport building and non-verbal communications as face-to-face interviews. Telephone interviews were not considered a viable option due to their inability to build rapport and assess non-verbal communications (Farooq & De Villiers 2017; Rowley 2012).

In this research, the interviewees were given a choice to select video or audio interview mode; all interviewees were comfortable with technology-mediated video calls. The technology-mediated video interviews improved the privacy and autonomy of the interviewer and the interviewees. For example, the interviews of busy stakeholders were conducted according to their convenience without any pressure to be in a professional environment as required for face-

to-face interviews. Moreover, the interviewees got chances to schedule and reschedule the interviews, which improved their involvement. The interviewer also got an opportunity to write the notes and probing questions on the interview guide because the lack of eye contact was not a significant issue during the video interviews process, unlike the face-to-face interviews where lack of eye contact can be distracting.

	Participant Stakeholder Group				
	Player	Umpire	Coach	Referee	Scorer
International	2	4		1	1
National	2	1	1	2	
State	3	1	2		
League/Club	3		1		

Table 3.3: Number of interviews conducted with participant stakeholders

In total, 32 semi-structured interviews were conducted; each interview lasted between 30 and 60 minutes. Tables 3.3 and 3.4 provide the actual number of interviews that were conducted. However, if multi-perspective is applied to the tables, the number of perspectives will increase to more than double the number of interviews. For the non-participant stakeholder groups, a similar recruitment approach for the participant stakeholder was used. In total, fourteen interviews were conducted to ensure that the non-participant stakeholders' quota of at least one interview conducted for each stakeholder category was achieved. Table 3.4 provides a summary of the interviews conducted with the non-participant stakeholder group.

	Non-Participant Stakeholder Group				
	Administrator	Manager	Educator	Journalist	Media
International	1	1		1	1
National	1		2		
State	1				
League/Club					

Table 3.4: Number of interviews conducted with non-participant stakeholders

3.3.4 Data collection and procedures

This qualitative study aimed to elicit information insights into the implementation of DRS technologies and detect suitable theoretical contributions to extant TAM 3 can optimise technology acceptance through more attuned stakeholder management practices. To achieve this aim, the data collection process constituted a collection of primary data from salient stakeholders. Due to the study's exploratory nature, the suitable data collection process was semi-structured interviews (Creswell & Creswell 2017). The following sections explain the steps taken to improve the validity and reliability of the data collection process through interviews.

3.3.4.1 Adhering ethical guidelines

Qualitative inquiries face a range of ethical issues that are highlighted during the data collection process, and those ethical issues were addressed to protect the rights of the interviewees (Creswell 2013; Creswell & Creswell 2017; Yin 2018). For this study, ethical issues were addressed during the data collection to ensure this research's reliability—this research adhered to the University of Tasmania's ethical guidelines. The approval obtained from the University of Tasmania's Social Sciences Human Research Ethics Committee (SSHREC), is part of the human research ethics committee (Tasmania) network. The guidelines provided by SSHREC reflect the national statement on ethical conduct in human research (Australian Government 2007). Before the application submission to the ethics committee, the study's risk level was determined based on the guidelines provided by SSHERC. The application was assessed as a minimum risk application which indicated no or minimal risk of harm or effect of the research and research process on the participants and the researcher. The ethical issues such as informed consent, the confidentiality of the information, anonymity of the interviewees, analysis, reporting and storage of data were communicated to the ethics committee after the discussion

with the supervisory team. Moreover, an information sheet was shared with the participants before the interview stating their rights of withdrawal, audio and video recording permission, statements to clarify the project aims, confidentiality and anonymity insurance statements, and information about the analysis and storage of data. The interviewees were also requested to fill a consent form before the interview to fulfil the ethical guideline of SSHERC regarding informed consent. The interviewees were asked before and during the interview if they wanted to disclose their identity in this research and the subsequent publications. Kvale (2007) indicates that a researcher must consider ethical issues during the data analysis. This includes the right of the interviewees to assess their statements for accurate interpretation. This ethical issue was addressed during the interview by repeating their ideas and words and asking if they were accurately interpreted.

3.3.4.2 Interview protocol

This phase included introducing the project and the investigator to the interviewee and developing the interview guide. The first stage was completed before the interview when the information sheet was shared with the interviewees through email. Also, the initial few minutes of the interview were spent introducing the project and the research aim in case the busy interviewees missed reading the email before the interview. Moreover, their right of withdrawal at any time and permission to record the interview was again taken at this time. This helped building rapport with the interviewee and made them comfortable before the interview. The second stage was to develop a set of questions to be asked during the interview. This step was completed with the help of a literature review and finding the significant areas of investigation. The areas of inquiry identified were:

- i. To determine how has the implementation of the DRS technologies been perceived positively and negatively by the salient stakeholders? i.e., What is the impact on player behaviour? What is the effect on the action of the umpires?

- ii. To determine how the DRS technologies impacted the decision-making processes of the players and umpires? i.e., Under what circumstances is the DRS used by the players; How does the DRS technologies influence the umpire's decision-making behaviour?
- iii. To determine how could the DRS technologies be modified to more efficiently serve the needs of the salient stakeholder groups? i.e., multi-perspective analysis of the optimal role that the DRS technologies can play in the game of cricket.

It is important to note that the interview guide questions were not structured; instead, they guide questions to keep the interviewer on track.

3.3.5 Administration of the data collection process

For the interviews, twenty-four participant stakeholders and eight non-participant stakeholders agreed to participate in the study. Initially, four face-to-face interviews were audio-recorded, while twenty-eight interviews were conducted through Zoom™ computer application which allowed audio and video recording. A timer permitted the management of the time and duration of the interview. The interview guide was used to manage the interview content and keep the interview on track. Notes were taken on the printed interview guide to formulate probing questions at the time of the interview.

3.3.6 Data organisation and storage

The obtained data from both data collection phases were organised and stored to attain useful information. The downloaded files and scanned documents were arranged according to the analysis needs with proper names. Moreover, a database was generated mentioning the name of the record, year, an overview of the document, the name of the folder it is saved in and the importance of the record for the analysis. This made the documents easily searchable and analysed more efficient. The data collected through interviews was also saved in electronic folders. A pre-and post-interview database was developed; the pre-interview database included

a list of the interviewees, time of the interview scheduled, recording type required during the interview (video, audio, or none) and the interviewees' contact details. The post-interview database included the interview duration, highlights of the major points discussed, and essential interview notes, including anything new or unusual discussed by the interviewees.

3.4 Methodological and analytical decisions

Nowell *et al.* (2017) discuss that the qualitative researcher must declare the methodological and analytical decisions taken throughout the research to make it trustworthy. Under this section, the investigator's analytical and methodological decisions to explore the components of TAM 3 that can be modified for DRS technologies and the related stakeholder management practices in the professional sports context are discussed.

3.4.1 Methodological choice

The qualitative frame of research was chosen to conduct this research because the aim of this research was to seek an understanding of the phenomenon from the lens of the participants experiencing it. The data collection analysis in a qualitative frame varies based on the required richness to find the answer (Cassell, Cunliffe & Grandy 2017; Creswell & Creswell 2017). As this research aimed to explore the extent to which TAM 3 needs to be modified for decision review technologies, the role of multiple stakeholders using DRS technologies was vital. Hence, the information communicated through interviews needed a careful examination to cluster it into meaningful information bundles. The qualitative content analysis was chosen among the various qualitative analysis techniques because it offered a structured way of reducing communicative material to the concepts (Elo *et al.* 2014; Mayring 2004). Other methods are ethnography, grounded theory, hermeneutic phenomenology, descriptive phenomenology, and thematic analysis. However, none of these methods provides an analysis

of the communicated data combined with the context of the phenomenon (Hsieh & Shannon 2005; Vaismoradi, Turunen & Bondas 2013). The qualitative content analysis can be performed based on the research strategy and the logic (Elo & Kyngäs 2008). This research study followed the inductive research strategy, which indicates that little was known about the phenomenon and the research aimed to explore the components of a new model. Elo and Kyngäs (2008) discuss that if the study's approach is inductive, the categories are derived from the data and then are merged into the broader categories to find the answer to the research question more systematically.

3.4.2 Analytical decisions

Hsieh and Shannon (2005) explain three approaches to qualitative content analysis: conventional, directed, and summative. This research employed directed content analysis. Unlike conventional content analysis, directed content analysis helps contribute and improve understanding of the pre-existing theories or existing phenomenon by adding further points to them. Conventional content analysis is used in studies where the exiting phenomenon is to be understood the first time. This research study undertook the acceptance issues with DRS technologies in a sports context and sought multiple stakeholders' views to modify TAM 3 to account for DRS technologies and the related stakeholder management practices. Moreover, this study did not aim to observe the use of certain words and then do their latent analysis, which is the aim of the summative content analysis (Hsieh & Shannon 2005). Instead, it only seeks the views of the stakeholders on the phenomenon under discussion. The directed content analysis approach aims to conceptually or theoretically extend the theory (Hsieh & Shannon 2005). As the research approach of this study was inductive, the investigator had to analyse the similarities and differences in data, describe them as categories and then develop a theoretical understating (Graneheim Lindgren & Lundman 2017), unlike the deductive approach where

the content is analysed to test the existing theories. Elo and Kyngäs (2008) relate the process to conduct a qualitative content analysis as preparation, organisation, and reporting phases. The preparation phase helps to choose either inductive or deductive content analysis, and the organisation phase is related to selecting the unit of analysis. As discussed previously, this study aimed to use the inductive approach and unit of analysis for the implementation of DRS technologies by the ICC. The next step was to engage with the data and perform content analysis. The following analytical decisions were taken while performing directed content analysis.

3.4.2.1 Familiarisation with the data

Familiarisation with the data means engaging in the data's breadth and depth (Braun & Clarke 2006; Maguire & Delahunt 2017). This was a crucial step because it enabled the researcher to identify the patterns in the data during the analysis stage. The two steps were taken to familiarise with the data: immediately transcribing the recorded interviews and writing early impressions and critical points on the interview guide. The next step was to cluster the data points and label them as tags. This required reading the transcripts several times as many points were discussed at various places that could be tagged the same. For example, one interviewee, in reply to the question about his views on the technology, said:

I do not think that it has undermined the role of an umpire; rather I think what it does effectively is for umpires to be absolutely at the top of the game.

I think "Umpire's Call" is such an important element of this. I think even though I want us to use technology; I do not think that the technology should be completely used to overrule the umpire.

This statement indicated the role of technology and suggested some points about the umpires as well. By reading and re-reading the transcripts allowed clustering the data points and tagging

them for further data analysis. This process included tagging the data points with meaningful labels that helped develop the codes and categories.

3.4.2.2 Development of codes

Code development is about arranging the data in a systematic way (Maguire & Delahunt 2017). The code is a meaningful phrase assigned to a part of data (Jonny 2009). It significantly reduces the large and complex data by dividing it into smaller groups (Maguire & Delahunt 2017). In the directed content analysis, coding links the data to the idea and concept under investigation (Elo & Kyngas 2008; Richards & Morse 2012). As such, the literature review presented in Chapter Two provided the base coding categories and rules used in this research; the first three codes generated in the study related to the unique characteristics of DRS technologies (see Table 3.5).

Code	Coding Rule
Undermining the role of the original decision-maker	This node captures any data related to the use of the DRS technologies- and the intent to use it.
No support for the original decision	This node captures any data related to the technological assistance in the decision-making process.
Changes the balance of stakeholder power in the decision-making process	This node captures any data related to the involvement of any stakeholder in the decision-making process.

Table 3.5: Codes and Coding Rules for the characteristics of DRS technologies

Subsequent codes were created to account for the extant TAM 3 variables in Table 3.6 based on the extant TAM 3 presented in Chapter Two.

Code	Coding Rule
Computer Self-Efficacy	This node captures any data related to beliefs that one has the capability to perform a specific task using the computer.
Perception of External Control	This node captures any data related to any belief that organisational and technical resources exist to support the use of the system.
Computer Anxiety	This node captures any data related to individual's apprehension or even fear when she/he is faced with the possibility of using computers.
Computer Playfulness	This node captures any data related to any degree of cognitive spontaneity in microcomputer interactions.
Perceived Enjoyment	This node captures any data related to any extent to which the activity of using a specific system is perceived to be enjoyable in its own right, aside from any performance consequences resulting from system use.
Objective Usability	This node captures any data related to the comparison of systems based on the actual level (rather than perceptions) of the effort required to complete specific tasks.
Experience	This node captures any data related to any experience gained using technology.
Perceived Ease of Use	This node captures any data related to any belief that the use of technology will be free of effort.
Subjective Norm	This node captures any data related to the individual perception that most people who are important to him think he should or should not use the system.
Image	This node captures any data related to the use of an innovation that is perceived to enhance one's image or status in one's social system.
Job Relevance	This node captures any data related to the capabilities of a system to enhance an individual's job performance.
Output Quality	This node captures any data related to any belief that the system performs the job tasks well.
Result Demonstrability	This node captures any data related to any beliefs that the results of using a system are tangible, observable, and communicable.
Facilitating Conditions	This node captures any data related to any control beliefs to the resource factors such as time and money and technological compatibility issues that may constrain usage.

Table 3.6: Codes and Coding Rules for extant TAM 3 variables

Data that remained uncoded against any of the coding rules (generated by the literature review in Chapter Two) were then scrutinised for evidence of multiple stakeholders' experience of DRS technology and its inherent factors. Once the original codes were developed and populated with primary data, the data were interrogated using a tagging process to match the acceptance processes evident to extant TAM 3. The tagging process was the re-adjustment of the textual data into distinct categories and labelling them for their importance. In contrast, in the coding process, the text itself was labelled according to its potential contribution to answering the research question. Maguire and Delahunt (2017) state that any study employing

an inductive research strategy code every line of data to link it to the main idea under investigation. To this end, this study developed three levels of codes (as proposed by Elo & Kyngäs 2008) and labelled them as sub-categories, generic categories, and main categories. The sub-categories or open code development was the first step in which the tagged paragraphs were re-read to identify what each line of the transcript was saying. For example, Table 3.7 demonstrates the coding and sub-category development process. The table contains text extracted from one interview; tagged as important for 'participant stakeholders'.

Sub-Categories (SC) SC 1 = Authority of the umpire SC 2 = Undermine the role of original decision-maker SC 3 = Balance between stakeholders	Participant stakeholders
^{SC1} I sometimes wish an umpire had a review up his sleeve as well. ^{SC2} I have spoken to international umpires about when teams waste all their reviews, putting pressure on an umpire. ^{SC3} If teams have exhausted their reviews but had the umpire had one in his back pocket that will be a good thing. I don't know if this has been discussed but certainly worth a discussion even though there would be a difference of opinion, I am sure.	

Table 3.7: Coding and sub-category development

It is evident that each line of the excerpt was read and re-read to label it meaningfully. Each line did not need to create a new sub-category, and this pattern was present in the whole transcript. It is important to note that it was rare that the name of the category was finalised for the first time; for example, the first category titled 'undermine the role of original decision-maker' was first labelled as 'undermine the role of stakeholders', however in later interviews, it was found that the 'undermining' was related to only the original decision makers. By adjusting the definition of this sub-category, it was then labelled as 'undermine the role of original decision-maker.'

For each level of code, the researcher established an inclusion criterion and definition, and the statement meeting the criteria was included in that category. For example, in sub-category two of Table 3.8, the definition was 'any statement indicating the roles of the participant or non-participant stakeholders undermined by the technology or its usage'. It improved the analysis process and made the analysis quick, traceable, and logical. The second stage of coding was to cluster the relevant codes into a higher-level code which was termed as a generic category. The excerpt shows that the interviewee talked about education and its benefits for all stakeholders. The sub-categories developed after reading each line of the transcript was clustered into a broader category that represented the whole sub-category.

Generic Category (GC) GC 1 = Educating the stakeholders	Participant and non-participant stakeholders
Sub-Categories (SC) SC 1 = Need for educating stakeholders SC 2 = Making stakeholders mindful SC 3 = Engaging stakeholders through education	
^{SC1} I sometimes wish an umpire had a review up his sleeve as well. ^{SC1} I have no problem bringing technology in, but umpire education is a critical thing. ^{SC1} You need your umpires to embrace it and feel confident within their abilities to handle these new systems. I think we tried to do that and do it pretty well. ^{SC2} Same applies to the other users of the technology. ^{SC3} Education will make them more conscious, aware and engaged in the game.	

Table 3.8: *Generic category development*

The final step was to cluster the generic categories presenting a similar idea into the main category. Table 3.9 illustrates the development process of main categories used to answer to the research question (and form the basis of the discussion in Chapter Six). Hsieh and Shannon (2005) state that the findings from the coding can be supportive or non-supportive to the theory in directed content analysis. The codes and categories developed in this stage helped the

theoretical model's analysis and development, which partially supported the theory and is further discussed in Chapter Six.

Participant and non-participant stakeholders	Participant stakeholders
^{SC1} I sometimes wish an umpire had a review up his sleeve as well. ^{SC1} I have no problem bringing technology in, but umpire education is a critical thing. ^{SC1} You need your umpires to embrace it and feel confident within their abilities to handle these new systems. I think we tried to do that and do it pretty well. ^{SC2} Same applies to the other users of the technology. ^{SC3} Education will make them more conscious, aware and engaged in the game.	^{SC1} I sometimes wish an umpire had a review up his sleeve as well. ^{SC2} I have spoken to international umpires about when teams waste all their reviews, putting pressure on an umpire. ^{SC3} I If teams have exhausted their reviews but had the umpire had one in his back pocket that will be a good thing. I don't know if this has been discussed but certainly worth a discussion even though there would be a difference of opinion, I am sure.
Sub Categories (SC) SC 1 = Need for educating stakeholders SC 2 = Making stakeholders mindful SC 3 = Engaging stakeholders through education	Sub Categories (SC) SC 1 = Authority of the umpire SC 2 = Undermine the role of original decision-maker SC 3 = Balance between stakeholders
Generic Category (GC) GC 1 = Educating the stakeholders	Generic Category (GC) GC 1 = Role of decision-makers (SC 1 and SC 2) GC 2 = Engaging with multiple stakeholders (SC 3)
Main Category (MC) MC1 = Stakeholder Management Practices	

Table 3.9: Main category development

3.5 Ensuring the quality of research

Nowell *et al.* (2017) discuss that qualitative research can only be judged as trustworthy if credible, transferable, dependable, and confirmable. Credibility, transferability, dependability, and confirmability are considered parallel to the validity and reliability in quantitative research (Lincoln & Guba 1985; Nowell *et al.* 2017).

3.5.1 Credibility

Credibility relates to the accurate presentation of the phenomenon discussed by the interviewee and interpreted and co-constructed by the investigator (Creswell & Miller 2000; Tobin &

Begley 2004). When discussed parallel to the validity, credibility is related to the legitimacy of the research; it also includes avoiding bias (Adams 2015; Kvale 2007). The following section discusses how the legitimacy of this research was established.

3.5.1.1 The legitimacy of research and process

This qualitative study's legitimacy was established at two stages: the pre-data collection and post-data collection phases. In the pre-data collection phase, the investigation's accurate theoretical conception and identification areas were given due importance by carefully reviewing the literature. Peer validation was also taken before moving forward in the research after reviewing the literature by discussing the investigation area with the supervisory team. In the post-data collection phase, the analysis part was crucial to determine the legitimacy of this research. Creswell and Miller (2000) discuss establishing the credibility of the research embedded in the constructivist paradigm. The investigator must search the disconfirming evidence, show the interviewees' prolonged engagement, and provide a detailed and thick description of the research process. To search for the disconfirming evidence, this study's investigator read and re-read the transcribed material and developed codes and categories from various aspects. Analysis of multiple levels improved the chances to find disconfirming evidence and reduced the researcher bias. The interviewee's engagement was thirty minutes on average, which indicates that twenty-one hours of interaction with the interviewee was recorded in this research. A description of the entire process adopted in this study is provided in the previous Section 3.3. It covers a detailed description of the selection process of research design, analytical and methodological decisions.

3.5.1.2 Bias control

The other part of establishing the credibility of the research is to reduce bias. Bias is the unwanted distortion in data and is inevitable (Walliman 2017); however, steps can be taken to

reduce its impact. Different types of bias can be either induced from the investigator side or from the interviewee side. From the investigator side, the most critical bias that could affect this research's credibility was the confirmation bias, in which the researcher deliberately interprets the data to support the existing theory or themes (Roberts 2020). To control this bias was particularly important because this research aimed to identify the stakeholder management practices necessary for DRS technology implementation and propose a modified TAM 3. This bias was controlled by developing the categories from the data and validating those categories through the discussion with the supervisory team. Moreover, the response summary was repeated during the interview to identify any bias in the interpretation. The other biases that could be induced from the interviewee side were acquaintances bias and social desirability bias (Brinkmann & Kvale 2018; King, Horrocks & Brooks 2018; Roberts 2020; Roulston & Choi 2018) in which the interviewee chooses to agree with the statements or deliberately skew the viewpoint to gain social desirability. The interview questions were worded neutrally to control this bias and needed their opinion with the logic and reasoning behind their argument.

3.5.2 Transferability

Transferability is parallel to the concept of generalisability (Nowell *et al.* 2017) and is related to the case-to-case transfer of the research findings and implications (Tobin & Begley 2004). Transferability can be ensured for the effective implementation of DRS technologies and the related stakeholder management practices in any other context other than professional sports following the same methodological and analytical decisions taken in this research. The aim of this research has been to modify TAM 3 for the implementation of DRS technologies and the related stakeholder management practices in the professional sports context. The ICC case was chosen as an exemplary case where stakeholder dissatisfaction has been apparent. This research focused on analysing the process of acceptance strategies rather than the cricket aspects. This

means that the context of cricket is minimised, and the usefulness of the findings can be transferred more readily to other fields where DRS technology is adopted (and are not confined to the professional sports context). The qualitative strategies were then used to obtain data through the semi-structured interview questions. The ICC's implementation of DRS technologies and the related stakeholder management practices were then examined as an exemplar domain. The data was collected and transformed into codes to answer the earlier developed research questions for enhancing TAM 3 to account for DRS technologies and the related stakeholder management practices.

3.5.3 Dependability

Nowell *et al.* (2017) and Tobin and Begley (2004) discuss that the research can be dependable when the reader can find it traceable and logical in the process. As indicated by Nowell *et al.* (2017), the researcher is responsible for providing detailed descriptions of the parts of the research they want to make dependable; this whole research process is dependable as detailed descriptions of research design, analytical and methodological decisions, and analysis are provided, and a thick and a detailed description of all research processes have been included.

3.5.4 Confirmability

Lincoln and Guba (1985) indicate that confirmability is established when credibility, transferability and dependability are achieved (Nowell *et al.* 2017). This includes the trail of the process of how the results were achieved. This research ensures its confirmability through a rich description provided in the section of analytical decisions made where the process of interpretation of the data into meaningful codes and categories is discussed.

3.6 Intra-coder reliability

A key step in ensuring the data analysis process's credibility, transferability, dependability, and confirmability was to ensure high-levels of intra-coder reliability were achieved through the data analysis process. Intra-coder reliability refers to a coder's consistent manner across a specific time, allowing the researcher to assess the consistency of the coding process (Castleberry & Nolan 2018). Given (2008) mentions that assessing the intra-coder reliability includes reviewing the codes for consistency, re-reading the codes, labels, and inclusion criteria. Each code was defined during the engagement with data, and an inclusion criterion was mentioned to ensure consistency. For example, Tables 3.5 and 3.6 includes the definition of each code and their inclusion criterion. At regular intervals throughout the data coding process (i.e., after the 8th, 16th, 24th and 32nd) interview was coded, audits of the coded data were conducted to check for consistency and coding errors; the data analysis process for this research employed checks to test the coding protocol's consistency and stability. No errors were detected at any stage. The coding protocols adopted in this study were generated from the literature review (i.e., the characteristics of DRS technologies and extant TAM 3) provided in Chapter Two.

3.7 Chapter summary

This chapter provided an account of the methods used to answer the research question posed at the end of Chapter Two. The constructivist approach adopted in this study demanded the gathering of secondary qualitative data (for the purposes of creating a historical timeline of critical incidents) and primary qualitative data (to detect any stakeholder management factors relevant to DRS technologies and TAM 3). The findings emanating from the method described in this chapter form the basis for the Findings chapter to follow.

Chapter Four

International Cricket Council's Decision Review System: An Historical Overview

4.1 Objective of the chapter

This chapter aims to present a chronological timeline of events regarding the ICC's implementation of DRS technologies into its international cricket competitions. In doing so, it will highlight the type of DRS technology implemented over time and the issues and resolutions evident in the ICC's stakeholder management process.

4.1.1 March 2008: The ICC confirms the DRS technology trial during an international cricket competition

In March 2008, the ICC decided to trial the DRS technology during an international Test series between two full-member nations. The trial was initially planned for the Test series between South Africa and England in 2008, but the Boards of the countries disagreed on the details of the DRS technologies used:

The proposed trial of umpire referrals won't take place during England's Test series against South Africa in July 2008 after the boards disagreed on the finer details. There was unease among the players about testing the DRS technology in such a high-profile series and continued concerns about undermining the on-field umpires. The next possible use of the technology, which includes Hawk-Eye, was set to take place between India and Sri Lanka later in July. The inventors of Hawk-Eye, led by Paul Hawkins, tested the accuracy of the DRS technology, and claimed that it had a margin of error of just 5mm. However, the ICC decided that only the tracking part of the software will be used, up to the point where the ball hits the pad, rather than the predictive element of the package. The decisions involving whether the batter has been hit in line with the off-stump or where the ball had pitched clearer, were left in the umpire's jurisdiction to judge the path of the ball

towards the stumps. When the DRS technology was finally trialled, each side was allowed three unsuccessful referrals per innings. Similar technology was used in the 2007 Friends Provident Trophy (domestic cricket competition in England) but did not find favour among players or umpires. None of the 11 referrals were overturned by the Third Umpire, and there was a feeling that the Third Umpire did not want to overrule his colleagues. (ESPNCricinfo 2008a)

Elements considered for DRS technologies Implementation	
For Umpires	On-field Umpires were permitted to refer 'line decisions' (i.e., Run Out or Stumping decisions that required no subjective interpretation of the laws) or 'boundary decisions' (i.e., objective evidence that the ball had touched or pitched over the designated boundary on the full) to the Third Umpire without the need for a player review. The ICC recommended that the Third Umpire use Hawk-Eye technology only for the purposes to determine the actual direction of the ball up until the point when the batter intercepted it. The predictive function of the technology is not to be used (ESPNCricinfo 2008b).
For Players	The players will be allowed to ask the on-field umpires to review any aspect of any other decision by consulting with the Third Umpire. Players are required to formally request reviews by making a 'T' sign with their arm(s) and/or arm and bat. The player review process must proceed as follows: the-field umpires give their decision unaided by any technology; the batter involved in that decision or the fielding captain formally requests a review of the umpire's decision; the on-field umpires formally request a review of the most recent decision from the Third Umpire; the Third Umpire interrogates the available information from the approved suite of technologies to determine whether there is sufficient evidence to overturn the original decision; the on-field umpire receives advice from the Third Umpire and then gives their final decision accordingly.

Table 4.1: Elements to be considered in DRS technology implementation (Source: ESPNCricinfo 2008)

The trial was rescheduled for the Test series between Sri Lanka and India in 2008 (ESPNCricinfo 2008b; SportStar 2019). According to The ICC Press Release (ESPNCricinfo 2008b), the DRS technologies implementation consisted of the elements mentioned in Table 4.1. A BCCI press release (ESPNCricinfo 2008b) confirmed that the technologies discussed in the following subsection could be used by the Third Umpire when considering a player review request.



Figure 4.1: T Signal used by the captain requesting a review of the decision made by the on-field umpire (Source: CricTracker Website 2018)

4.1.1.1 Hawk-Eye

Hawk-Eye technology provides a ball-tracking tool that supports the Third Umpire to adjudicate LBW queries (Hawk-Eye 2021). It relies on several high-definition cameras that record each play from multiple angles when the ball leaves the bowler's hand. The cameras record an array of information, including the ball's speed, path, bounce, and data uploaded frame by frame to computer software. This software uses the stored data to predict the ball's direction by triangulating and combining it to produce a three-dimensional representation of the ball's path until the point where it makes contact with the batter. The approved DRS technologies for the trial were to be used for ball-tracking purposes only and not for predicting the ball's future trajectory (ESPNCricinfo 2008b; ESPNCricinfo 2011a; Hawk-Eye 2018;

Singh & Dureja 2018). This required the Third Umpire to personally predict the ball's future path, looking at the calculated images, as shown in Figure 4.2:



Figure 4.2: Path calculation through Hawk-Eye technology. The image on the left shows an actual impact of the ball, whereas the image on the right provides a calculated path produced by the Hawk-Eye technology used in the first DRS technology trial (Source: Cricket Master Website 2020)

4.1.1.2 Sound Trakker

Sound from the stump microphones with the replays at average speed and in slow motion. The live sound is picked up using the stump microphone, gets filtered and relayed using an attached oscilloscope that traces the relevant sound waves. Cameras record the visual and replay it in slow motion with the sound attached. If the ball has touched the bat, the sound graph gets ticketed for impact - a single, sharp spike on the graph confirming the ball is hitting the bat. The Third Umpire then correlates the sound with the ball's slow-motion replays, passing the bat to make their decision (Shenoy 2016).



Figure 4.3: *Sound Trakker Technology used in the DRS technology trial (Source: Cricket Master Website 2020)*

4.1.1.3 Slow-Motion, Super Slow-Motion or Ultra-Motion cameras

Slow Motion cameras allow the umpires to record videos using a high-speed camera and capture moments that the human eye can't see by playing them eight times slower than standard. A 'Slow-Mo' featured camera can take shots up to 32 times slower than a Slow-Motion camera (NACIC 2020; Samsung 2020).

4.1.2 July 2008: Trial of DRS technology in international cricket competitions

Several systems and process anomalies became evident during the DRS technology trial, and the DRS technologies received mixed reviews from stakeholders regarding its reliability (ESPNCricinfo 2012).

4.1.2.1 Controversy #1: Leg Before Wicket

The Indian team's opening batter was controversially given 'Out' Leg Before Wicket after the Sri Lankan team requested a review of the on-field umpire's original Not Out decision (see Figure 4.4). The controversy arose because the DRS technology failed to account for an important aspect of the Laws of Cricket that only the first interception of the ball by the batter be considered in the decision; the DRS technology included the second interception in error.



Figure 4.4: The controversial Leg Before Wicket (Source: Cricket Master Website 2020)

4.1.2.2 Controversy #2: Caught Behind the Wicket

Another controversial decision related to Sri Lankan batter Tillakaratne Dilshan arose when he was declared 'Out' Caught Behind the Wicket by the on-field umpire. The batter asked for a review thinking the ball had no contact with the bat before being caught by the opposing team. The Third Umpire had to rely upon the DRS technology to determine whether the ball had touched the bat. Based on the evidence produced by the DRS technology, the original on-field umpire's decision was overturned because the Third Umpire was unable to differentiate the sound of the bat hitting the ground or ball as both happened at the same time. The controversy

started since the fielding side considered being deprived of a decision as, in their opinion, the ball had clearly hit the bat.



Figure 4.5: *The controversial Caught Behind the Wicket (Source: Cricket Master Website 2020)*

In addition to the technological issues evident in the trial, there was also a process issue that caused controversy. The provision of three player reviews for each side in each innings of the match underpinned an unforeseen tactical use of the DRS technology; the ICC intended the DRS technology to overcome blatant errors by the on-field umpires; however, it soon became evident that players were using the DRS technology to review umpire decision depending on other factors (i.e., opposition player ability, state of the game etc.) and not merely when an on-field umpiring decision was considered to be clearly incorrect (ESPNCricinfo 2009b). After the Test Match, the captain of the Indian side, Anil Kumble, noted: "The DRS technology is still an experiment and that it has to be reviewed" (SportSatr 2019). His counterpart, the Sri Lankan captain Mahela Jayawardene, supported the use of continued use of the DRS

technology: "If the DRS technology wasn't used in this Test, we would probably have had four bad decisions go against us," he said (SportSatr 2019).

4.1.3 March 2009: The ICC extends the scope of DRS technology

In March 2009, the ICC announced that 'Hot Spot' images, as shown in Figure 4.6, will be available for use to the Third Umpire as part of its ongoing technology trial during the Tests between Australia and South Africa. The DRS technology was available to the Third Umpire in the event of a player referral (ESPNCricinfo 2009b). The Hot Spot technology was initially developed in the military for tank- and jet fighter tracking. The technology was then adapted for television by BBG Sports, an Australian company, in conjunction with Sky Sports (ESPNCricinfo 2009b).



Figure 4.6: Hot Spot detects contact between bat and ball and displays the point of contact using a bright spot (Source: Cricket.com.au Website 2017)

The Hot Spot technology uses two infra-red cameras positioned at either end of the ground. These cameras can sense and measure heat from friction generated by a collision, such as a ball on the pad, ball on bat, ball on the ground or ball on the glove. A black-and-white negative

image of the incident gets generated into a computer which shows the ball's precise point of contact. The Third Umpire can assess more information when Hot Spot technology is available while reviewing a decision and assisting the on-field umpires. A week before the Tests in South Africa, the ICC match referee Alan Hurst dismissed any suggestions that Hot Spot would be used in any trials, arguing that there were only three such systems available to broadcasters in the world (ESPNCricinfo 2009b). However, the ICC cricket committee said that "we hope Hot Spot can play a part over the next two Tests in Durban and Cape Town in establishing whether technology has a role to play in helping achieve those aims" (ESPNCricinfo 2009b). It also meant that when the ICC cricket committee met in May 2009 to consider the issues of the DRS technology trial, they had a better understanding of how well it worked in a match situation.

Regarding the inclusion of Hot Spot in the DRS technologies, South Africa captain Graeme Smith said that he remained unconvinced about the new referral technology but was pleased to have Hot Spot available for the remainder of the Test matches. "If you're only half-heartedly using technology, it's just going to create more frustration," he said. "I think if you're going to do something, let's do it properly, let's not do half-heartedly, and certainly I think both teams will feel the same way. It's very difficult to pick up those slight edges, or whether you've nicked it or not, whether you've hit your pad unless you've got the proper technologies" (SMH 2009). Australian captain Ricky Ponting criticised umpires for not using the infra-red imaging System during the first Test. "It's frustrating when all the technology is not there". Ponting said. "We must remember it is still a trial. When the Third Umpire came in, there were some negatives and some positives. We shouldn't expect every decision to be perfect, but hopefully, it eradicates the really obvious mistake," he added (ABC News Website 2009).

4.1.4 November 2009: The BCCI and other cricketing boards oppose using DRS technology due to cost

The DRS technology came under additional criticism, with broadcasters and national boards complaining at the implementation costs between US\$20,000 and US\$60,000 per match day (Borooah 2016). The DRS technology was removed from the India-Sri Lanka and South Africa-England Test series 2009-10, and its use in the Australia home series in the same season came into question. After a series of trials at the international levels, the ICC announced that the referral system would be implemented from October 2010. There was further communication that the ICC would not be responsible for the cost burden of the DRS technology, leaving cricketing boards and broadcasters to bear it all (ESPNCricinfo 2009b; ESPNCricinfo 2011e; ESPNCricinfo 2011f).

This decision prompted discussion among national cricketing boards, who believed that the ICC should have handled the cost for a system they wanted to implement. Gerald Majola, chief executive of Cricket South Africa, was more vocal about it and told ESPNCricinfo in an interview that "if ICC is pushing for the DRS technology, they must pay for it". He added that this matter should be between the broadcasters and the ICC as we informed the ICC that we would not be paying for it (Brown & Gollarudi 2009). On the other hand, Ratnakar Shetty, the Indian board's Chief Administrative Officer, stated no explanation was required about the BCCI's decision not to use the DRS technology for the Test series against Sri Lanka. Team India initially supported the implementation of the DRS technology but was discouraged by their players due to reservations about the effectiveness of referrals. The confusion over the cost of implementation may have resulted in the DRS technology being terminated before it was used in earnest. Hawk-Eye, ultra-slow-motion cameras and Hot Spot were among the technological devices used in the trials at a sizeable cost. The DRS technology also required an additional Hawk-Eye technician at each venue if it were to be used for the DRS technology,

rather than standard television coverage. There was other rental equipment with the system that impacted the budgets of broadcasters (ESPNCricinfo 2009b; ESPNCricinfo 2011e). The ICC dispatched a senior official to hold discussions with Cricket Australia and the host broadcaster to implement the DRS technology for the Test series against West Indies and Pakistan that year without any guarantees. All this was an attempt to implement DRS technology without any cost implication to the member cricketing boards (ESPNCricinfo 2009b; ESPNCricinfo 2011f).

4.1.5 Controversy #3: The ICC investigate referral complaint made by the ECB

In January 2010, during the final Test between England and South Africa, the English team requested a Caught Behind review after the South Africa captain Graeme Smith was given Not Out by the on-field umpire (BBC 2010). The controversy started after the Third Umpire agreed with the Not Out decision by the on-field umpire as he could not hear any noise when the ball passed the bat even though the noise was audible on replays. However, England alleged that the Third Umpire did not have the speaker volume on the stump microphone turned up high enough and so was unable to hear the ball hitting Smith's bat (BBC 2010). Technological advances such as Snickometer and Hot Spot, which are much more effective at breaking down caught behind decisions, were not used for the series because host South African broadcaster SABC deemed the tools were too expensive (BBC 2010; ESPNCricinfo 2011g). The ECB chairman at the time Giles Clarke described the International Cricket Council's implementation of the DRS technologies as 'a shambles' (BBC 2010). England bowler Ryan Sidebottom who was the bowler when England requested the review said the DRS technology still has a future at the highest level, but may have been rushed out prematurely. "It's a good system, but it needs more work doing on it," he said. "Rather than using it in big Test matches, they could have trialled it in county games and umpires can get more experience. It's a good system as long as it is done properly" (BBC 2010). However, former England captain Michael Vaughan said

Third Umpire, rather than the DRS technology itself, was at fault. "The system works, but the Third Umpire didn't turn his stump mic up," he said, referring to a controversial decision made by the Third Umpire (BBC 2010). The ICC's Chief Executive at the time Haroon Lorgat said: "There is a large amount of controversy, speculation and potentially unfair criticism circulating at present, so it is important to establish the exact facts before reaching any final conclusions. These are early days for the DRS technology, and so far, we are pleased with its effectiveness and the value it can add to the game in support of umpires. However, there had always been 'room for further improvement' in the technology" he added (BBC 2010).

4.1.6 October 2010: DRS technologies available for the 2011 World Cup

In October 2010, the ICC announced that the DRS technologies would be available for the 2011 World Cup. While ball-tracking would be in place for all games, Hot Spot would be available only for the semi-finals and final rounds of the World Cup, to be held from Feb to April 2011 in India, Sri Lanka, and Bangladesh. Hot Spot would not be used with the DRS technologies during the 2011 World Cup, reducing the DRS technologies to run without a valuable tool: the set of technology will comprise of ball-tracker, super slow-motion camera, and stump microphone (Talya 2011; Ugra 2011a):

The Hot Spot infrared cameras fall under the category of restricted equipment as they form part of military hardware used in jetfighters, tanks, and warships. Whenever these cameras are taken out of the country, they require a temporary export licence from the Australian Defence Department. The licence includes the return of the cameras to Australia within a seven-day period after being used. An ICC spokesperson admitted that the ICC was "unaware of the Australian government regulations to this effect." The cameras were first brought into India during the 2009 Champions League Twenty20 event. The logistical hurdle of moving the cameras were both expensive and sensitive security equipment, in and out of Sri Lanka and India could have led to the decision. India remained the only country in cricket that

had not accepted the DRS technology. Ironically, the only part of the DRS technology considered by the Indians to be the most trustworthy was the Hot Spot camera. (ESPNCricinfo 2010; Ugra 2011e)

The Hot Spot cameras did not meet the minimum requirement of the DRS technologies but could provide the most accurate images of any contact by the ball with bat or pad, by using infrared thermal imaging. However, an ICC spokesperson confirmed to ESPNCricinfo that "the supplier of the Hot Spot technology advised that it was not willing to supply its cameras for the tournament so, accordingly, they will not be used at any stage in the tournament. The use of these cameras is not a minimum requirement for the DRS technologies" (Ugra 2011b; Ugra 2011c).

4.1.7 Controversy #4: The 2011 World Cup

On 27th February 2011, during the 2011 ICC World Cup match between India and England at Chinnaswamy Stadium in Bangalore, the English batter Ian Bell was controversially given 'Not Out' Leg Before Wicket after the Indian team requested a review of the on-field umpire's original Not Out decision (ESPNCricinfo 2011b). The incident raised the issue because the Third Umpire referred the decision back to the on-field umpire even after the projected ball-tracking showed the ball was hitting the stumps. The Third Umpire was not convinced with the projection given that the ball impacted the batter more than 2.5-meters in front of the stumps and as per the DRS technology rules went back to the on-field umpire asking him to make the final judgement. The on-field umpire stayed with his original decision of Not Out (ESPNCricinfo 2011b). The ICC playing conditions ruling states that if a 'Not Out' decision is being reviewed and the distance from impact to the stumps is greater than 2.5-meters then the Third Umpire passes this information to the on-field official along with:

- the distance from the wickets of the point of impact with the batter,

- the approximate distance from the point of pitching to the point of impact,
- and whether the ball is predicted to hit the stumps

The playing condition goes on to state that: "In such a case the on-field umpire shall have regard to the normal cricketing principles concerning the level of certainty in making his decision as to whether to change his decision" (ESPNCricinfo 2011b). The Indian captain MS Dhoni was unaware of how the DRS technology and human intervention arrived at such a decision that allowed Bell to remain Not Out. "Adulteration is quite bad, whether it is natural or technology," he said. "I think the adulteration of technology with human intention was the reason why we didn't get that wicket. Hopefully next time, it will be either technology or human intention in the DRS technology." When he was asked whether he had known about the 2.5-meters rule, MS Dhoni didn't hide his frustration: "Well, if the Hawk-Eye says it is going to hit the stump, and it's going to hit the middle stump, then there is no reason why the distance really matters," he said (ESPNCricinfo 2011b).

The 2.5-meters ruling also surprised the England captain, Andrew Strauss, as he was made aware of the rule only during the review process. When asked about the ruling and the decision, he said: "Apparently, if you are that far down the pitch, it needs to be hitting middle stump to be given Out. I didn't know that was part of the rules. Obviously, Bell was lucky to get away with it" (ESPNCricinfo 2011b; Ugra 2011d). The ball tracker comes into play from the time the ball pitches and to the point of impact. The passage of that information is passed on to the computer through multiple camera frames. This helps the tracker to pick up on trajectory and make its own assessment. The greater the distance between the point where the ball pitches and the point where it makes contact with either bat or pad, the more camera frames there are for the tracker to trace the predictive path. The shorter the distance, the less information is available

to the technology to make an accurate prediction; this is why the 2.5-meters rule was brought into the game (Borooah 2016; ESPNcricinfo 2011b; Steen 2011).



Figure 4.7: Indian captain MS Dhoni unsuccessfully reviews an LBW appeal against England's Ian Bell during the 2011 World Cup (Source: ABC News Website 2011)

The Bell incident has raised the issue of whether the Third Umpire can instruct the on-field umpire to change his decision instead of merely passing on the information. One argument in favour was that the Third Umpire has both access and time to look at the replay and is detached from the emotions on the field. The other argument against states that giving the Third Umpire these powers will reduce the importance of the on-field umpires. In an interview with a news channel in India CNN-IBN, the ICC General Manager at the time, Dave Richardson, said that MS Dhoni should be aware of the rules before passing on any judgement about the DRS technology. "There are a set of rules along with Hawk-Eye technology to assist in making the decision when the DRS technology is implemented. Most of the time, a player is not fully aware of all the rules. If MS Dhoni is made aware of these rule's specifications, then I am sure

that he will accept the decision made" (ESPNCricinfo 2011c). In a letter to the ICC chief executive Haroon Lorgat, the BCCI objected to the ICC general manager's comments about MS Dhoni after the Indian captain questioned the DRS technology during the World Cup. The board also reiterated its concerns over the use of the DRS technology in the World Cup by saying that the DRS's "inadequacy has been exposed" during the tournament:

Lorgat defended the DRS technology, telling ESPNCricinfo that Dhoni's characterisation of it wasn't exactly accurate. Haroon Lorgat, said, "The controversial clause in the DRS technology that reprieved Ian Bell was introduced into the rules because of players' doubts over the accuracy of the ball tracker technology. The rules and protocols for DRS technologies were designed by experts with much thought. The Bell example applies the 2.5-meters rule, which in my view, accommodates the scepticism that some have with predictive technology. If the ball needed to travel more than 2.5-meters, then it is for the umpire to make the final decision as he sees it." Lorgat went on to say that regardless of the technology, the umpire had remained at the front and centre of decision-making on the field. "The umpire still has the authority to say, despite what I've heard and seen, I still don't think that is enough reason for me to change my decision. It is for them to decide. DRS technology is "no more than a tool" to help the decision-making of the umpire and has aided them, that does not overrule the umpire. That is a support structure we have put in place. You can't deny the progress of technology; either we embrace it, or we reject it. I believe we reject it at our own peril". (ESPNCricinfo 2011c)

4.1.8 February 2013: Introduction of the Real-Time Snickometer

A further enhancement was added to the DRS technologies by increasing the certainty to capture the witness of any contact made by the ball. The introduction of the Real-Time Snickometer added additional clarity to the existing DRS technologies, and the new technology will be part of the decision-making process. The new process will eliminate any controversy over the Hot Spot cameras' lack of ability not detect any contact (Gollapudi 2013). Snickometer

so far was used as a viewing aid and not part of the DRS technologies as it required a physical process run by a technician to overlay the pictures with the stump microphones sound provided by the. This would lead to inconsistencies in producing the final results. The new DRS technology aimed to run a fully automated process (Gollapudi 2013). The issue with Hot Spot known as the 'motion blur' is the inability to detect a contact made by the ball primarily when the fast bowlers are operating, due to the speed of the ball, and has been a constant source of the vulnerability of the DRS technologies (Gollapudi 2013).



Figure 4.8: *The Real-Time Snickometer detects contact between ball and bat (Source: BBG Sports Website 2019)*

The inventor of Hot Spot (Australian company BBG Sports) conducted trials on the new audio technology and believe that the new Real-Time Snickometer is the ideal complement to the Hot Spot when used simultaneously. It enhanced the DRS technology, making the review system more reliable, faster, and consistent (Gollapudi 2013). The head of BBG Sports, Warren

Brennan, talked about the new technology that "I am hopeful that it would improve fine-edge detection dramatically. On most occasions, you are going to have the Real-Time Snickometer and Hot Spot agreeing with another. So, the Third Umpire will now have two points of reference. There can be more consistency that way". The Real-Time Snickometer was tested for twelve months in trials across Australia and the UK (Gollapudi 2013).

4.1.9 July 2013: Take DRS technology off players and leave it to the umpires

In 2013, Australian wicketkeeper and vice-captain Brad Haddin stated: "Decision referrals should be taken out of the hands of players and left as a tool of international umpires" (Brettig 2013a). Haddin also said, "It's a consistent challenge to take emotion out of decision referrals. I personally think the umpires might as well use the reviews. I don't think they need to be in the player's hands, to be honest" (Brettig 2013a). "That's what the system was brought in for, the howler. The system is the same for both teams; we just haven't used it very well. That's the bottom line. We have to take emotion out of the decision and go on with what we see. If you think it's Out, challenge it," the Australian said (Brettig 2013a). Haddin's views on the use of the DRS technologies mirrored those made by the former Australian umpire Daryl Harper, who said, "Decisions should be left with the appointed decision-makers rather than turned into a tactical device by the players. If this current DRS technology is the best we can come up with, then something is wrong," Harper said (Brettig 2013a). "If the reviews were taken out of the player's hands and given to the umpires, then eventually the stronger performing umpires would emerge and be identified by the lesser number of reviewed decisions" (Brettig 2013a). Similar thoughts were shared by the former Australian Captain Ian Chappell where he said, "Instead of putting a limit on the number of referrals available and leaving them in the hands of the players, the use of the DRS technology should be solely at the discretion of the umpires. The arbiters in the middle should be encouraged to make decisions, and then, only if the video umpire sees

a glaring error should he intervene. This way, the umpiring standards could be raised, and only the howlers, and not the 50-50 decisions, would be overturned. No system is infallible, and that's part of the beauty of the game. The players make mistakes, which in turn require the umpires to come to a decision, which can occasionally result in an error. The human element is an important part of cricket's magic" (Chappell 2013).

4.1.10 July 2013: The DRS technology is not the solution, it's part of the problem

Adam Gilchrist, the former Australian wicketkeeper, stated that the DRS technology denied the game of spontaneity and failed to remove the mistakes it was created to eradicate: "The spontaneity and drama, the magic and intrigue that Test cricket always possessed has been lost. The reality and finality of seeing the umpire's finger raised was erased because everyone now looks to the batter or fielding captain to see their response" (Gilchrist 2013). "In my opinion, the game is poorer for that. I don't say that because of an umpiring decision cost Australia a Test Match. I understand the proposed benefits of technology eradicating umpiring errors, but the errors are still occurring" (Gilchrist 2013). He further commented that "I have always found it frustrating that a player can question an umpire's decision under the DRS technology. One of the strongest elements of the spirit of any sport is not questioning the umpires or referees. We now have a situation where players can do that, albeit limited times. That doesn't sit comfortably with me. Yes, technology is here, but perhaps a Test match like this has gone a long way to indicate that the umpires need to have full control as to when a decision is reviewed, rather than the players. For the first time, I'm starting to understand India's reluctance to go with the system. It's not a remedy that seems to have cured the problem. In fact, it may even have become more of a problem. I feel technology took something away from the game, for both the teams and for the spectators" (Gilchrist 2013).

4.1.11 September 2013: ICC announces DRS technology changes

Later in 2013, the ICC announced a major change to the agreed number of reviews and will be topped up to two after every 80 overs in a Test innings. Previously, teams were only permitted to have a maximum of two unsuccessful reviews per innings (AAP 2013). "It was agreed that a trial will be conducted whereby a team's referrals will be topped-up to two reviews after 80 overs of an innings. This trial will start from 1st October 2013 in all Test matches in which the DRS technology is used, with the results being monitored and considered by the ICC's Working Group," The ICC said in a release (AAP 2013). After the conclusion of the Ashes series between Australia and England, this change came in the place where both teams were dissatisfied at times during the series with the DRS's implementation. The ICC took the initiative of sending their general manager of cricket, Geoff Allardice, over to England to meet the team's midway through the series to address their concerns. The issues centred around the Third Umpire's use of DRS technology and how the replay official interpreted their verdict (AAP 2013).

4.1.12 November 2013: Australian Captain Michael Clarke claims DRS technology 'distorts the game'

Australian captain Michael Clarke questioned the wisdom for disputed verdicts by saying that using a DRS technology distorted the actions of the umpires and players. He went on to say that he would prefer the DRS technology to be thrown out entirely if its consistency did not improve. He also declared he did not wish to see Hot Spot return to the array of technological tools for Third Umpires until it's reliability can be improved beyond the level seen so far (Brettig 2013b). "The referral system where captains have two unsuccessful referrals at their disposal can distort the process. I don't like the tactics involved, where umpires and the teams know how many referrals are left and change their decisions accordingly. It should be consistent for all players" (Brettig 2013b).



Figure 4.9: The Australians stand in disbelief after running out of reviews when the on-field umpire gave a decision of Not Out to an obvious catch (Source: Dailymail Website 2013)

Clarke touched the most controversial moment of the Ashes series in 2013 when Stuart Broad hit Ashton Agar to slip fielding position via wicketkeeper Brad Haddin's gloves. But he chose not to walk, and the DRS technology allowed him to remain at the crease even though the sound was clear to everyone around except for the standing on-field umpire. "The ultimate problem with the Broad dismissal was not that he didn't walk or that the on-field umpire made an error, but it was the complicated DRS technology meant that the Third Umpire didn't have the opportunity to overrule the on-field decision" (Brettig 2013b). The wider principle Clarke argued for is that if technology shows the batter to be Out, then that should be the decision irrespective of the original decision or the number of referrals. "I believe that if it's clearly shown that the batter has hit the ball and he was caught, then the technology should be used to ensure he is Out," he wrote. "If he's hit in front of the wickets and the technology shows he is LBW, then he should be Out regardless of how many referrals remain. He further said that "As a captain, I'd just like the technology to be used to make more correct decisions, without all the

complications of how many referrals remain or don't remain. There shouldn't be a numerical limit. If this means passing referrals back into the hands of the three umpires, on and off the field, then so be it. My final word on the matter of technology, and the use of technology by the umpires, continues to be as inconsistent as it has been; I would rather it is not used at all. My opinion is that if the technology isn't perfect, it shouldn't be used at all," he wrote (Brettig 2013b). "The inventor and owner of Hot Spot (Warren Brennan) came out and admitted it doesn't pick up all edges. OK, that's fine: Hot Spot should not be used until it is more reliable. Once the technology has been tested and is shown to be correct, then the ICC should rule that every team has to use it. We should have the same rule for everyone," the Australian said (Brettig 2013b).

4.1.13 January 2015: Indian cricketing legend Sachin Tendulkar wants standardised DRS technology

In 2015, Sachin Tendulkar supported the use of the DRS technologies in international cricket as long as a uniform system was implemented across all the cricketing nations. Tendulkar said that he supported the referral system as long as both Hot Spot and Snickometer were used in tandem. "I don't know right now what BCCI's stance is. From an individual point of view, I can definitely say that we can't have bits and pieces in different parts of the world. One part of the world is using Snickometer. The other part of the world is using Hot Spot. Somewhere else, we are using something else. It's got to be standardised." Tendulkar recognised that the technology could never be perfect but stressed that it needed to be consistent. "Why should we settle for only 50% result? Why not get as close to 100%? It is impossible to get 100% right. There will be some errors here and there. It really does not matter if Zimbabwe and Bangladesh are playing, or England and Australia are playing. An international match is an international match. It is unfair on lesser teams who do not have the full package. We can definitely use technology as long as it is standardised" (Gollapudi 2014).

4.1.14 Controversy #5: The ICC admits Third Umpire mistake in 2015 World Cup

The International Cricket Council admitted that the Third Umpire made an error after reviewing England's final wicket in the World Cup match against Australia. There was some debate about whether the final wicket should have been deemed a 'Dead Ball' after an LBW appeal on James Taylor was overturned by the Third Umpire, and as a consequence, James Anderson should not have subsequently been given Run Out (Natsport 2015).



Figure 4.10: English batter James Taylor talks to the on-field umpire after his partner James Anderson was given Run Out (Source: The National News Website 2015)

The controversy occurred after the English batter James Taylor was given Out LBW by the on-field umpire. The batter went for review, and the delivery was found to be missing leg stump. However, Taylor had attempted to take a run with his partner at the crease James Anderson, who was given Run Out due to a direct hit from the Australian fielder Glenn Maxwell. The Third Umpire adjudged Anderson Run Out after the decision was referred to him by the on-field umpire. According to the ICC rules, the ball was deemed to have been dead following the umpire originally giving Taylor Out, which technically should have nullified Anderson's Run

Out. While the change of decisions would not have made any impact on the result, the error was not acceptable (Natsport 2015). The ICC released a statement after the match, which read: "The PCT spoke to the England team management and acknowledged that the game ended incorrectly, and an error was made". The ICC's DRS technology rules state that: "the ball should have been deemed dead when a batter is given Out LBW; no further runs or dismissals were possible" (Natsport 2015).

4.1.15 November 2015: Serious flaws in DRS technologies indicate a need for a review

The former Australian captain Ian Chappell criticised the DRS technology and believed it is used as a tactic rather than eliminating errors from the game. Ian Chappell said in his column on the ESPNcricinfo website that "Firstly, any system that has a finite number of reviews can't guarantee the correct decision will always be reached. Therein lies another problem with the DRS technology that the players often use as a tactic. Umpires' decisions shouldn't be part of a team's tactics. It's a ludicrous concept, but that's what the DRS technology has spawned" (Chappell 2015):

Decision reviews being in the hands of the players highlights another flaw in the DRS technology. It encourages cricketers to question the umpire's decision. That flouts what used to be the first tenet of a young cricketer's education: "The umpire is always right, don't dispute his decision." The administrators have legislated to undermine what used to be an important aspect of a young cricketer's upbringing. Because there isn't universal acceptance of the DRS technologies, we also have the ludicrous situation where Tests are contested under different laws, depending on where you are in the world and who is playing. This makes a mockery of fairness in Test cricket. Then there's the question of who controls the DRS technology. The ICC should pay for the equipment, so each Test has a full complement of DRS technologies. At the moment, the system is dependent on what the host broadcaster can afford, and consequently, there's a discrepancy in the amount and standard of equipment used in different parts of the world. The host

broadcaster is also involved in the DRS technology process when it should be controlled solely by cricket umpires. Television companies are in the entertainment business and shouldn't be part of the decision-making process. (ESPNCricinfo 2015)

"If the umpires were in sole charge of the DRS technology, then you'd expect only howlers to be overturned, the way DRS technology should work, and the fifty-fifty decisions wouldn't attract any scrutiny. Fifty-fifty decisions don't cause problems on the cricket field. The players accept that one day those will go in their favour, and on another day, they will go against them. That's the way it should be. However, the really bad decisions that eventually cost one team dearly can cause friction between two sides," the Australian added further to his frustration with the DRS technologies (Chappell 2015).

4.1.16 July 2016: Umpire's Call review for the DRS technology

In 2016, the Umpire's Call ruling in the DRS technologies for Leg Before Wicket dismissals was amended by the ICC. The change came into effect on October 1st, 2016, or, according to an ICC release, "from the start of any series using DRS technology that commences just prior to this date" (Middleton 2016). The existing protocols for overturning an Umpire's Call verdict on an LBW decision require the ball-tracking software to rule that more than half of the ball would hit an area that was between the middle of the off and leg stumps and below the bottom of the bails. Under the previous Umpire's Call conditions, for a Not Out LBW decision to be overturned, the ball needed to have hit the pad in line with the stumps and the projection needed to show more than half of the ball striking the wicket. For such decisions to be overturned, more than half the ball still needs to hit the pad in line with the stumps, but not as much as before, and more than half the ball still needs to hit the stumps (ESPNCricinfo 2016a), as shown in Figure 4.11. On a practical level, the change widens the area by half a stump width in each

direction. The change resulted in the zone increasing to the outside of the off and leg stumps, but still below the bottom of the bails (ESPNcricinfo 2016a), as shown in figure 4.12.

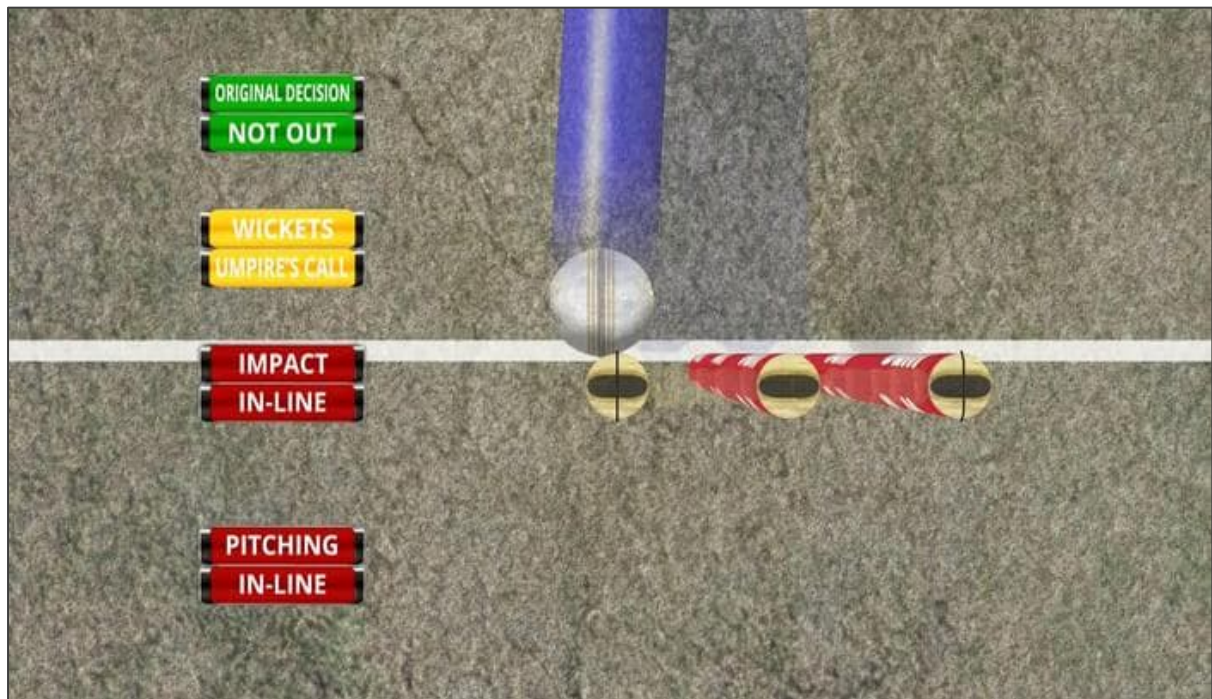


Figure 4.11: Under the new 'Umpire's Call' rules, this decision will change from 'Not Out' to 'Out' (Source: Fox Sports Website 2016)

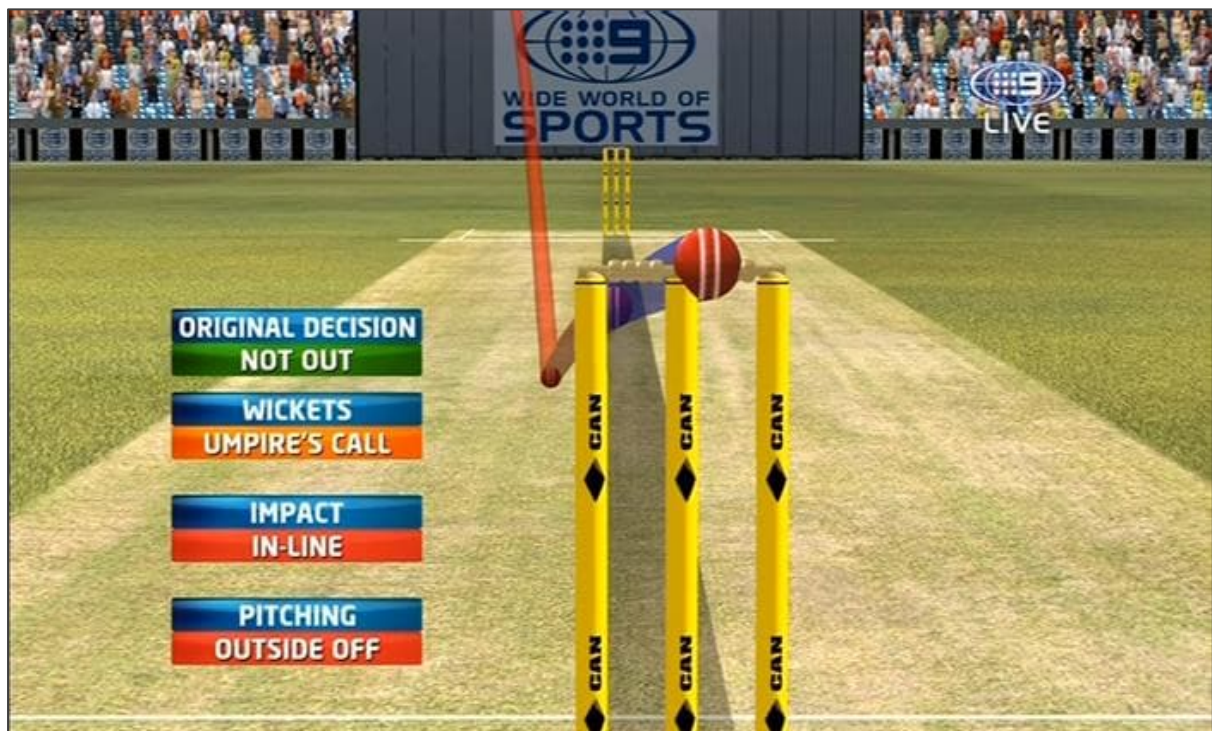


Figure 4.12: Under the new 'Umpire's Call' rules, this decision will remain 'Not Out' (Source: Fox Sports Website 2016)

Indian cricketer Sachin Tendulkar expressed his view about Umpire's Call for the LBW in the DRS technology, asking the ICC to remove it altogether and termed it as harsh and unfair on players. He said, "One thing I don't agree with the ICC, is the Umpire's Call in DRS technology they have been using for quite some time. It is the LBW decision where more than 50% of the ball must be hitting the stumps for the on-field decision to be overturned. The only reason they (the batter or the bowler) have gone upstairs is that they are unhappy with the on-field decision. So, when the decision goes to the Third Umpire, let the technology take over; just like in Tennis – it is either in or out, there's nothing in between. Once you have decided to use technology, then you rely on it," he added (Indian Express 2020). Another Indian Cricketer, Harbajan Singh, supported Tendulkar's thought on Umpire's Call by saying, "Agree with Sachin 100%. If the ball is touching the stump, it should be given Out. It does not matter how much part of the ball hits the wicket. A few rules should be changed in the game for the betterment of the game, and this is certainly one of them" (Vishwanathan 2020).

4.1.17 October 2016: When will we see a uniform application of the DRS technology everywhere?

Former Australian captain Ian Chappell wanted the ICC to have a uniform application of the DRS technology and stop having the differences in technology used from one series to the next: "It has been bad enough that one nation refused to use the DRS technology, but the fact of the matter is that not having the same standard of technology all around the world makes it an illogical System. The ICC must provide the same high standard technology for all matches. Having the rich countries utilise the best technology and the developing nations using minimal equipment means that the game is played under different laws in different parts of the world," Chappell added: "The ICC must also control all the umpiring protocols rather than relying on television broadcasters to be part of them. If the ICC wants to recoup the cost of controlling the decision-making process, then it could do so by factoring it into the sale of television rights.

And I'd love to see a situation where the DRS technology, with the very best technology and the ICC officials in complete control of the process and applied to every Test match. I just don't think it's going to happen in my lifetime," (Chappell 2016).

4.1.18 June 2017: Teams to retain review for Umpire's Call verdict

From the 1st of October 2017, the ICC decided that teams would not lose a review for the Umpire's Call verdict under DRS technology. If an LBW decision goes to the Third Umpire and ball-tracking technology determines that the on-field Umpire's Call could not be overturned (i.e., when a decision is considered too close to overturn), the team initiating the review would not lose from their allotted quota (Cameron 2017). Previously, teams were allowed two unsuccessful reviews every 80 overs during Test innings and one unsuccessful review in ODI innings. Few other changes made to existing DRS technology included the use in T20 International matches and ball-tracking along with edge-detection technologies set as a minimum standard for the DRS technology in international cricket (Cameron 2017).

4.1.19 March 2019: Indian captain Virat Kohli says, "DRS technology is not consistent at all."

The Indian captain Virat Kohli queried the accuracy of the DRS technology after losing a review against Australia. The captain made his displeasure publicly by saying, "It's (DRS technology) just not consistent at all" following an overturned review for Caught Behind decision against Australian batter Ashton Turner during the fourth ODI in Mohali (Muthu



Figure 4.13: *The Real-Time Snickometer detects contact before and after the ball goes past the bat (Source: The Cricket Times Website 2019)*

2019). In the 44th over of Australia's innings, the incident took place when Turner tried to hit the ball bowled by Indian bowler Yuzvendra Chahal. The ball seemed to have connected with the bottom part of Turner's bat. Having collected the ball, the Indian wicketkeeper Rishabh Pant urged the Indian captain to take the review after the on-field umpire ruled it Not Out. The decision was referred to the Third Umpire, where Ultra-Edge technology showed spikes before the ball had passed the bat and just after the ball passed the bat (Muthu 2019).



Figure 4.14: The image on the left shows an actual impact of the ball, whereas the image on the right provides a calculated path produced by the Hawk-Eye technology (Source: Punjabkesari Website 2019)

The Third Umpire turned down India's review and ruled it Not Out, something Kohli disagreed with, displaying his displeasure by shaking his head. Kohli called the DRS technology decision 'contentious' at the post-match presentation ceremony, adding "it was a bit of a surprise". "It's the DRS technology becoming more of a talking point every game. It's just not consistent at all, and that was a game-changer moment as well," he said (Muthu 2019). Kohli's mention of DRS technology becoming 'a talking point' was a reference to the third ODI in Ranchi, where a glaring error in ball-tracking came into the spotlight when Australian captain Aaron Finch was given Out LBW off Kuldeep Yadav's bowling (Muthu 2019). Kuldeep's delivery hit the Australian captain on the back pad in the 32nd over, and the on-field umpire ruled the batter Out LBW. Finch asked for a review where the replay showed the ball pitching on leg stump when it had pitched on the middle stump (Muthu 2019). The ball might have gone on to hit the stumps even if it had pitched on leg stump, but the noticeable mistake was on the part of the

ball-tracking technology used. This technical error became the reason for another controversy (Muthu 2019).

4.1.20 August 2020: Third Umpire to make the original decision for front foot No Balls

In 2020, the ICC announced that front foot No Balls were to be called by the Third Umpire only – and not the on-field umpire as it had been for the entire history of the game up to that point. It marked the first time the technology was to be used in Test cricket and is set to be reviewed after the three-match series before a decision is made on its future in the five-day format. Front-foot no-ball technology has been used in the February 2020 Women's T20 World Cup and made its debut in Test cricket (Ferris 2020a). The decision to hand over the No Ball responsibility to the Third Umpire pleased former Australian captain Ricky Ponting, who questioned whether the on-field umpires were checking for front-foot indiscretions following several missed calls in the 2018-19 summer (Ferris 2020a). Ponting's comments came after several no-balls were missed by the on-field umpires on day four of the first Test between Australia and India at the Adelaide Oval in December 2018. "I've said this for a lot of years; I honestly don't think the umpires look at the front line anymore," Ponting said at the time. "Some of the ones we've seen have been 4-6 inches over the line. I don't think the umpires are looking, and I certainly don't think they were looking at those ones because they were blatantly obvious ones. And as we know that they'll only ever look at them if a wicket falls, which, as far as I'm concerned, is not right. Part of umpiring is to get the no-ball decisions right as well. I'm not asking for everything to be spot on, but if you're six inches over the line, then surely you can call it" (Ferris 2020a).

4.1.21 Controversy #6: Virat Kohli complains after failed DRS technology protocol in a T20 game against Australia in 2020

The DRS technology was at the centre of another controversy when Australian batter Matthew Wade was given a reprieve against India at the Sydney Cricket Ground. The controversial moment arrived after the Indian captain's attempted review was declared "null and void" by the Third Umpire, who deemed the captain had witnessed a replay of the venue's big screen (Ferris 2020b). Confusion reigned during the third T20 International match when the Australian opening batter was struck on the front pad by Indian bowler Thangarasu Natarajan and given Not Out by the on-field umpire. Kohli tried to communicate with his bowler and wicketkeeper as to whether he should review the umpire's decision and, after long deliberation, looked as though he decided against it. But when the captain turned to walk back to his mark, he saw a replay of the delivery on the big screen and asked for a review (Ferris 2020b).



Figure 4.15: The Third Umpire declared a review taken by Indian captain Virat Kohli as 'null and void' since the replay was broadcast on the stadium screen prior to the Player Review being requested (Source: Fox Sports Website 2020)



Figure 4.16: The image on the left shows the Indian captain having a conversation with the on-field umpires after his referral was rejected by the Third Umpire; the image on the right provides calculated path produced by the Hawk-Eye technology (Source: Fox Sports Website 2020)

First, Wade was heard protesting to the square-leg umpire on Fox Cricket's stump mic "They referred it off the big screen". Midway through the review process, the Third Umpire cancelled the assessment, by saying "We can't go through this review anymore," said Wilson. "There are replays on the big screen. This is a null and void review" (Ferris 2020b). By showing a replay, the integrity of the review was lost. Had Kohli reviewed without seeing the replay, Wade would have been Out, with ball-tracking technology confirming the ball pitched and struck the batter in line and would have hit the stumps. India retained their review; the match continued (Ferris 2020b). Later, Virat Kohli said, "That LBW was a strange one because we were still discussing whether the ball was going down, and within the 15-second time, they showed the replay on the screen. We decided to go for the review, but the umpire said they have already shown the replay on the screen. I asked the on-field umpires what we do in this situation, and they said

nothing could be done as it was a mistake from the television crew. I thought from a management perspective; a message has to be passed that this can't be done at the highest level because, in a very important game, it can be very costly. It was a little bit of a miss from the television guys upstairs, and I am sure it won't be repeated" (Karhadkar 2020).

4.1.22 Controversy #7: Tim Paine queries DRS technology verdict with umpires

Australian captain Tim Paine confronted match umpires by questioning their decision-making consistency after his second innings Caught Behind dismissal against India during the second Test match at the Melbourne Cricket Ground. The captain complained about being adjudged Caught Behind by the Third Umpire, who used the Real-Time Snickometer spike as evidence after Hot Spot indicated no mark on the bat. The original decision by the on-field umpire was Not Out. The Indian captain requested a referral, and the on-field decision got overturned by the Third Umpire. This process brought the controversy that a batter had his Not Out verdict overturned without any hint of a mark on the bat using Hot Spot cameras (Brettig 2020).



Figure 4.17: *There was no detection of contact between bat and ball by Hot Spot technology in Tim Paine's Caught Behind decision (Source: Wionews Website 2020)*



Figure 4.18: The Real-Time Snickometer detected contact between the ball and the bat by Hot Spot for Tim Paine's Caught Behind decision (Source: Sporting News Website 2020)

"I've spoken to them. It wasn't very productive. But I've spoken to them and raised my only concern," Paine said. "My concern yesterday was not with the technology; I just think the decision was made too quickly. He didn't look at enough replays to see the full evidence that there was probably a gap between bat and ball; the line itself had started before it went past my bat, and it finished again. So, there were just lots of things that didn't marry up. I saw some photos of it, all sorts of things; I don't think he (Third Umpire) took the time to look at the evidence. The technology itself, I thought, was okay," the Australian added (Brettig 2020). However, ICC's DRS technology protocols have dictated for several years that the umpire is entitled to source an edge from Snickometer evidence even if there is no Hot Spot. Paine was aware of the protocols as he explained the exact reasons, he had sought further discussion about the Third Umpire decision but felt that the decision had been made too hastily (Brettig 2020).

4.1.23 January 2021: Ex-ICC Elite Umpire calls for end to Umpire's Call

Daryl Harper, a former ICC elite international umpire, stood in more Test matches than any other Australian umpire. Harper umpired a total of 95 tests, and the last of his Test was in 2011. He remained formally in the game as a Cricket Australia match referee until 2020. "If there was one change I could make, it would be to eradicate Umpire's Call. The idea of the ball only hitting part of the stumps and therefore not always being out is incongruous," the Australian suggested. "I've had enough of Umpire's Call. Let's just ban Umpire's Call. Get rid of the controversy and just go with it. Any contact with the ball on the stump will dislodge a bail. No 48 per cent, 49 per cent. The fact it's been going for twelve years, and the public are still mystified, and the players are still mystified, would suggest that there are some deficiencies in either the communication or the understanding. So there needs to be some serious work done from the ICC's end. Because we shouldn't be talking about umpiring decisions," Harper said about the DRS technology. Despite this call, the most recent ICC conference reaffirmed the role of the 'Umpire's Call' as a critical element that balances the captain's power to challenge the original decision with respect for the role that the original decision-maker plays under the Laws of Cricket (O'Loughlin 2021).

4.2 Chapter summary

The objective of this chapter was to present a timeline of DRS technology implementation issues that have arisen from the secondary data analysis process described in Chapter Three. The timeline of events described in this chapter served two purposes; firstly, to provide a context for the technology implementation issues experienced by the various stakeholders to the ICC's international cricket competition since its inception in 2008. Secondly, as described in Chapter Three, this chapter provided a basis for determining the set of interview questions to be posed to the various stakeholders during the primary data collection process. The

following chapter will present the findings of the primary qualitative data analysis process in terms of the theoretical issues idiosyncratic to the DRS technology and multiple stakeholder management practices associated with its acceptance in the international cricket competition context.

Chapter Five

Research Findings

5.1 Objectives of the chapter

This chapter presents the results of the primary and secondary data analysis process described in Chapter Three. Firstly, the characteristics of DRS technologies identified in Chapter Two are discussed in terms of salient stakeholder perceptions of its role in the International Cricket context. Secondly, the chapter discusses how the stakeholders' acceptance of the ICC's DRS technologies is supported by extant TAM 3. Lastly, the chapter discusses how the stakeholder management practice requirements of DRS technologies inform the modification of extant TAM 3.

5.2 Characteristics of DRS technology

As noted in Chapter Two, DRS technology comprise of three characteristics: (i) they undermine the role of the decision-maker, (ii) they provide no support for the original decision, and (iii) they change the balance of stakeholder power in the decision-making process. Analysis of the data indicated that all three elements of DRS technologies characteristics were detected in the data coding process, as discussed below:

5.2.1 Undermining the role of the original decision-maker

The data analysis indicated stakeholder agreement that the specific decision-maker whose role is undermined by the introduction of DRS technologies into the International Cricket Competition was that of the on-field umpire:

At first, there was a sense of humiliation if you had to change a decision that you made and was proven wrong, so revoking the decision was always a

difficult thing to do, but I think it's not a bigger deal now. It's part of the game.
(Stakeholder Interviewee: Umpire 1)

Some umpires didn't take it (DRS technology) on board, and they felt offended that they have been overruled, and DRS technology makes them feel that they appear to be not doing a good job. (Stakeholder Interviewee: Player 3)

The umpire is always right, and you don't argue with his decision used to be the first lesson a young cricketer was taught. This admirable exercise in discipline and self-control is now not applicable as DRS technology introduction encourages a form of player dissent. (Ian Chappell, Ex-Australian Captain, cited in India TV News Website 2020)

For our main stakeholders, players, coaches and team officials, decision-making is the key for an umpire to be effective, and DRS technologies have taken that away from the umpires. (Stakeholder Interviewee: Scorer 1)

If DRS technology does what it should do and be doing is to support the decision-makers, then DRS technology will hopefully see the way.
(Stakeholder Interviewee: Administrator 2)

This confirms the claim in Chapter Two that only the role of the on-field decision-maker in the international cricket context has been undermined. No other roles in the game (i.e., the Third Umpire, the Match Referee, the players etc.) were identified as being adversely affected by the introduction of the DRS technology.

5.2.2 DRS technology undermining the role of the original decision-maker

Analysis of the data indicated that the role of the on-field umpire was undermined in one specific manner by the introduction of the DRS technologies into the International Cricket Competition. The DRS technology was perceived to contradict the ethos of the game as set out in the 'Preamble to the Laws of Cricket' and the Laws proper (see Appendix H):

The Preamble to the Laws of Cricket includes accepting an umpire's decision, respecting the umpires' authority, and thanking them at the end of the match.
(Marylebone Cricket Club, cited in MCC Lords Website 2021)

Umpires are increasingly undermined in international cricket, and some players seem to think they are running the game rather than the officials. (David Lloyd, Cricket Commentator, cited in Daily Mail 2021)

Surely, it's time to put any review system in the hands of the umpires so that it stops being a tactic, rids the game of the howler, and on most occasions, brings a satisfactory outcome. (Ian Chappell, Ex-Australian Captain, cited in ESPNcricinfo 2013)

In terms of the umpire and coach stakeholders, the issue was related to undermining the on-field umpire's legitimacy:

When DRS technology is there, the players do not care about the quality of the decision made by the on-field umpire and, in the next few seconds, send it upstairs, hoping to get it overturned. (Stakeholder Interviewee: Umpire 2)

When we (on-field umpires) obviously know the decision, but we still expect a review from the players. That's frustrating. (Stakeholder Interviewee: Umpire 3)

Umpires were "not massive fans" when the DRS technology was first introduced. If a batter came forward and was hit on the pad, then it was Not Out. Then all of a sudden, you saw these things in front of you. That was the hard bit. When I first joined, a lot of senior umpires felt as though they were being shown up. (Ian Gould, Ex-English Cricket Umpire, cited in BBC 2020)

I think all decisions will go to the Third Umpire in the coming future, which will absolutely supersede the guys in the middle. (Stakeholder Interviewee: Coach 3)

Very good umpires were virtually giving up on their duties at square leg, knowing there was no point in even contemplating making a judgment: it could only get them into trouble; why take the risk? We saw players starting to pressurise umpires if they did not call for a replay. And we saw increasing demands (sometimes on the field) for the system to be extended to close catches as well. The heart of the game, the finality of the umpire's verdict, is being eaten away. (Steen 2011)

In terms of the player and commentator stakeholders, the issue was related to the use of DRS technologies strategically rather than overturning incorrect decisions:

If someone makes a very bad decision and DRS technology was brought in to get rid of the howlers, but now players are using it as a different tool contrary to the philosophical reason it was used in the first place. If they (players) treat it (DRS technology) what it is, then it is an aid to make the game better and decision-making more accurate and not as a weapon to use it (DRS technology) just for their advantage thing. (Stakeholder Interviewee: Player 4)

We players use it (DRS technology) as a tactic at times. (Stakeholder Interviewee: Player 1)

What may start out as fun – a club bowler or fielder making the shape of a T when he disagrees with a decision – will become a whole process of insidious undermining. (Scyld Berry, Cricket Journalist, cited in ESPNcricinfo 2010)

I do not think the players must check every straightforward decision. I do not want to see the players going upstairs every time there is an appeal and ask the Third Umpire to check it. In other words, why have an umpire rather than had a robot sitting there? I want DRS technology to assist and be complementary to the umpiring rather than overtaking. (Stakeholder Interviewee: Administrator 2)

I wasn't a fan of the DRS technology until I realised I could get 50 more LBWs. It used to be so easy for left-handers to stick their leg down the middle and get away with it. (Graeme Swann, Ex-English Cricketer, cited in Steen 2011)

Overall, the stakeholder interviewees agreed that the introduction of the DRS technologies undermined the role of the on-field decision-maker in the International Cricket Competition. There was a variation of concern by the stakeholder interviewees regarding undermining the role of the on-field umpire. The umpire and coach stakeholder interviewees related the issue to the sense of legitimacy. The player stakeholder interviewees related the issue to the use of

intent-based on the sense of strategy and tactics. The Match Referee stakeholder interviewees related the issue to the confusion regarding the use of DRS technology.

5.2.3 DRS technologies provide no support for the original decision

The data analysis indicated stakeholder agreement that the introduction of DRS technologies into the International Cricket Competition was aimed to assist the specific stakeholder, the on-field umpires, in the decision-making process, but despite the fact, the original decision-makers are expected to make judgements without any support by the system:

The DRS technology is flawed because it puts undue pressure on the umpires in the middle, whose decisions can now be changed with the use of DRS technologies. I would have never brought this in DRS technologies. They are taking the authority away from the on-field umpires, and the whole thing is causing more problems than it's worth. (Dickie Bird, Ex-English Cricket Umpire, cited in Saltau 2009)

Players are there to play a match and win the match and take wickets or score runs. I think umpires are there to make sure that the game runs smoothly and the right decisions are made. And I think if everybody just did their job and let the umpires do their job and use DRS technology to support them, then we can expect better results. (Stakeholder Interviewee: Match Referee 3)

The on-field umpire is paramount out there, and he is the decision-maker and not the DRS technology. DRS technologies must only be used to support the decision-maker and not overtake their role. (Stakeholder Interviewee: Player 5)

Anything that can be used which would help the umpires in doing a better job and help them make better decisions it's something that is worth using, and unfortunately, DRS technology doesn't do that. (Stakeholder Interviewee: Media Representative 1)

This confirms the claim in Chapter Two that DRS technologies provide no support for the original decision during the original decision-making process. The original decision-maker is required to make the decision without the use of any technology.

5.2.4 DRS not providing support for the original decision

Analysis of the data indicated that the introduction of DRS technologies into the International Cricket Competition affected the decision-making process in one specific manner by not providing any support to the on-field umpires while making the original decision. The DRS technology was perceived to contradict the statement by the ICC claiming that the system is to assist match officials with their decision-making (Decision Review System, cited in ICC Website 2021):

The Decision Review System is a technology-based process for assisting match officials with decision-making. (Decision Review System, cited in ICC Website 2021)

Early days there was obviously some questioning about the accuracy and validity of the technology, but as that has gone year and year and improved, I think now it's an expected part of the game at the higher level. (Stakeholder Interviewee: Administrator 1)

In terms of the umpire stakeholders, the issue was related to the lack of technological support:

I think we should have a mechanism where the on-field should have access to the technology whenever they want to use it. I don't think we will ever get 100% decisions using the technology, yet we still want umpires to be perfect. (Stakeholder Interviewee: Match Referee 2)

The traditional role of umpires went out of the window a long time ago. At the international level, we umpire totally different to what any other umpire does around the world. We umpire with the DRS technology, go into the game with the DRS technology, and know the teams will use it against us. It is what it is. (Stakeholder Interviewee: Umpire 1)

So, we then have decisions overturned using DRS technology that really, we have no chance of making that decision. That can be frustrating because it goes against us (on-field umpires) as it goes down as incorrect in our assessment. (Stakeholder Interviewee: Umpire 4).

In terms of the Match Referee stakeholders, the issue was related to the tactical use of DRS technologies by the players:

At the end of the day, what should be the objective is to get the howler right. And if an umpire is not sure and thinks that maybe I was wrong if that happens, I am actually not opposed to the on-field umpire going for a review. (Stakeholder Interviewee: Match Referee 2)

I think DRS technology has a lot of good to offer, but at the same time, I think it also has kind of taken the sensation out of the game. When you are playing cricket, or when you are playing any other sports, you want to live in that moment, you want to enjoy the moment, and errors are part of any game, and basically, you suck it up and deal with those errors or issues which makes the game more exciting. If we were to use the DRS technology, then it must be used by the umpires and not the players. (Stakeholder Interviewee: Match Referee 2)

By not having the DRS technology implemented properly and not supporting the on-field umpires, we are not treating the game professionally. (Stakeholder Interviewee: Match Referee 1)

I think that the DRS technology has been rushed into elite cricket, and it's supposed to be supporting the on-field umpires well. (Stakeholder Interviewee: Journalist 1)

Initially, the system was inducted to get rid of the howlers, and yes, you have every right to appeal because that is something that is part of the game. Still, when you start knowing in your mind and doubting that the decision by the umpire is not correct, then that's where it becomes detrimental for the game. That part I do not like. (Stakeholder Interviewee: Technology Specialist 1)

Overall, the stakeholder interviewees agreed that the introduction of DRS technologies into the International Cricket Competition does not provide any support to the on-field umpires in

making the original decision during the decision-making process. The stakeholder interviewees expressed a variation of concern regarding the DRS technology assistance in the decision-making process. The umpire stakeholders related the issue to the lack of technological support; the Match Referee stakeholder interviewees related the issue to the players' tactical use of DRS technologies. The Technology Specialist stakeholder interviewees related the issue to the intended use of DRS technology.

5.2.5 Changes the balance of stakeholder power in the decision-making process

The data analysis indicated stakeholder agreement that the introduction of the DRS technologies into the International Cricket Competition changed the balance of stakeholder power in the decision-making process by empowering the player stakeholders:

Instead of leaving the DRS technology in the players' hands, its use should solely be at the discretion of the umpires. The arbiters in the middle should be encouraged to make decisions, and then, only if the Third Umpire sees a glaring error should he intervene. This way, the umpiring standards could be raised, and only the howlers, not the 50-50 decisions, would be overturned. No system is infallible, and that's part of the beauty of the game. The players make mistakes, which in turn require the umpires to come to a decision, which can occasionally result in an error. (Ian Chappell, Ex-Australian Captain, cited in ESPNcricinfo 2013)

Sometimes I get frustrated when I see players just reviewing a decision for the sake of it; fingers crossed, you never know. The system was not designed for that reason, and players/teams have exploited that a little bit. It's beyond the spirit of the game. (Stakeholder Interviewee: Commentator 1)

My biggest complaint about the DRS technology is that it invites players to challenge something they absolutely have no way of getting right most of the time. (Stakeholder Interviewee: Umpire 2)

It is frustrating when we obviously know the decision made by the umpire is the right decision, but still, we expect a review from the players just for the sake of it. (Stakeholder Interviewee: Scorer 1)

This confirms the claim that only the introduction of DRS technologies into the International Cricket Competition changed the balance of stakeholder power in the decision-making process. DRS technologies allowed "challenging" as a part of the decision-making process by player stakeholder's who were not the decision-makers or part of the decision-making process in the game.

5.2.6 DRS technologies changing the balance of stakeholder power in the decision-making process

Analysis of the data indicated that the introduction of DRS technologies into the International Cricket Competition changed the balance of stakeholder power in the decision-making process by fully supporting the challenging mechanism initiated by a stakeholder who is not the original decision-maker. The DRS technology was perceived to contradict the ethos of the game as set out in the 'Preamble to the Laws of Cricket' and the Laws proper (see Appendix H) by allowing players to challenge the original decision by the on-field umpire:

A player may request a review of any decision taken by the on-field umpires concerning whether or not a batter is dismissed. (Decision Review System, cited in ICC Website 2021)

The DRS technology was introduced to remove the 'howler', but it remains a clutch of 'TV gimmicks' of 'spurious accuracy' and should be taken out of players' hands. (Lucas 2010)

In terms of the umpire stakeholders, the issue was related to the lack of stakeholder engagement in the decision-making process:

I am a massive fan of creating the 'we' mentality and not us or them. Keeping a balance of the review system between the players and the match officials would be best for the game rather than segregating it to one stakeholder. I will be happy to add an extra layer of collaborative approach where both players and match officials have a review. Not to forget, it's the player's game, but we play our part in it as decision-makers. (Stakeholder Interviewee: Umpire 4)

I sometimes wish an umpire had a review up his sleeve as well. I have spoken to International Umpires about when teams waste all their reviews, which puts pressure on an umpire. The umpire review will add another layer to the existing DRS technologies. (Stakeholder Interviewee: Umpire 5)

In terms of the Match Referee stakeholders, the issue was related to the implementation of DRS technologies:

If the teams have used their reviews and then the umpire makes an error, you've got these hundreds of thousands of dollars of technology around you, and you can't use it, and it seems pointless. We've got the Third Umpire especially sitting there, and you can't use him, and you say that's not quite right. (Stakeholder Interviewee: Match Referee 3)

If teams have exhausted their reviews but had the umpire had one in his back pocket as well, then that will be a good thing. I don't know if this has been discussed, but certainly worth a discussion even though there would be a difference of opinion, I am sure. (Stakeholder Interviewee: Match Referee 1)

An umpire must have some discretion over what happened, and if it's a 50/50 call, we must keep the umpire's opinion as a central element in decision-making. (Stakeholder Interviewee: Match Referee 2)

In terms of the coach stakeholders, the issue was related to the challenging mechanism of DRS technologies:

You may say challenging is a way of doing it, but maybe it's not the best way of doing it. The umpires should control the DRS technologies and not the teams; that would be the ideal situation, but it's nearly impossible now (Stakeholder Interviewee: Coach 2)

And you've got these hundreds of thousands of dollars with the DRS technology around you, and you can't use it, and it seems pointless. We've got this Third Umpires especially sitting there, and you can't use it, and you say that's not quite right. The key is giving the right amount of challenges; if you use DRS technology, use it; don't half use it. (Stakeholder Interviewee: Player 3)

Overall, the stakeholder interviewees agreed that the introduction of DRS technologies into the International Cricket Competition changed the balance of stakeholder power in the decision-making process by empowering the player stakeholders, who were not the decision-makers or part of the decision-making process, to challenge the decisions of the on-field umpires. There was a variation of concern by the stakeholder interviewees regarding the technology assistance in the decision-making process. The umpire stakeholder interviewees related the issue to the lack of stakeholder engagement in the decision-making process, the Match Referee stakeholder interviewees related the issue to DRS technologies implementation, and the player stakeholder interviewees related the issue to the challenging mechanism of DRS technologies.

5.3 Extant TAM 3 and its role in DRS technology acceptance in the ICC context

As noted in Chapter Two, extant TAM 3 has been predicated on research into the direct relationship between the user and a given new technology, but some of its tenets may still be relevant to multiple stakeholder management practices necessary for effective DRS technology acceptance. Analysis of the data indicated that four external variables of TAM 3 were relevant for DRS technology acceptance (i.e., job relevance, result demonstrability, experience, and perceptions of external control); each will be discussed in the following sections:

5.3.1 TAM 3's 'Job Relevance'

Consistent with TAM 3, the key external variable of job relevance was relevant to stakeholders' acceptance of DRS technologies in the international cricket context. Job relevance relates to the capabilities of a system that are applicable to an individual's job and can enhance the degree of their performance (Ho *et al.* 2020). The data analysis indicated stakeholder agreement that the DRS technologies are perceived as a relevant tool that assists the decision-making by correcting any obvious errors made by the on-field umpire:

Anything [DRS technology] that reduces the margin of error as much as possible, I think that should be included, and DRS technology has proven that. We actually have recognised that the game with DRS technology is much better than without it. (Stakeholder Interviewee: Player 4)

I like the technology aspect. I like the right decision being arrived at. I think that's good. (Stakeholder Interviewee: Coach 3)

I was actually on the other end of the stick too when I made a decision (on-field) that was overturned, and I was really happy about that because there were two noises and I thought bat then pad, but it was actually the bat hitting the ground and the ball hitting the pad. It was in a final and at a critical stage. I was so glad that the decision got overturned, even though it was a sense of 'Oh crap, I have got a decision overturned', but I was really pleased that thank God, this game hasn't been affected by an umpiring error (Stakeholder Interviewee: Umpire 6)

If the on-field umpire makes a terrible decision, DRS technology was brought in to get rid of that howler. (Stakeholder Interviewee: Umpire 3)

I think DRS technology has brought a lot of good to the game by improving the standard of decision-making, which is the most critical thing for all the main stakeholders. (Stakeholder Interviewee: Technology Specialist 1)

I think the standard of umpiring is getting better with DRS technologies. There are errors made and get rectified using the DRS technology. (Stakeholder Interviewee: Player 5)

If we have DRS technology like this all the time, you will find that the most effective and best umpires will come through over some time. (Stakeholder Interviewee: Umpire Educator 1)

The good umpires will always put up their hand and say I have made a mistake, but I am so glad that the DRS technology has overturned it. (Stakeholder Interviewee: Match Referee 2)

I also think that the DRS technologies have added another layer of drama and theatre to cricket. I like that. That's my view on it now. So as a stakeholder, I am happy that we arrive at the correct decision, and I like the drama and theatre that it adds to view cricket. (Stakeholder Interviewee: Media Representative 1)

The data analysis indicated that even though the DRS technologies changed the decision-making process contrary to the ethos of the game (and that it underpinned a fundamental shift in stakeholder power relationships), there was stakeholder consensus that it improved the enactment of the competition in two main ways. Firstly, it provided a framework where the players could review decisions made by the on-field umpire, reducing perceptions that incorrect decisions were affecting the result of a given game. Secondly, it provided a powerful review framework for umpires and coaches to review on-field umpires' decision-making processes for the purposes of their professional development.

5.3.2 TAM 3's 'Result Demonstrability'

Consistent with TAM 3, the key external variable of result demonstrability was relevant to stakeholders' acceptance of DRS technologies in the international cricket context. Result demonstrability relates to the degree to which an individual believes that the results of using a system are tangible, observable, and communicable (Ho *et al.* 2020). The data analysis indicated stakeholder agreement that the stakeholder's feedback proved beneficial in improving the initial implementation flaws of DRS technology, attaining desired and better outcomes.

The technology evolves, so it is better to bring it on a pretty low base level and engage all the stakeholders to get their thoughts and implement all the required feedback. The whole process gets multi-layered. (Stakeholder Interviewee: Match Referee 3)

The ICC has kept adding tools to DRS technologies over the years after getting feedback from different stakeholders, and every addition has been helpful. (Stakeholder Interviewee: Administrator 2)

Technology is a massive aid to the decision-making process. It provides better and impartial results. What best you can do for the game is have DRS technologies in it. (Stakeholder Interviewee: Scorer 1)

I think faith in DRS technology is an essential factor. I have watched cricket, and I have seen ball tracker and thought that doesn't make sense. I think

everybody had experienced that when DRS technology was in its early days and used to be a layer of frustration. Now we cannot even imagine playing cricket without DRS technology. (Stakeholder Interviewee: Educator 2)

The ICC made a slight but significant alteration to the 2.5-meters rule, the most controversial segment of DRS technology used in the 2011 World Cup, to enable more consistent application by the umpires. The change to the DRS technology rule expanded the umpire's scope for interpretation of LBW following its contrary usage in the first two weeks of the competition. (Sharda Ugra, Cricket Journalist, cited in ESPNcricinfo 2011e)

The data indicated that even though the DRS technologies changed the decision-making process contrary to the ethos and stated aims of the game by the governing body, there was stakeholder consensus that it improved the enactment of the competition in the following three main ways: (i) **Changes to the ball-tracking:** As noted in Section 4.1.1.1, the ICC changed ball-tracking to predict the ball's future path after the first impact rather than the Third Umpire to do it by looking at the calculated images, based on the feedback and expectation of the stakeholders; (ii) **Changes to the 2.5-meters rule:** As noted in Section 4.1.7, the ICC had to tweak the 2.5-meters rule halfway through the World Cup tournament in order to bring consistency based on the feedback and expectation of the stakeholders (Ugra 2011f). The 2.5-meters rule came as a surprise to a lot of stakeholders exposing lack of education and engagement with the stakeholders by the game's governing body at the time; and (iii) **Changes to the Umpire's Call:** As noted in Section 4.1.16, the ICC made changes based on the feedback and expectation of the stakeholders to the Umpire's Call for situations when the DRS technology is not definitive and reverts to supporting the original decision made by the on-field umpire. According to Chalkley, MacMahon and Ball (2013), the LBW decision is the most critical umpiring decision in cricket that attracts controversy. The LBW law includes a unique component of predicting what would have happened had the ball not been intercepted, requiring the Third Umpire to predict the ball's trajectory. All the previously mentioned

changes were related to the LBW law, where it is critical to calculate all judgements; otherwise, even a tiny error can ultimately alter a decision. The ICC made all the changes based on the stakeholders' feedback and expectation, resulting in getting rid of the flaws and providing better results.

5.3.3 TAM 3's 'Experience'

Consistent with TAM 3, the key external variable of 'experience' was relevant to stakeholders' acceptance of DRS technologies in the international cricket context. 'Experience' relates to any prior involvement in using technology or any experience gained by using technology (Ho *et al.* 2020). The data analysis indicated stakeholder agreement that involving more stakeholders in the decision-making process has provided the various stakeholders with a sense of involvement and helped them better understand the DRS technologies and their purpose. The main beneficiary of all these experiences and satisfactory outcomes has been cricket:

I was talking to the current Australian Cricket team Captain Tim Paine during a Sheffield Shield game; he was talking about how he and some of the senior players visited the DRS technology van studio to talk about the DRS technology and get a bit more insight into the practical application of the tool works to try to help them in their reviewing. He found that very useful, and I don't see why all the players will not find this sort of tutorial of any benefit. (Stakeholder Interviewee: Coach 4)

It happened for the first time in a match where the discussions of the Third Umpire with the director and the whole decision-making process went onto the loudspeakers on the ground. Even though the flow might have been upset, in terms of stopping the game, I was told later by a number of people that how interesting they thought it was, how embraced they got in his decision-making and process, how he went about his thoughts, they can actually see the rhythm of what he was trying to do. I think there are ways to overcome the flow issue as long as you involve everyone. (Stakeholder Interviewee: Match Referee 3)

I have no problem bringing technology in, but umpire experience and exposure to technology is a critical thing. You really need your umpires to embrace it and feel confident within their abilities to handle these new systems. I think we tried to do that and do it pretty well. I will use the example of Live to Air this year, which was something new for us, and I think it's gone pretty well so far. I think as long as umpires are educated and become confident in these new implementations, it's a good thing. (Stakeholder Interviewee: Umpire 7)

I think we got to accept that if our Third Umpires are retrained well, to ask the right question and get the proper information as quickly as they can and not to second guess themselves and have four thousand different replays if they can see then get on with life rather than stop then stop the flow of the game for any longer. (Stakeholder Interviewee: Administrator 3)

I was a fan of having a specialised Third Umpire operator in the early days, and I am still of that feeling. However, I understand ICC, who are keen to ensure that all their elite match officials are comfortable in both roles on-field and the Third Umpire. The Third Umpire role gives you a great appreciation of the demands of the on-field role and helps develop skills for both environments. (Stakeholder Interviewee: Umpire 6)

The data indicated that even though the DRS technologies changed the decision-making process contrary to the ethos of the game and that it underpinned a fundamental shift in stakeholder power relationships, there was stakeholder consensus that it improved the enactment of the competition in one main way. The scope of the Third Umpire role was enhanced, making it a specialised role, and providing more authority to it. The role was not limited to only the line calls but involved in adjudicating more complex decisions while working independently of the TV director or broadcasters.

5.3.4 TAM 3's 'Perceptions of External Control'

Consistent with TAM 3, the key external variable of perceptions of external control or facilitating conditions was relevant to stakeholders' acceptance of DRS technologies in the

international cricket context. Perceptions of external control relates to resource factors such as time, money and technology compatibility issues that may constrain usage (Ho *et al.* 2020). The data analysis indicated stakeholder agreement that the broadcasters' variable and inconsistent use of the DRS technology due to higher cost has contributed to various DRS technology implementation issues across all cricketing nations.

I think in most sports, you expect the elite section of sport to take the lead in innovation who are obliged to do the testing at a higher level and then spread it down the track to all the other levels. They did enough tests before they started to know there was value and will change decisions and affect games in a way not affected before. ICC are the leaders, and it's their job. They could have given it to a County in England or State Cricket in Australia, but it might have been a money thing or a lot of other reasons that they couldn't. I am glad that the ICC took the initiative, and even after 12 years, we don't have it in first-class cricket as we couldn't afford to have it due to the cost factor. (Stakeholder Interviewee: Umpire Educator 1)

Now players everywhere making a lot of noise about having DRS technologies as part of their local competitions, and I believe that it's just a matter of time before DRS technology is embedded into every cricket competition around the world, depending on the money, obviously if broadcasters are willing to pay for that type of expenditure. (Stakeholder Interviewee: Player 8)

Suppose every major domestic competition around the world gets it. In that case, they might develop their own extras and add-ons that will add to the system's overall effectiveness, and that might spread provided everyone else is using it. People may give a theoretical view of what may or may not work, but they actually got to see it in practical terms to decide whether it's worth going with it or not, bearing the expense. (Stakeholder Interviewee: Umpire Educator 2)

I would agree with that (in future, we should start at the lower level and involve and educate all stakeholders). At a low level, games are like laboratories for experiments, and the cost can be less. If it works appropriately and the way you have designed, you definitely look at how you will use it in bigger stages. The opposite is also true. You might have thought that you got

a big difference-maker, and you try it, and it falls at the place where you don't want to. (Stakeholder Interviewee: Commentator 1)

The data indicated that even though the DRS technologies changed the decision-making process contrary to the ethos and stated aims of the game by the governing body, it underpinned a fundamental shift in stakeholder power relationships. There was stakeholder consensus that consistent technology across all the playing nations and competitions can optimise the benefits of DRS technologies. Removing the variable use of the DRS technology was perceived as important in minimising some of the issues the stakeholders had witnessed and bringing reliability and confidence in using DRS technologies.

5.4 Additional Factors in DRS technology acceptance in the ICC context

Analysis of the data indicated that two emergent factors were critical in the ICC's implementation of the DRS technologies in the international cricket competition context. The two emergent factors are related to 'stakeholder engagement' and 'stakeholder education'. Each of these emergent factors is described in the sections below:

5.4.1 Multiple stakeholder engagement

The data presented in Chapter Four provided examples where specific issues raised by stakeholders were dealt with using multiple stakeholder engagement and consensus-building by the governing body. The data analysis indicated an ability for the veracious stakeholder to express their dissatisfaction with the DRS technologies as they existed at the time. In terms of the player stakeholders, the issues were related to the reliability of the DRS technology and the Third Umpire's use of the DRS technologies:

As a captain, I'd just like the DRS technology to be used to make more correct decisions, without all the complications of how many referrals remain or don't

remain. There shouldn't be a numerical limit. If this means passing referrals back into the hands of the three umpires, on and off the field, then so be it. My final word on the matter of DRS technology, and its use by the umpires, continues to be as inconsistent as it has been; I would rather it is not used at all. My opinion is that if the DRS technology isn't perfect, it shouldn't be used at all. (Michael Clarke, Ex-Australian Captain, cited in Brettig 2013b)

For the first time, I'm starting to understand India's reluctance to go with the system. It's not a remedy that seems to have cured the problem. In fact, it may even have become more of a problem. I feel DRS technology took something away from the game for both the teams and for the spectators. (Adam Gilchrist, Ex-Australian Player, cited in ESPNcricinfo 2013)

I have watched cricket overseas when they haven't got the DRS technology or are very limited and cannot investigate a decision properly. I wish that International Cricket Council – ICC would fund every test match to have the same DRS technology available across the board. Particularly, when there is a test championship in place, the teams that go through to the final may have benefited from a very well-supported DRS technology. It just compromises the competition if you ask me. (Stakeholder Interviewee: Player 10)

The ultimate problem with the Broad dismissal was not that he didn't walk or that the on-field umpire made an error, but the complicated DRS technology meant the Third Umpire could not overrule any of the on-field decisions. (Stakeholder Interviewee: Player 3)

In terms of the other stakeholders, the issue was related to technology use, and the limitations to the Third Umpire's authority to intervene after an incorrect decision is made by the on-field umpire:

I mean, just about everybody on the ground said that's hitting outside leg, and Hawk-Eye showed it was actually pitching in line, and then you start to wonder, were we wrong or is DRS technology wrong? (Harsha Bhogle, Cricket Commentator, cited in Journal of Sports Economics 2016)

I've always been very upfront and honest, and I've said that our system is not perfect. There are times when things just go wrong. People bang on about the fact that DRS technology should be 100%. Well, it's just never going to get

to that stage. There are so many little intangibles that can create problems. (Warren Brennan, Inventor Hot Spot, cited in The Guardian 2013)

The referral system where captains have two unsuccessful referrals at their disposal can distort the process. I don't like the tactics involved, where on-field umpires and the teams know how many referrals are left and change their decisions accordingly. It should be consistent for all players. (Michael Clarke, Ex-Australian Captain, cited in Brettig 2013b)

If the DRS technology couldn't support the system and not all tools are deployed, it is very hard to ascertain whether a batter has nicked a ball onto his pads if we don't have Snikco[meter] or Hot Spot. It will help on-field umpires be more effective and help them make the right decisions if there is consistency with the tools used. (Stakeholder Interviewee: Umpire 2)

As referees, we need to decide if the on-field umpire got a decision right or wrong, and sometimes we don't have the right DRS technology to determine correctly. (Stakeholder Interviewee: Match Referee 2)

As noted in Section 4.2.11, the data similarly indicated that the governing body took measures to respond in a direct and timely manner:

It was agreed that a trial would be conducted whereby a team's referrals will be topped up to two reviews after 80 overs of an innings. This trial will start from 1st October 2013 in all Test matches in which the DRS technology is used, with the results being monitored and considered by the ICC's Working Group. (ICC statement release, cited in AAP 2013)

The Home Board will ensure a separate room is provided for the Third Umpire. He has access to television monitors and direct sound links with the television control broadcast director to facilitate as many replays as necessary to assist him in making a decision. (Borooah 2016)

I'm pretty confident that he probably did hit it; assuming that is the case, I can do nothing more than apologise to the players & the fans that, unfortunately, Hot Spot let everybody down yesterday on that decision. (Warren Brennan, Inventor Hot Spot, cited in The Guardian 2013)

The CEC (chief executives' committee) agreed a Working Group be constituted to consider how the ICC should best use technology in on-field

umpire decision-making in the future. The group's considerations will be wide-ranging and include a review of the objectives and philosophies of using DRS technologies, protocols and procedures, and the role and training of the Third Umpires. (ICC statement release, cited in AAP 2013)

Analysis of the data indicated that a level of stakeholder acceptance was achieved by implementing the feedback received via multiple stakeholder engagement process that was actioned in a timely manner:

I will be honest that where the ICC has landed with their current DRS technology protocols, they are in a pretty good place at the moment. It took some time to evolve, and I really applauded how they have gone about it. They have listened to feedback and made appropriate changes like the impact zone to the stump's half or inside and the time limitation in case of a review etc. We cannot afford to have players having a mother's club meeting for five minutes to decide whether or not to use a review. The existing time limit creates the theatre and the drama. (Stakeholder Interviewee: Administrator 1)

The DRS technology evolves, so it is better to bring it on a pretty low base level and engage all the stakeholders to get their thoughts and implement all the required feedback. The whole process gets multi-layered. (Stakeholder Interviewee: Administrator 2)

The ICC has kept on adding tools to DRS technology over the years after getting feedback from different stakeholders, and every addition has been helpful. (Stakeholder Interviewee: Administrator 1)

I don't think that we can go backwards, and it would be a retrograde step if we got rid of DRS technology. I am not saying that DRS technology cannot be improved by engaging all the stakeholders and imperative getting their thoughts. (Stakeholder Interviewee: Match Referee 3)

It is ok to get the views of other stakeholders, but ICC needs to have a practical demonstration of constant stakeholder's feedback and engagement. We don't have DRS technology locally, so if there is any further innovation, we don't have any say for it. (Stakeholder Interviewee: Umpire 5)

In a practical sense, we need to be a little bit innovative and stay ahead of the game, plus keep listening to the stakeholder feedback. I think more and more

we are doing it, and governing bodies are getting better at it. The end product reflects that sort of approach. I can only see benefits with a few or not at all negatives with having the collaborative approach and not isolating it to one group or the other. Certainly, the match officials have the technical expertise, and the players have the passion, and if we merge those two driving forces together, I think we can come up with a pretty good map to drive us into the future. (Stakeholder Interviewee: Match Referee 1).

Therefore, the data analysis provides support for the role that multiple stakeholder engagement has as a factor for multiple stakeholder acceptance of DRS technologies – primarily through an ongoing dialogue with all the salient stakeholders followed by timely and robust responses by the governing body.

5.4.2 Multiple stakeholder education

The data presented in Chapter Four provided evidence of knowledge deficits experienced by stakeholders that were overcome with the governing body's specialised education and training programs. The data analysis detected two important multiple stakeholder education factors: 'specialised education and training of umpires' and 'general education of stakeholders via transparency'. In terms of 'specialised education and training of umpires', the data indicated the importance of clearly identifying the knowledge deficits and implementing corrective action immediately. In terms of the player stakeholders, the issue was related to the deficit of knowledge regarding the DRS technologies:

Adulteration is quite bad, whether it is natural or technology. I think the adulteration of technology with human intention was why we didn't get that wicket. Hopefully, next time, it will be either technology or human intention in the DRS technologies. (MS Dhoni, Ex-Indian Captain, cited in ESPNcricinfo 2011d)

Apparently, if you are that far down the pitch, it needs to be hitting the middle stump to be given Out. I didn't know that was part of the rules. Obviously,

Bell was lucky to get away with it. (Andrew Strauss, Ex-English Captain, cited in ESPNcricinfo 2011d)

Cricketers, commentators, and spectators could all benefit from a simple explanation of why a 2.5-meters rule might have been used to disregard the results of Hawk-Eye in favour of the field umpire's decision. This is that explanation. (Stakeholder Interviewee: Player 2)

When a decision gets reviewed, you can see everything as it unfolds on the big screen. When I saw it pitch in line and hit the stumps, I thought that was it. I wasn't aware of the rule of how far you had to be down the wicket. I got waved back on by the fourth official, and I moved on from there. I wasn't aware that the distance down the wicket was a factor. I didn't even know that rule existed. As soon as I saw it pitch in line and hit, I thought that was enough. To be honest with you, it's strange if you see Hawk-Eye saying it's going to hit the stumps. It's a little bit strange. But that's the rule, I guess, and we're not going to be able to change that for this World Cup. (Ian Bell, Ex-English Player, cited in ESPNcricinfo 2011d)

In terms of the umpire stakeholders, the issue was related to the deficit of knowledge and training regarding the DRS technology and the decision-making process adopted by the Third Umpire:

We on-field umpires had no training and when the DRS technology first came into use. Umpires were trained without using any technology for years, and suddenly, they had to use it and umpire differently. It was really difficult for us, but now they are trained to use the system. (Stakeholder Interviewee: Umpire 1).

I started out as a Third Umpire with no training, no preparation. I was told that there's the equipment, just do it. (Simon Taufel, Ex-Australian Cricket Umpire, cited in Kishore 2019)

I think we got to accept that if our Third Umpires are trained well, to ask the right question and get the proper information as quickly as they can and not to second guess themselves and have four thousand different replays if they can see clearly then get on with life rather than stop the flow of the game for any longer. (Stakeholder Interviewee: Match Referee 2)

The data indicated that the governing body responded to these stakeholder issues by enhancing the resources and training for the Third Umpire and making it a specialised role.

There are a set of rules along with Hawk-Eye technology to assist in making the decision when the DRS technology is implemented. Most of the time, a player is not fully aware of all the rules. If MS Dhoni is made aware of these rule's specifications, I am sure that he will accept the decision. (Dave Richardson, Ex-ICC General Manager, cited in ESPNcricinfo 2011)

It would be nice to see more captains and players do an umpire's course and stand and see what it is like to be an umpire because we think differently, and I respect the fact that players think differently. If you are going to step into the shoes of making a decision, decide whether it's right or wrong. It's important to start engaging with umpires and the officiating community about why we make decisions that we do, what goes into a decision, and how they can look at those rationally than emotionally. (Simon Taufel, Ex-Australian Cricket Umpire, cited in Kishore 2019)

All elite and international panel umpires are share information and experiences amongst themselves, and all that information is then pushed to the first-class and national levels. ICC plays a major role in providing and sharing educational resources, video captures, and assessment reports from different games to bring consistency. (Stakeholder Interviewee: Umpire 6)

This confirms that the role of the 'specialised education and training of umpires' for Stakeholder Education is to highlight the lack of knowledge deficit and then implement a corrective action instantly to overcome the identified knowledge deficit.

In terms of 'education of stakeholders via transparency', analysis of the data indicated that the ICC took measures to ensure that the use of DRS technologies was more open to scrutiny by initiating the 'Live to Air' process (i.e., broadcasting the Third Umpire's use of DRS technologies so that they were visible to all stakeholders):

Discussions between the on-field and Third Umpires will be aired. This initiative is part of ICC's efforts to make umpiring more understandable to viewers and spectators. (BBC Report 2014)

In fact, they have made it a wonderful theatre. The spectators can get involved while they wait for decisions to come up on the big screen within the stadium, and it frees the on-field umpires to concentrate on other aspects of the game that they might otherwise not be able to follow fully. (Speed 2005)

If there is a delay, the general public wants to know the reason for the delay. And I think it educates the public what process is going through in reaching the final decision. That educates the public. It definitely makes it easier to understand how the decision is made. I am fully in agreement with this. (Stakeholder Interviewee: Administrator 2)

The game will continue to test you in those areas, and you've got to work out strategies. More resources need to be given to match officials in the same way they do with the players. We need support-mechanism programmes to be able to cope with the challenges of elite sport in very public environments. (Simon Taufel, Ex-Australian Cricket Umpire, cited in Kishore 2019)

In the T20 environment, one of the concerns with the 'fast-food' version of the game is stakeholders sitting at home and watching that vision instead of being there in person at the ground. They are getting used to the luxury of life having the DRS technology coming into their lounge room via their television screen. Suppose we can create that environment at the ground on the bring screen and have the voice to air (the Third Umpire talking through his process) rather than holding the game and making them sit idle. In that case, we can pull the cricket lovers from their lounge to the grounds in person. (Stakeholder Interviewee: Administrator 1)

This confirms that the role of the 'general education of stakeholders via transparency' for multiple stakeholder education is to demonstrate a transparent decision-making process to all the stakeholders with the strict process and interpretations of information. Analysis of the data also indicated that a level of multiple stakeholder acceptance was achieved by implementing education of stakeholders via transparency:

There have been plenty of times I have seen situations where I wasn't sure if they got the tracking right, and all of a sudden, the ball is going in a different direction to what it was going. That is a human error rather than a system error, probably as the people who are running the system, which I can't speak for them. (Stakeholder Interviewee: Player 2)

I think umpires' call is exactly the right thing to do because on-field umpires should be making decisions. So, when they have a fair-catch scenario or obstructing the field, and the on-field umpire makes a soft signal, that's a decision. The second thing is, they're then not sending the decision upstairs to give the benefit of the doubt to the batter. They're actually saying the on-field umpire has made a decision on the field. Can you conclusively prove and demonstrate that the on-field umpire decision is not right? (Simon Taufel, Ex-Australian Cricket Umpire, cited in Kishore 2019)

If you're an international umpire, the one thing that's going to tell you one way or the other is a 90-foot screen at every ground. You've got to put your ego away. And I think this group of modern on-field umpires have thrown it away because they understand DRS technologies. They understand the one that just flicks the glove they're never going to see. But if the additional review makes the game a better place, well, so be it. (Ian Gould, Ex-English Cricket Umpire, cited in ESPNcricinfo 2020)

This provides support for the role of multiple stakeholder education as a factor for DRS technology acceptance by providing: (i) ongoing 'specialised education and training of umpires' to all the umpire stakeholders enabling them to use the system professionally and allowing them to become independent decision-makers without any external influences and (ii) ongoing 'general education of stakeholders via transparency' to educate other stakeholders interested in the decision-making process in a transparent manner.

5.5 Chapter summary

This chapter presented the results of the primary and secondary data analysis process described in Chapter Three. Firstly, it confirmed the characteristics of the DRS technology as outlined in Chapter Two (see Baldwin 2014; Borooah 2013; Collins & Evans 2008; Leveaux 2009; Nlandu

2012; Ryall 2012; Steen 2011). Secondly, it provided evidence of the extent to which extant TAM 3 factors (i.e., job relevance, result demonstrability, experience, and perceptions of external control) supported multiple stakeholder acceptance of DRS technologies. Lastly, it discussed the engagement and education processes evident in the governing body's management of the salient stakeholders to the DRS technology implementation process over time; the implications of the finding presented in this chapter for the basis of the Discussion chapter that follows.

Chapter Six

Discussion and Conclusion

6.1 Objectives of the chapter

The objectives of this chapter are four-fold. Firstly, it provides a discussion of the characteristics of DRS technologies and related multiple stakeholder management practices in the professional sports context. Secondly, the current external variables that support the DRS characteristics and related multiple stakeholder management practices in extant TAM 3 and the additional external variables required are noted. Thirdly, the chapter addresses the specific research question in this thesis by proposing a modified TAM 3 that accounts for the characteristics of DRS technologies and the related multiple stakeholder management practices. Lastly, the chapter concludes with a discussion of the limitations of the thesis and opportunities for future research.

6.2 Characteristics of DRS technology in the professional sports context

As noted in Chapter Two, DRS technology comprises three characteristics that necessitate multiple stakeholder management practices; namely, they: (i) undermine the role of the original decision-maker, (ii) provide no support for the original decision, and (iii) change the balance of stakeholder power in the decision-making process.

6.2.1 Undermining the role of the original decision-maker

The research confirms that DRS technologies undermine the role of the decision-maker in the International Cricket Competition context. Specifically, this research found that only the role of the on-field umpire (i.e., the actor with authority to make a binding decision for all other stakeholders to follow) was undermined and that the authority of all other decision-makers in

the game (i.e., the Third Umpire, the Match Referee, the Captains, the players etc.) remained unaffected. The DRS technologies were perceived to undermine the role of the on-field umpire in three main ways (i.e., it contradicted the ethos of the game, it undermined the legitimacy of the on-field umpires' position as the final arbiter in contested or controversial incidents, and it underpinned stakeholder gamesmanship in the use of the DRS technologies process). Common to all of these factors is an apparent lack of effective and holistic stakeholder management practices in the ICC's implementation of the DRS technologies and processes. In terms of contradicting the ethos of the game and undermining the legitimacy of the on-field umpire as the final arbiter, the ICC's priority in introducing DRS technologies to the International Cricket Competition appeared focused on the reduction (and elimination) of poor/incorrect (i.e., 'howlers') from the game. This focus on a specific outcome circumvented a careful analysis of the competition stakeholders' needs (in terms of consultation, processes and how the original decision was to be supported or overturned). In terms of stakeholder gamesmanship, the implementation of the DRS technologies resulted in the captains (and their deputies) engaging in the decision-review process for purposes not aligned with its original intent – i.e., as guesswork (or a 'Hail Mary') to dismiss the opposition's more talented batter or to slow play down when the opposition side is dominating play etc.

6.2.2 No support for the original decision

The research confirms that DRS technologies provided no support for the original decision in the International Cricket Competition context. Specifically, this research found that the DRS technologies contradict the claim made by the ICC (i.e., games global governing body responsible for managing all the regulations and ethos of cricket) that the system was devised to assist the match officials with their decision-making. Still, the untimely execution of the DRS technologies process after the original decision has been made deprives the core decision-

maker (i.e., on-field umpire) in the game of any technology assistance during their decision-making. The DRS technologies were designed not to support the original decision in one main way (i.e., the system only activates after the on-field umpire adjudicates a decision without any support by technology). Common to this factor is an apparent lack of effective and holistic stakeholder management practice in the ICC's implementation of the DRS technologies and processes. In terms of the untimely execution of the DRS technologies process for a decision in ICC's DRS technologies implementation, highlight the following issues: (i) the lack of engagement of the core decision-maker (i.e., on-field umpire) who is not provided with any assistance by technology while making the original decision; (ii) the engagement of a player stakeholder in the decision-making process not part of the original decision but can initiate the DRS technologies execution by using a review; and (iii) the lack of engagement of the Third Umpire who is a decision-making stakeholder and has access to all the DRS technologies but cannot intervene during the original decision-making process until a player stakeholder has requested a review.

6.2.3 Changes the balance of stakeholder power in the decision-making process

The research confirms that the introduction of the DRS technologies into the International Cricket Competition changes the balance of stakeholder power in the decision-making process by empowering what were traditionally passive actors in the game. Specifically, this research found that the DRS technologies and processes allowed player stakeholders to "review or challenge" a decision made by the on-field umpire against the ethos of the game as set out in the 'preamble to the Laws of Cricket' and the Laws proper. The DRS technologies were perceived to change the balance of stakeholder power in the decision-making process in two main ways: (i) it contradicted the ethos of the game by allowing players to challenge the original decision by the on-field umpire, and (ii) it enabled the player stakeholders to execute

the use of technology endorsing its tactical use. Common to all of these factors is an apparent lack of effective and holistic stakeholder management practices in the ICC's implementation of the DRS technologies and processes. In terms of contradicting the ethos of the game, the 'preamble to the Laws of Cricket specifically states to accept an umpire's decision, whereas the DRS technologies promote the challenging mechanism to an umpire's decision. In terms of empowering the players by engaging them in the decision-making process, the players started using the technology for tactical reasons by putting their team's benefit ahead of the original intent of using technology while completely ignoring the on-field decision. The core decision-makers (i.e., the on-field umpire) seemed to have accepted the idea of bringing player stakeholders into the decision-making process only to find out that the players intent of using technology was not necessary to have better decision-making in the game.

6.3 Extant TAM 3 variables related to multiple stakeholder acceptance of DRS technologies

The findings of this research establish that four external variables of the TAM 3 were relevant in multiple stakeholders' acceptance of DRS technologies in the International Cricket Competition context (i.e., job relevance, result demonstrability, experience, and perceptions of external control).

6.3.1 Job Relevance

As noted in Chapter Two, the TAM 3 external variable of 'job relevance' was critical to PU in terms of (i) demonstrability of relevance and (ii) perceived relevance by the user of the technology. The findings of this thesis indicate that these factors remain critical but extend extant TAM 3 to focus on the needs of multiple stakeholders to the DRS technology. Consistent with the Stakeholder Identification (Elsbach & Cable 2019; Laplante, Laplante & Voas 2016; Mitchell *et al.* 2021; Perrault 2017; Raha, Hajdini & Windsperger 2021) and Stakeholder

Engagement (Almughairy 2021; Anastasi 2018; Baugh 2015; Lindgreen 2019; Masefield *et al.* 2021), the findings of this thesis indicated that the ICC overcame barriers to the DRS technology acceptance by (i) identifying all of the relevant stakeholders to the DRS technologies decision-making process within the game, and (ii) defining each stakeholders' role (or 'job') in the new DRS-based decision-making process. In terms of stakeholder identification, the ICC identified five stakeholders to the DRS-based decision-making process (i.e., on-field umpires, captains, Third Umpires, DRS technologies and broadcasting technicians, and Match Referees). For example, on-field umpires were given the role of making the original decision without the influence or input from any other stakeholder group. The captain's role was to challenge the decision if, in their opinion, the decision made by the on-field umpire was incorrect. The Third Umpire role could only activate and could intervene after the captains initiated the challenge. The DRS technologies and broadcasting technicians were limited to only providing any information requested by the Third Umpire and constrained from any other involvement. The Match Referees were responsible for overseeing the entire process, ensuring every stakeholder performed within their assigned role, and there was no overlapping of duties. This mechanism provided all salient stakeholders with a sense of involvement, assisted them in understanding their and other stakeholders' roles, and recognised the benefits of using the DRS technologies.

Once these distinct stakeholder groups were identified and defined, the ICC subsequently used multiple stakeholder engagement strategies to determine how the relevance of DRS technologies and processes could be increased for each stakeholder groups' role. The ICC formed a cricket committee comprised of former cricketers, umpires, and Match Referees responsible for assessing all the feedback received and recommended any changes to existing DRS technology protocols. The feedback considered by the committee on the use of technology included (and was not limited to) all the self-assessment reports submitted by both the on-field

and the Third umpires, captains and Match Referees reports, any surveys conducted by the ICC with multiple stakeholders targeting the use of DRS technologies, and the ICC annual conference reports. All these self-assessments, reports, and surveys have sections where individual stakeholders could express their experiences. The committee considered every feedback or criticism raised by the stakeholders to optimise DRS technology acceptance. There were instances where the ICC cricket committee contacted third parties to review the existing DRS technologies to bring uniformity to their use across every cricketing competition. For instance, the committee commissioned engineers at the Massachusetts Institute of Technology (MIT) to independently assess the performance of all the technologies that were part of the DRS technologies at the time. The assessing process was for the researchers to present their findings on each of the technologies and provide better suitability for use (ESPNCricinfo 2011h; ESPNCricinfo 2011i; ESPNCricinfo 2016b). By engaging multiple stakeholders on a regular basis, the ICC generated feedback and applied it in a timely manner to improve the DRS technologies acceptance.

6.3.2 Result Demonstrability

As noted in Chapter Two, the TAM 3 external variable of 'result demonstrability' was critical to PU in terms of (i) demonstrability of tangible and easily recognisable results, and (ii) perceptions that use of DRS technologies is advantageous. The findings of this thesis indicate that these factors remain critical but extend TAM 3 to focus on the needs of multiple stakeholders to the DRS technology. Consistent with the Stakeholder Identification (Elsbach & Cable 2019; Laplante, Laplante & Voas 2016; Mitchell *et al.* 2021; Perrault 2017; Raha, Hajdini & Windsperger 2021) and Stakeholder Engagement (Almughairy 2021; Anastasi 2018; Baugh 2015; Lindgreen 2019; Masefield *et al.* 2021), the findings of this thesis indicated that the ICC overcame barriers to the DRS technology acceptance by (i) identifying all of the

relevant stakeholders with concerns using the DRS technologies and (ii) addressing all of the issues raised by the stakeholders and providing them with their desired outcomes. In terms of stakeholder identification, the ICC identified three stakeholders using the pre-established engagement methods defined in Section 6.2.1 who had issues using the DRS technologies (i.e., on-field umpires, players, and Third Umpires). For example, after the initial trial of the DRS technologies, the on-field umpire stakeholders raised a concern with the ICC about the captains taking too long at times to request a review. The captains were involved in detailed discussions with their players before challenging the on-field decision, and these delays were affecting the game's flow. On the other hand, the player stakeholders were not pleased with the 2.5-meters rule during the 2011 World Cup. They came to know about it after the controversial incident involving England captain Ian Bell during the tournament. As noted in Section 4.1.7, the player stakeholders were unaware that such a rule existed. This highlighted the lack of education and engagement between multiple stakeholders and the governing body.

Similarly, the Third Umpire stakeholders found difficulty with ball-tracking to predict the ball's trajectory after the first impact by looking at the calculated images during the initial trial of the DRS technologies. Chapter Four provides a detailed timeline of the various DRS technology issues faced by multiple stakeholders and the measures taken by the ICC to overcome them. Once the concerns of the distinct stakeholder groups were found, the ICC took strategic measures to address all those issues promptly. For example, the ICC applied a time restriction on the challenge mechanism using a fifteen-second window to challenge the on-field umpire original decision. The captains were to make any review request within that time limit. Any review requests after the fifteen seconds were rejected by the on-field umpires. To address the player stakeholders' issue, the ICC used multiple stakeholder engagement and consensus-building to satisfy multiple stakeholders. The stakeholder engagement helped ICC determine the gap between the technological use and the concerned stakeholders and take measures to fill

the gap by acknowledging and implementing all the relevant feedback. The knowledge deficits experienced by multiple stakeholders were addressed with the governing body's specialised education and training programs. To resolve the Third Umpire stakeholders' issue, the ICC changed ball-tracking to predict the ball's future path after the first impact rather than the Third Umpire calculating it. The ICC made all the changes (i.e., review time limit, the 2.5-metres rule and ball-tracking) based on the stakeholder's feedback and expectation, resulting in removing the perceived flaws and providing expected results. The resolving of multiple stakeholder issues highlighted the advantages of using the DRS technologies amongst the concerned stakeholder groups. The initiatives by the ICC encouraged and provided multiple stakeholders with confidence to experience and understand the benefits of using the DRS technologies. By engaging multiple stakeholders on a regular basis, the ICC generated feedback and applied it in a timely manner to improve the DRS technologies' acceptance.

6.3.3 Experience

As noted in Chapter Two, the TAM 3 external variable of 'experience' was critical to behavioural intent in terms of users understating of technology and its purpose based on their involvement. The findings of this thesis indicate that this factor remains critical but extend extant TAM 3 to focus on the needs of multiple stakeholders to the DRS technology. Consistent with the Stakeholder Identification (Elsbach & Cable 2019; Laplante, Laplante & Voas 2016; Mitchell *et al.* 2021; Perrault 2017; Raha, Hajdini & Windsperger 2021) and Stakeholder Engagement (Almughairy 2021; Anastasi 2018; Baugh 2015; Lindgreen 2019; Masefield *et al.* 2021), the findings of this thesis indicated that the ICC addressed barriers to the DRS technology acceptance by (i) identifying all the relevant stakeholders that required more exposure and understanding of the DRS-based decision-making process in the game (ii) providing the various stakeholders with a sense of involvement and helping them have a better

understanding of the technology and its purpose. In terms of stakeholder identification, the ICC identified two stakeholders to the DRS-based decision-making process (i.e., players and the viewing public). For example, as noted in Chapter Four, the most aggrieved stakeholder group were the player stakeholders. Each had constant issues with having technology in the game; on the other hand, the viewing public demonstrated confusion about certain decisions in the game, especially the Third Umpire decisions, since there was no way of viewing the protocols the Third Umpire followed to reach a specific conclusion. Once these distinct stakeholder groups were identified and defined, the ICC subsequently used multiple stakeholder engagement strategies to reduce the gap between technology use and the concerned stakeholders. For example, the player's stakeholders welcomed the idea of being part of the DRS-based decision-making process, initiating the review process and getting the help of technology to remove incorrect on-field decisions. The viewing public gained access to the 'Live to Air' process, allowing them to know the decision-making steps taken by the Third Umpire, reducing any confusion on certain umpiring decisions.

6.3.4 Perceptions of External Control

As noted in Chapter Two, the TAM 3 external variable of 'perceptions of external control' was critical to PEOU in terms of (i) providing support and training and (ii) educating all users about the technology. The findings of this thesis indicate that these factors remain critical but extend extant TAM 3 to focus on the needs of multiple stakeholders to the DRS technology. Consistent with the Stakeholder Identification (Elsbach & Cable 2019; Laplante, Laplante & Voas 2016; Mitchell *et al.* 2021; Perrault 2017; Raha, Hajdini & Windsperger 2021) and Stakeholder Education (Bourne 2011; Dentoni & Veldhuizen 2012; Pacagnella 2015; Pedrini & Ferri 2019), the findings of this thesis indicated that the ICC overcame barriers to the DRS technology acceptance by (i) identifying all of the relevant stakeholders who experienced knowledge

deficit with the DRS technologies implementation and (ii) provide training and support to identified multiple stakeholders for consistent use of technology. In terms of stakeholder identification, the ICC identified three stakeholders using the pre-established engagement methods defined in Section 6.2.1, who required specialised education (i.e., Third Umpires) and general education (i.e., the players). For example, the role of the Third Umpire was limited and not used to its full potential due to the variant use of DRS technology by the different cricketing boards. Also, as noted in Section 6.2.2, the players were not fully aware of the rules and protocols of the DRS technology, which made it difficult for them to understand the decisions made by the on-field or Third Umpires at times.

Once these stakeholder groups were identified and defined, the ICC subsequently implemented a corrective action to overcome the identified knowledge deficit by consistently using the DRS technology across all international cricket competitions and then providing multiple stakeholders with specialised education and training programs to understand the technology and its purpose. This also increased reliability and confidence in using DRS technologies amongst multiple stakeholders and optimised its benefits across cricket playing nations. The multiple stakeholder education and training strategies had two critical factors: 'specialised education and training of umpires' and 'general education of stakeholders via transparency'. The ongoing 'specialised education and training of umpires' enabled all the umpire stakeholders to use the system professionally and become independent decision-makers without any external influences. The training resources helped enhance the Third Umpire's decision-making process, make it a specialised role, and provide more authority. The role was no longer limited to only marginal decisions and involved in adjudicating more complex decisions while working independently of the TV director or broadcasters. The ongoing 'general education of stakeholders via transparency' educated other stakeholders interested in the decision-making process transparently. It also demonstrated a transparent decision-making process to all the

stakeholders with the strict process and interpretations of information. The ICC took measures to ensure that the DRS technology and review process was open to scrutiny by initiating the 'Live to Air' process (i.e., broadcasting the Third Umpire's decision review process visible to all stakeholders).

6.4 Enhancing TAM 3 with stakeholder management practices

In addition to the findings noted above, the results of this study also indicate that TAM 3 would benefit by explicitly recognising the importance of multiple stakeholder engagement (i.e., communicating, involving and improving relationships with stakeholders – see Anastasi 2018; Almughairy 2021; Baugh 2015; Lindgreen 2019; Masefield *et al.* 2021) and multiple stakeholder education (in particular specialised training for specialised roles and generalised training for other stakeholders with transparency in the process).

6.4.1 Multiple stakeholder engagement

Multiple stakeholder engagement involves communicating and connecting with the full array of salient stakeholders (Almughairy 2021; Anastasi 2018; Baugh 2015; Lindgreen 2019; Masefield *et al.* 2021). The multiple stakeholder engagement process requires the results in improved relationships amongst stakeholders, ensuring the stakeholder's participation in the decision-making process (Cascetta *et al.* 2015). The two main levels of multiple stakeholder engagement are 'involvement' (i.e., 'identifying' and 'analysing') and 'participation' (i.e., 'prioritising' and 'gathering feedback'). Involvement encompasses informing and consulting to increase stakeholders' knowledge, while participation includes a higher level of engagement to reduce potential stakeholder conflicts (Deegan & Parkin 2011; Pacagnella 2015; Pedrini & Ferri 2019). The multiple stakeholder engagement indicates that stakeholders have been provided with the opportunity to voice their views, influence plans, and know what has been decided (Elsbach & Cable 2019; Laplante, Laplante & Voas 2016; Raha, Hajdini &

Windsperger 2021; Turner & Zolin 2012). The overall goal of engagement is to achieve transparent results, with greater input from stakeholders and stakeholder support for the decisions made (Cascetta *et al.* 2015; Elsbach & Cable 2019; Mitchell *et al.* 2021; Perrault 2017). Section 6.2 demonstrated that multiple stakeholder engagement is vital for TAM 3 and comprises the following elements:

- *Identify*: This study demonstrates that explicitly identifying and incorporating of multiple stakeholder groups and their needs into TAM 3 underpins an increased understanding of the relevance of DRS technology according to the roles of each stakeholder group. In this study, for example, the identification of the ‘on-field umpire’ and “Third Umpire’ stakeholder groups was necessary to understand their relative reliance on the DRS technology as play unfolded. By explicitly incorporating the identification of multiple stakeholder groups and their needs into TAM 3, the criticisms of Agerfalk and Eriksson (2006), Dube, Van and Zuva (2020), Zhang *et al.* 2020 (i.e., that extant TAM 3 is not fully capable of addressing multiple stakeholders in an organisational setting) can be addressed. The multiple stakeholder identification process will enable managers to identify the specialised and non-specialised stakeholder roles in the organisation as per the decision-making authority and use of technology. Also, the managers will be able to ensure every stakeholder performs within their assigned role and there is no *unnecessary* overlapping of duties.
- *Analyse*: This study demonstrates that explicitly incorporating the analysis of multiple stakeholder groups and their needs into TAM 3 underpins the appropriate level of communication required of salient stakeholder groups. In this study, for example, the analyses of the stakeholder groups enabled the researcher to understand the needs each group had regarding DRS technology and how they could best be used in their role. By explicitly incorporating the analysis of multiple stakeholder groups and their needs into

TAM 3, the criticisms of Raza and Standing (2010), Straub (2017), Pardo and Coronado (2016) (i.e., that TAM 3 ignores the other stakeholders and their roles in the acceptance process) can be addressed. Incorporating a stakeholder analysis process into TAM 3 will prompt managers and stakeholders to consider any sequential and/or reciprocal interdependence in any new technology's acceptance and use in an organisational setting. Similarly, it will enable an organisation to more effectively adopt new technology into their decision-making processes by identifying and removing any redundancy and/or unnecessary overlaps that may be present.

- *Prioritise:* This study demonstrates that explicitly incorporating stakeholder prioritisation into TAM 3 underpins the effective assessment of each stakeholder group's involvement with a given (new) technology, based on its relevance in decision-making authority and use of technology. In this study, for example, the prioritisation of stakeholder groups enabled the researcher to understand when the DRS technologies were relevant to the decision review process and, therefore, how the process for DRS technology use needed to unfold during the game. By explicitly incorporating stakeholder prioritisation into TAM 3, the criticisms of Mingers and Walsham (2010), Farre *et al.* (2019), Freiesleben *et al.* (2021) (i.e., that TAM 3 requires more inclusive and ethical methods to ensure that none of the stakeholders are beyond the boundary of the model) can be addressed. The prioritising process will help stakeholders determine if the relevance of the technology and protocols require any adjustments for any stakeholder group. The managers will then be able to determine the level of communication and education with specialised and non-specialised stakeholders accordingly.
- *Gather Feedback:* This study demonstrates that explicitly incorporating multiple stakeholders' feedback into TAM 3 underpins managerial engagement with salient

stakeholders and enable them to share their experiences using the technology. In this study, for example, the gathering of feedback from each stakeholder group regarding their perceptions of DRS technology and its use was associated with increased level of satisfaction and acceptance. By explicitly incorporating multiple stakeholder feedback into TAM 3, the criticisms of Govindaraju, Chandra and Siregar (2012), Pardo and Coronado (2016), Ajibade (2018) (i.e., that TAM 3 emphasises the gathering of feedback from the individual user of a given technology, and not necessarily those stakeholders that are indirectly impacted by the implementation of new technologies) can be addressed. The stakeholder feedback gathering and analysis process will allow managers to engage all stakeholders in a consensus-building process. The stakeholders will be able to provide any feedback, concerns, or pre-deployment experience with any new technology. The multiple stakeholder engagement process will allow the organisation to determine the gap between the technological use and the salient stakeholders and take measures to fill the gap by implementing all of the gathered feedback.

The modified TAM 3 for DRS technologies and the related stakeholder management practices proposed in this study adds a specific external variable named 'multiple stakeholder engagement' and will prompt (i) communications with multiple stakeholders, (ii) prioritise various specialised and non-specialised stakeholders' roles, (iii) support and help stakeholders in understanding the technology and its purpose, (iv) build trust amongst multiple stakeholders, (v) anticipate any stakeholder needs, and (vi) gather and analyse feedback. The overall engagement mechanism will underpin increased reliability and confidence of multiple stakeholders in using the new technology, provide them with a sense of involvement, help them understand the role and roles of other stakeholders, and recognise the benefits of using the technology (see Figure 6.1).

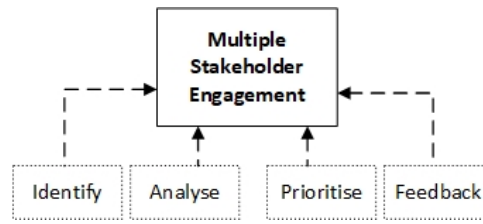


Figure 6.1: The elements of multiple stakeholder engagement as an external variable of TAM 3

As stated in Chapter One, TAM 3 does not explicitly account for multiple stakeholder management practices (Ajibade 2018; Almaiah, Al-Khasawneh & Althunibat 2020; Dube, Van & Zuva 2020; Govindaraju, Chandra & Siregar 2012; Farre *et al.* 2019; Freiesleben *et al.* 2021; Li 2020; Maruping *et al.* 2017; Okafor, Nico & Azman 2016; Pardo & Coronado 2016; Ratjeana *et al.* 2020; Raza, Umer & Shah 2017; Raza & Standing 2010; Zhang *et al.* 2020). This study found some of the external variables of TAM 3 to have multiple stakeholder engagement (i.e., job relevance, result demonstrability, and experience) in ICC's implementation of DRS technology. This study also demonstrated that multiple stakeholder engagement needs embedding as a separate entity leading into TAM 3. All the elements of multiple stakeholder engagement (i.e., analyse, identify, prioritise and feedback) need to occur as part of the entire process connecting to the external variables of TAM 3, which includes job relevance, result demonstrability, and experience (as depicted in Figure 6.2).

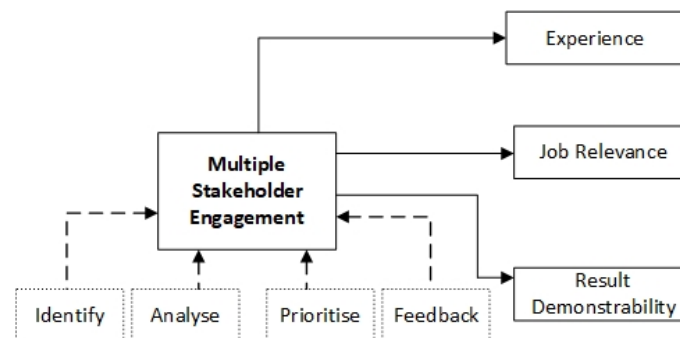


Figure 6.2: The external variables of TAM 3 with elements of multiple stakeholder engagement

6.4.2 Multiple stakeholder education

Multiple stakeholder education involves transparent training of all the stakeholders as per their roles and job specifications (i.e., which stakeholders need specialised training to fully utilise the new technology in the workplace; which stakeholders require generalised training to understand the nature of the new technology in the workplace). The multiple stakeholder education provides an understating of the roles and interdependencies of other stakeholders, ensuring that every stakeholder works within their given parameters and boundaries without any interference with different stakeholder roles. Section 6.3 discussed the role of multiple stakeholder education and transparency in TAM 3 and comprised the following elements:

- *Specialised training:* This study demonstrates that explicitly incorporating stakeholders' specialised training into TAM 3 will support the development of any specialised roles that frequently interact with technology and play an essential part in the decision-making process. By explicitly incorporating specialised training into TAM 3, the criticisms of Zahid *et al.* (2013), Almaiah, Al-Khasawneh & Althunibat (2020), Ratjeana *et al.* (2020) (i.e., TAM 3 does not consider factors such as education and training as external variables influencing acceptance and willingness to use technology) can be addressed. The specialised training process will allow the managers to segregate the specialised roles in the organisation and educate the group of potential stakeholders fulfilling the identified roles before implementing any technology. The users performing specialised roles will get familiar with the new technology and have some hands-on experience before the final implementation.
- *Generalised training:* This study demonstrates that explicitly incorporating stakeholders' generalised training into TAM 3 will support the development of other stakeholders with non-specialised roles and/or that have only indirect interaction with

technology or the decision-making process. By explicitly incorporating generalised training into TAM 3, the criticisms of Hess, McNab and Basoglu (2014), Barrio-Garcia (2015), Ratjeana *et al.* (2020) (i.e., TAM 3 only considers training or retraining a specific group of stakeholders when a new technology is purchased or any existing technology is updated) can be addressed. The generalised training process will allow the managers to train the specialised and non-specialised stakeholders in case new technology gets employed, or any updates are performed to any existing technology. It is equally essential to the non-specialised and the specialised stakeholder's roles to understand how new or updated technology interacts within the entire system.

- *Transparency:* This study demonstrates that transparency in any new technology training and actual use will provide a clear understanding of the technology to all salient stakeholders. By explicitly incorporating transparency into TAM 3, the criticisms of Barrio-Garcia (2015), Raza, Umer and Shah (2017), Almaiah, Al-Khasawneh and Althunibat (2020) (i.e., TAM 3 fails to consider factors such as institutional IT policies or any organisational rules and regulations to regulate and compel staff to use a system) can be addressed. The transparency process will demonstrate a carefully planned and implemented education and training process for all salient stakeholders. The multiple stakeholder education and training strategies both for specialised and general education of stakeholders via transparency enable all the managers to overcome any identified knowledge deficiency amongst the stakeholder groups. It enables the specialised or general user of the system to understand the technology and its purpose, use the technology professionally and become independent decision-makers without any external influences. The whole process increases reliability and confidence in using the new technology amongst multiple stakeholders and maximises its benefits across the entire organisation.

This study specifically identified two of the external variables inherent to TAM 3 benefit from multiple stakeholder education (i.e., result demonstrability and perceptions of external control) in adopting new technology (see Figure 6.3).

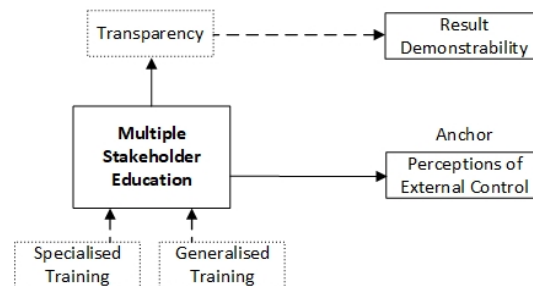


Figure 6.3: The external variables of TAM 3 with elements of multiple stakeholder education

This study posits that multiple stakeholder education needs to be embedded as a separate factor for TAM 3. All the elements of multiple stakeholder education (i.e., specialised training for specialised roles and generalised training for other stakeholders with transparency in the process) need to occur as part of the entire process connecting to the external variables of TAM 3, which includes result demonstrability and perceptions of external control. This section demonstrates how the extant TAM 3 would benefit from the inclusion of multiple stakeholder management practices embedded in it. The study has also demonstrated that the two external variables of stakeholder engagement and stakeholder education are critical to TAM 3, as depicted in Figure 6.4.

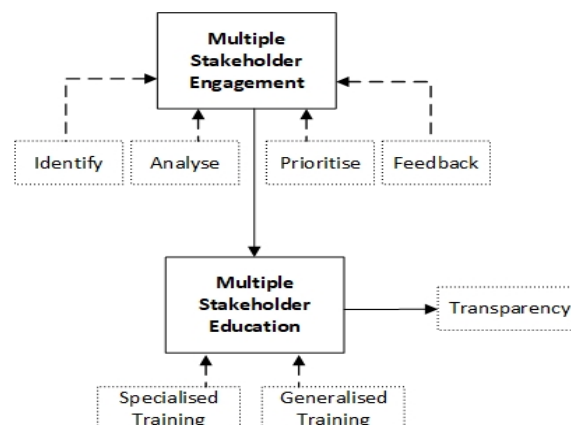


Figure 6.4: The proposed elements of multiple stakeholder engagement and multiple stakeholder education for TAM 3

6.5 A proposed modified TAM 3

As noted in Chapter One, extant TAM 3 has been criticised for its lack of explicit stakeholder management practices. This study has demonstrated that multiple stakeholder engagement and multiple stakeholder education (in particular specialised training for specialised roles and generalised training for other stakeholders with transparency in the process) enabled the ICC to overcome the DRS technology acceptance issues experienced by multiple stakeholders. This study found elements of multiple stakeholder engagement and multiple stakeholder education in overcoming the stakeholder issues in ICC's implementation of DRS technology, as shown in Figure 6.5:

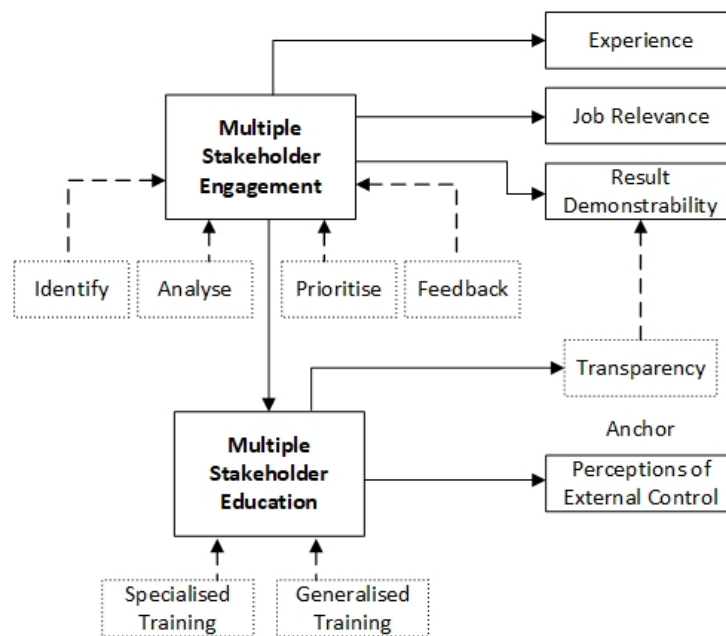


Figure 6.5: The elements of multiple stakeholder engagement and multiple stakeholder education

Given the discussion above, Figure 6.6 presents a modified TAM 3 that is inclusive of the emergent DRS technology and related stakeholder management practices studied in this thesis.

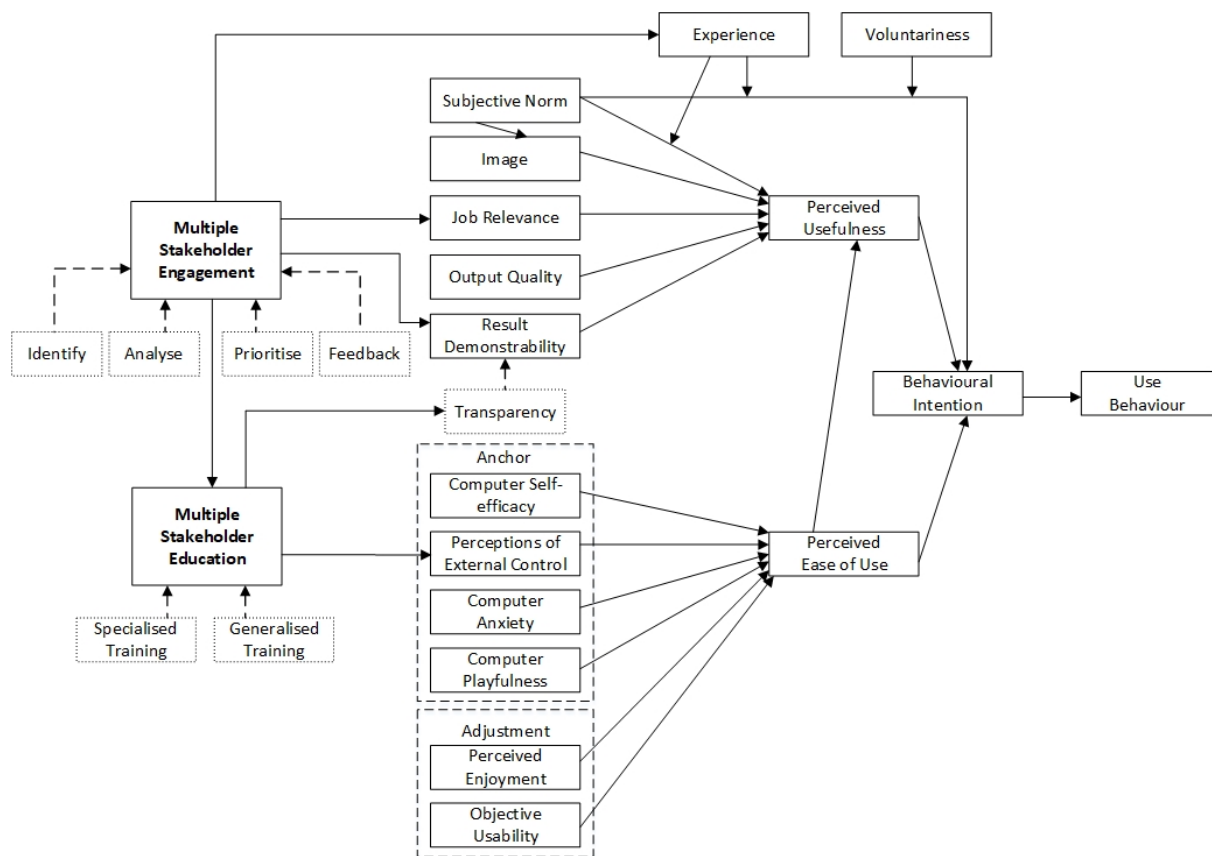


Figure 6.6: A modified TAM 3 for DRS technologies and related stakeholder management practices

6.6 Limitations of the research

The findings and discussions of this thesis need to be assessed regarding the three inherent limitations of this research: 'single case study design', 'DRS technologies' and 'the qualitative method design'. Firstly, in terms of the single embedded case study design, the findings of the thesis were based on the implementation of the DRS technology by the ICC in the international cricket context. The limitations of this approach include potential misrepresentation and poor interpretation of critical incidents in the ICC's implementation of its DRS technologies that may be occasioned by interviewees' biases as the utilisation of semi-structured interviews carries the risk of distorting primary data validity due to potential interviewer bias and/or poor recall (Denzin & Lincoln 2018; Yin 2014). As noted in Chapter Three, however, several measures were adopted to increase the accuracy of secondary data account, including the prompting of interviewees to describe critical incidents not included in the original line of

questioning. It is equally important to acknowledge that locating interviewees was critical in this study as they provided access to credible information and granted access to documentary records. One major consideration for the choice of the interviewees interviewed in this study was the need to draw from diverse stakeholder groups' responses. This study also employed face-to-face interviews, which offered interviewees some flexibility in reconstructing the past, present and future as the interviewer can always repeat questions for more clarification (Mills, Durepos & Wiebe 2009).

Secondly, this study focused exclusively on the ICC's implementation of DRS technologies focusing on international cricket competitions. Therefore, it is likely that the proposed modification to TAM 3 presented in this thesis may not be representative of the stakeholder management practices required for the implementation of other technology types. Lastly, the qualitative method adopted in this research means that the findings and implications of this thesis must be considered specific to the process of DRS technologies implementation and stakeholder management in the international cricket competition context. Therefore, whilst the findings of this research may be transferable to other professional sporting contexts of a similar nature to the international cricket competition (e.g., international tennis, European football, rugby competitions etc.), they cannot be readily generalised to other or professional employment settings (e.g., accounting, information technology, legal services etc.).

6.7 Future research opportunities

Given the limitations noted above, three key areas of future research are identified. Firstly, the modified TAM 3 for DRS technologies and related stakeholder management practices proposed in this thesis would be well-served by further validation and/or modification by empirical studies in an array of other professional sports settings and organisational contexts. Studying the TAM 3 related to the professional sports context presents a significant research

opportunity to confirm or modify extant TAM 3 for this growing and important context. More research is required regarding the vast array of stakeholder perceptions to accept new technology, which will help keep up with the ever-changing demands of professional sports. Secondly, there is also the opportunity to explore the modified TAM 3 for DRS technologies and related stakeholder management practices across various technologies. The proposed model can also be further developed by testing for technologies other than DRS technologies to gain the status of a legitimate research tool. Lastly, future research on the relationship between DRS technology acceptance, multiple-stakeholder practices and their relationship to TAM will benefit significantly through the application of quantitative research methods. Quantitative (and mixed-methods approaches) will be able to establish any statistical significance in the strength of relationships between elements of the modified TAM 3 for DRS technologies and related stakeholder management practices, and present generalisable findings for academics and practitioners alike.

6.8 Chapter summary

In conclusion, this chapter presented the answer to the research question of this study by proposing a modified TAM 3 for DRS technologies and the related stakeholder management practices that account for multiple stakeholder acceptance. Firstly, the chapter discussed the characteristics of DRS technologies in a professional sports context. Secondly, it provided an insight into the current external variables supporting the DRS technologies characteristics and related stakeholder management practices in extant TAM theory and the additional external variables required for DRS technologies acceptance. Lastly, the chapter presented the limitations, opportunities for future research and general recommendations for implementing the proposed modified TAM 3.

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Appendix A - Case Study Protocol

1. Proposed Title of the Research Project

Enhancing TAM for DRS technologies and related stakeholder management practices: a case study of international cricket

2. Research Objectives

- i. To determine how has the implementation of the DRS technologies been perceived positively and negatively by the salient stakeholders? i.e., What is the impact on player behaviour? What is the effect on the action of the umpires?
- ii. To determine how the DRS technologies impacted the decision-making processes of the players and umpires? i.e., Under what circumstances is the DRS used by the players; How does the DRS technologies influence the umpire's decision-making behaviour?
- iii. To determine how could the DRS technologies be modified to more efficiently serve the needs of the salient stakeholder groups? i.e., multi-perspective analysis of the optimal role that the DRS technologies can play in the game of cricket.

3. Significance of the Phenomena of Interest

DRS technologies are not designed to support the decision-maker or decision-making process while making the original decision and are only involved after an original decision gets reviewed or challenged. Extant TAM 3 is currently not equipped to include the characteristics of DRS technologies and related stakeholder management practices. This study finds it necessary to identify and modify TAM 3's external variables to examine the factors influencing DRS acceptance by multiple stakeholders in a professional sports context. This study aims to modify TAM 3 for DRS technologies and the related stakeholder management practices in the professional sports context.

4. Specific Research Question

This study aims to modify TAM 3 for DRS technologies and the related stakeholder management practices in the professional sports context; hence providing the answer to the following research question: *To what extent does TAM 3 need to be modified to account for DRS technologies and related stakeholder management practices?*

To address this research question, the research was designed as a two-stage process; firstly, a detailed history of the ICC's DRS technology implementation will be constructed from secondary data. Secondly, critical incidents from the historical account of the ICC's implementation of DRS technologies will be used to form the base interview questions asked of the expert stakeholders in Stage Two.

5. Selection of the Research Respondents

This thesis seeks to interview stakeholders of the ICC's international cricket competition as its key interviewees; the stakeholders are broadly divided into 'participant' and 'non-participant' stakeholder groups (where participation relates to whether they engage in playing the sport directly or not). The participant stakeholders will include players, umpires, coaches, match referees and scorers, while non-participant stakeholders will be administrators, managers, educators, journalists, and media.

6. Recruitment of the Research Participants

After the review of the literature following criteria was developed to select the interviewees for this study:

1. Participants can be multi-perspective; and
2. At least one of their designations must be in a professional capacity

A rigorous approach will be followed to recruit the interviewees required to represent the salient stakeholder groups. Firstly, the leading cricketing authority in Australia (i.e., Cricket Australia) will be contacted by email explaining the project and requesting the stakeholders' direct contact or sharing the project details with interested interviewees. The information sheet, consent form and ethics documents for the potential participants will be attached with the initial email to the cricketing body (see Appendices C, D, E, F and G).

7. Data Collection and Management Techniques

A list of stakeholders will be obtained from Cricket Australia, and the stakeholders will then be contacted through email. The time and medium of the interview will be decided as per the convenience of the interviewees. The stakeholders recruited for this study will have performed a variety of professional cricket roles. As such, they provide multiple perspectives that increases the number of views collection via the interviews. The interviews will cover each role of stakeholders by asking the same set of questions for each role separately. However, multiple interviews may be conducted to cover each role individually for the same stakeholder if required. The experience levels will be categorised as ICC (International), Cricket Australia (National), and State Cricket.

This qualitative study aims to elicit information insights into the implementation of DRS technologies and detect suitable theoretical contributions to extant TAM 3 that can optimise technology acceptance through more attuned stakeholder management practices. To achieve this aim, the data collection process constitutes a collection of primary data from salient stakeholders. Due to the study's exploratory nature, the suitable data collection process will be semi-structured interviews.

In this study, mixed of face-to-face and telephone interviews will be conducted. The main reason to use a mixed technique will be the physical location of the interviewees in many cases. There will be many targeted interviewees who will be located worldwide; contacting them involved cost and time-related issues; for the local interviewees, interviews will be conducted face-to-face. The interviewees will be given a choice to select video or audio interview mode. The technology-mediated video interviews will improve the privacy and autonomy of the interviewer and the interviewees. Moreover, the interviewees will have the opportunity to schedule and reschedule the interviews, which will improve their engagement.

8. Data Analysis Technique

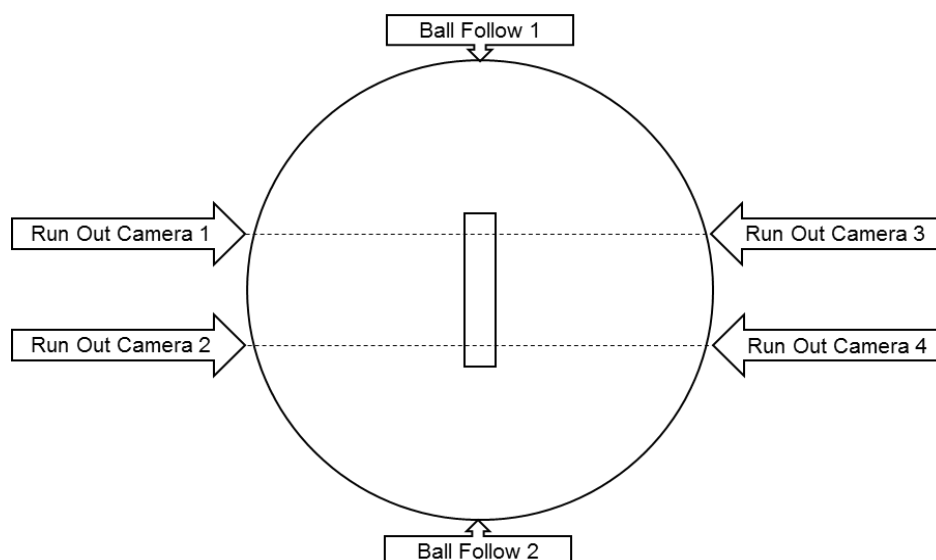
Data gathered during the interview process will be imported into the NVivo (version 12.0) software program, where concepts pertinent to the specific research question (and their interrelationships) will be coded for further analysis. The literature review included in Chapter Two will provide the base coding categories used in this research; the initial codes generation will be related to the theoretical characteristics of DRS technologies. Subsequent codes will be created to account for the extant TAM 3 variables based on the extant TAM 3 theory.

Appendix B – The ICC’s Process for DRS Technology Use

1. General

Minimum requirements for use of DRS.

The camera specification set out below shall be mandatory as a minimum requirement.



The table below summarises the minimum requirements for DRS to be used, and the regulations around the appointment of the Third Umpire:

	DRS
Minimum Requirement	Cameras - Specification detailed in paragraph 1.2.1. Technology - Approved ball-tracking technology. - Approved sound-based edge detection technology.
Third Umpire Jurisdiction	Umpire Reviews and Player Reviews
Replays that can be used	Any replay, stump microphone audio or technology
ICC Technical Officer	The ICC shall appoint an independent technology expert (ICC Technical Officer) to be present at every series to assist the Third Umpire and to protect the integrity of the DRS process.

The ICC shall ensure that a separate room is provided for the Third Umpire and that he/she has access to the television equipment and technology so as to be in the best position to facilitate the referral and/or consultation processes referred to in paragraphs 2 (Umpire Review) and 3 (Player Review) below.

2. Umpire Review

In the circumstances detailed in paragraphs 2.1, 2.2, 2.3 and 2.4 below, the on-field umpire shall have the discretion to refer the decision to the Third Umpire or, in the case of paragraphs 2.2, and 2.4, to consult with the Third Umpire before making the decision. In line with Law 21.5, the Third Umpire shall already have reviewed television replays of the bowler's front foot landing and, where not satisfied that any of the three conditions in Law 21.5.2 have not been met, shall immediately have advised the bowler's end umpire (who shall in turn have immediately called and signalled No Ball). Accordingly, paragraphs 2 (Umpire Review) and 3 (Player Review) of this Appendix shall operate on the basis that this element of the referral and/or consultation process has already been undertaken.

Save for requesting the umpire to review his/her decision under paragraph 3 (Player Review) below, players may not appeal to the on-field umpires to use the Umpire Review. Breach of this provision may constitute dissent and the player may be subject to disciplinary action under the ICC Code of Conduct for Players and Player Support Personnel.

2.1 Run Out, Stumped, Bowled and Hit Wicket Decisions

- 2.1.1 The relevant on-field umpire shall be entitled to refer an appeal for Run Out, Stumped, bowled or Hit Wicket to the Third Umpire.
- 2.1.2 An on-field umpire wishing to refer a decision to the Third Umpire shall signal to the Third Umpire by making the shape of a TV screen with his/her hands.
- 2.1.3 In the case of a referral of a Bowled, Hit Wicket or Stumped decision, the Third Umpire shall first check the fairness of the delivery (all modes of No Ball except for (i) in respect of the front foot, which shall already have been checked in accordance with Law 21.5; and (ii) the bowler using an Illegal Bowling Action, subject to the proviso that the Third Umpire may review whether the bowler has used a prohibited Specific Variation under Article 6.2 of the Illegal Bowling Regulations). If the delivery was not a fair delivery the Third Umpire shall indicate that the batsman is Not Out and advise the on-field umpire to signal No Ball. See also paragraph 2.5 below.
- 2.1.4 Additionally, if the Third Umpire finds the batsman is Out by another mode of dismissal (excluding LBW), or Not Out by any mode of dismissal (excluding LBW), he/she shall notify the on-field umpire so that the correct decision is made.
- 2.1.5 If the Third Umpire decides that the batsman is Out, a red light shall be displayed; if the Third Umpire decides that the batsman is Not Out, a green light shall be displayed. Should the Third Umpire be temporarily unable to respond, a white light (where available) shall remain illuminated throughout the period of interruption to signify to the on-field umpires that Umpire Reviews are temporarily unavailable, in which case the decision shall be taken by the on-field umpire. As an alternative to the red/green light system, the replay screen (where available) may be used for the purpose of conveying the Third Umpire's decision, in line with the ICC Big Screen Policy.

2.2 Caught Decisions, Obstructing the Field

- 2.2.1 Where the bowler's end umpire is unable to decide upon a Fair Catch or a Bump Ball, or if, on appeal from the fielding side, the batsman obstructed the field, he/she shall first consult with the striker's end umpire.

- 2.2.2 Should both on-field umpires require assistance from the Third Umpire to make a decision, the bowler's end umpire shall firstly take a decision on-field after consulting with the striker's end umpire, before consulting by two-way radio with the Third Umpire. Such consultation shall be initiated by the bowler's end umpire to the Third Umpire by making the shape of a TV screen with his/her hands, followed by a Soft Signal of Out or Not Out made with the hands close to the chest at chest height. If the Third Umpire advises that the replay evidence is inconclusive, the on-field decision communicated at the start of the consultation process shall stand.
- 2.2.3 The Third Umpire shall determine whether the batsman has been caught, whether the delivery was a Bump Ball, or if the batsman obstructed the field. However, in reviewing the television replay(s), the Third Umpire shall first check the fairness of the delivery for all decisions involving a catch (all modes of No Ball except in respect of the front foot, which shall already have been checked in accordance with Law 21.5; and the bowler using an Illegal Bowling Action, subject to the proviso that the Third Umpire may review whether the bowler has used a prohibited Specific Variation under Article 6.2 of the Illegal Bowling Regulations) and whether the batsman has hit the ball. If the delivery was not a fair delivery or if it is clear to the Third Umpire that the batsman did not hit the ball, he/she shall indicate to the bowler's end umpire that the batsman is Not Out caught, and in the case of an unfair delivery, advise the bowler's end umpire to signal No Ball. Additionally, if it is clear to the Third Umpire that the batsman is Out by another mode of dismissal (excluding LBW), or Not Out by any mode of dismissal (excluding LBW), he/she shall notify the bowler's end umpire so that the correct decision can be made.
- 2.2.4 The Third Umpire shall communicate his/her decision.

2.3 Boundary Decisions

- 2.3.1 The bowler's end umpire shall be entitled to refer to the Third Umpire for a decision on:
- 2.3.1.1 whether a four or six has been scored;
 - 2.3.1.2 whether a fielder had any part of his person in contact with the ball when he touched the boundary;
 - 2.3.1.3 whether the fielder had any part of his person in contact with the ball when he had any part of his person grounded beyond the boundary; or
 - 2.3.1.4 the number of runs scored when a boundary results from an overthrow or the wilful act of a fielder (see Law 19.8).
- 2.3.2 A decision shall be made immediately and cannot be changed thereafter.
- 2.3.3 If the television evidence is inconclusive as to whether or not a boundary has been scored, the default presumption shall be in favour of no boundary being awarded.
- 2.3.4 Where the bowler's end umpire wishes to use the assistance of the Third Umpire in this circumstance, he/she shall communicate with the Third Umpire by use of a two-way radio and the Third Umpire shall convey his/her decision to the bowler's end umpire by the same method.
- 2.3.5 The Third Umpire may initiate contact with the on-field umpire by two-way radio if TV coverage shows a boundary line infringement or incident that appears not to have been acted upon by the on-field umpires.

2.4 Batsmen Running to the Same End

- 2.4.1 Where both batsmen have run to the same end and the on-field umpires are uncertain over which batsman made his ground first, the on-field umpires may consult with the Third Umpire.
- 2.4.2 The procedure set out in paragraph 2.3.4 shall apply.

2.5 Short Run

- 2.5.1 The Third Umpire shall automatically review any call of Short Run by an on-field umpire under Laws 18.3 and 18.4, communicating with the relevant on-field umpire by two-way radio and conveying his/her decision by the same method.

2.6 No Balls

- 2.6.1 If the bowler's end umpire is uncertain as to the fairness of the delivery following a dismissal (other than in respect of the front foot, which shall already have been checked by the Third Umpire), either affecting the validity of the dismissal or which batsman is dismissed, he/she shall be entitled to request the batsman to delay leaving the field and to check the fairness of the delivery with the Third Umpire. Communication with the Third Umpire shall be by two-way radio.
- 2.6.2 The Third Umpire shall check all modes of No Ball (other than in respect of the front foot, which shall already have been checked) except for the bowler using an Illegal Bowling Action (subject to the proviso that the Third Umpire may review whether the bowler has used a prohibited Specific Variation under Article 6.2 of the Illegal Bowling Regulations). The Third Umpire shall apply Law 21.5 when deciding whether a No Ball should have been called (and must therefore be satisfied that none of the three conditions in Law 21.5 have been met before calling a No Ball).
- 2.6.3 If the delivery was not a fair delivery, the bowler's end umpire shall indicate that the batsman is Not Out and signal No Ball (except in the case of a dismissal for obstructing the field, which may still be effected despite a No Ball being called, in which case the bowler's end umpire shall indicate that the relevant batsman is Out and additionally call a No Ball).
- 2.6.4 If a No Ball is called following the check by the Third Umpire, the batting side shall benefit from the reversal of the dismissal and the one run for the No Ball, but shall not benefit from any runs that may subsequently have accrued from the delivery had the on-field umpire originally called a No Ball. Where the batsmen crossed while the ball was in the air before being caught, the batsmen shall remain at the same ends as if the striker had been dismissed, but no runs shall be credited to the striker even if one (or more) runs were completed prior to the catch being taken.
- 2.6.5 Cameras On or Over the Field of Play
- 2.6.6 The on-field umpires shall be entitled to refer to the Third Umpire for a decision as to whether the ball has at any time during the normal course of play come into contact with any part of the camera, its apparatus or its cables above the playing area, as contemplated in Law 20.1.3.
- 2.6.7 Where an on-field umpire wishes to use the assistance of the Third Umpire in this circumstance, he/she shall communicate with the Third Umpire by use of a two-way

radio and the Third Umpire shall convey his/her decision to the bowler's end umpire by the same method.

2.6.8 A decision shall be made immediately and cannot be changed thereafter. If the television evidence is inconclusive as to whether or not the ball has come into contact with any part of the camera, its apparatus or its cables above the playing area, the default presumption shall be in favour of no contact having been made.

2.6.9 The Third Umpire may initiate contact with the on-field umpire by two-way radio if TV coverage shows the ball to have been in contact with any part of the camera or its cables above the playing area as envisaged under this paragraph.

3. Player Review

The following paragraphs shall operate in addition to and in conjunction with paragraph 2 (Umpire Review). Circumstances in which a Player Review may be requested:

A player may request a review of any decision taken by the on-field umpires concerning whether or not a batsman is dismissed, with the exception of 'Timed Out' (Player Review).

No other decisions made by the umpires are eligible for a Player Review with the exception of Fair Catch/Bump Ball (even after the Third Umpire has been consulted and the decision communicated).

Only the batsman involved in a dismissal may request a Player Review of an Out decision and only the captain (or acting captain) of the fielding team may request a Player Review of a Not Out decision.

A decision concerning whether or not a batsman is dismissed that could have been the subject of an Umpire Review under paragraph 2 is eligible for a Player Review as soon as it is clear that the on-field umpire has chosen not to initiate the Umpire Review.

1.1 The manner of requesting the Player Review

1.1.1 The request shall be made by the player making a 'T' sign with both forearms at head height.

1.1.2 The total time elapsed between the ball becoming dead and the review request being made shall be no more than 15 seconds. The only exception permitted shall be when an Umpire Review for Fair Catch or Bump Ball (as permitted in paragraph 2.2 above) is required to answer an appeal for a caught decision, in which case either team is able to request a Player Review of that caught decision within 15 seconds of the decision being communicated. The bowler's end umpire shall provide the relevant player with a prompt after 10 seconds if the request has not been made at that time and the player shall request the review immediately thereafter. If the on-field umpires believe that a request has not been made within the 15 second time limit, they shall decline the request for a Player Review.

1.1.3 The captain may consult with the bowler and other fielders, and the two batsmen may consult with each other prior to deciding whether to request a Player Review. Either the captain or the two batsmen may, in respect of a decision relating to an

appeal for LBW, question the bowler's end umpire as to whether his/her decision has been made on the basis that the striker made no genuine attempt to play the ball with the bat. Under no other circumstance is any player permitted to query an umpire about any aspect of a decision before deciding on whether or not to request a Player Review. If the on-field umpires believe that the captain or either batsman has received direct or indirect input emanating other than from the players on the field, then they may at their discretion decline the request for a Player Review. In particular, signals from the dressing room must not be given.

- 1.1.4 No replays, either at normal speed or slow motion, shall be shown on a big screen to spectators until the 15 second time limit allowed for requesting a Player Review has elapsed. The only exception to this provision is where a Player Review of a caught decision is requested after the Umpire Review of a Fair Catch or Bump Ball has concluded, as detailed above (due to the fact that replays may have been shown on the big screen during that Umpire Review process).
- 1.1.5 Where either on-field umpire initiates an Umpire Review, this does not preclude a player seeking a Player Review of a separate incident from the same delivery. The request for a Player Review may be made after the Umpire Review, provided the request is still within the 15 second time limit.
- 1.1.6 A request for a Player Review cannot be withdrawn once it has been made.

3.2 The process of consultation

- 3.2.1 On receipt of an eligible and timely request for a Player Review, the relevant on-field umpire shall make the sign of a shape of a TV screen with his/her hands in the normal way
- 3.2.2 The relevant on-field umpire shall initiate communication with the Third Umpire by confirming;
 - 3.2.2.1 That a Player Review has been requested,
 - 3.2.2.2 The mode of dismissal for which the relevant on-field umpire adjudicated the appeal,
 - 3.2.2.3 The decision that has been made (Out or Not Out), and
 - 3.2.2.4 For LBW appeals, where relevant, if the bowler's end umpire believed that the striker made no genuine attempt to play the ball with the bat (the default presumption of the Third Umpire in the absence of any information on this point from the bowler's end umpire shall be that a genuine attempt to play the ball with the bat was made).
- 3.2.3 A two-way consultation process shall begin to investigate whether there is anything that the Third Umpire can see or hear which would indicate that the on-field umpire should change his/her original decision.
- 3.2.4 The Third Umpire shall not withhold any factual information which may help in the decision-making process. In particular, in reviewing a dismissal, if the Third Umpire believes that the batsman may instead be Out by any other mode of dismissal, he/she shall advise the on-field umpire accordingly. The process of consultation described in this paragraph in respect of such other mode of dismissal shall then be conducted as if the batsman has been given Not Out.
- 3.2.5 The Third Umpire shall initially check all modes of No Ball except for the bowler using an Illegal Bowling Action (subject to the proviso that the Third Umpire may review

whether the bowler has used a prohibited Specific Variation under Article 6.2 of the Illegal Bowling Regulations), where appropriate advising the on-field umpire accordingly.

- 3.2.6 If despite the available technology, the Third Umpire is unable to decide with a high degree of confidence whether the original on-field decision should be changed, then he/she shall report that the replays are ‘inconclusive’, and that the on-field decision shall stand. The Third Umpire shall not give answers conveying likelihoods or probabilities.
- 3.2.7 In circumstances where the television technology (all or parts thereof) is not available to the Third Umpire or fails for whatever reason, the Third Umpire shall advise the on-field umpire of this fact but still provide any relevant factual information that may be ascertained from the available television replays and other technology.
- 3.2.8 The on-field umpire shall then make his/her decision based on the information provided by the Third Umpire, any other factual information offered by the Third Umpire and his/her recollection and opinion of the original incident.
- 3.2.9 The on-field umpire shall reverse his/her decision if the nature of the supplementary information received from the Third Umpire leads him/her to conclude that his/her original decision was incorrect.

3.3 Review of LBW Decisions

- 3.3.1 In assessing whether a batsman is Out LBW in accordance with Law 36, the Third Umpire shall first judge whether the delivery is fair (as set out in Law 36.1.1), and second, whether or not the ball has touched the bat before being intercepted by any part of the striker’s person (as set out in Law 36.1.3).
- 3.3.2 If the batsman is still eligible to be Out, the ball-tracking technology shall then present three pieces of information to the Third Umpire relating to the path of the ball:
 - 3.3.1.1 The point of pitching (where applicable) (PITCHING)
 - 3.3.1.2 The position of the ball at the point of first interception (IMPACT)
 - 3.3.1.3 Whether the ball would have hit the wicket (WICKET)
- 3.3.3 This Decision Review System (DRS) and Third Umpire Protocol includes a category of Umpire’s Call, which shall be the conclusion reported where the technology indicates a marginal decision in respect of either the point of first interception or whether the ball would have hit the stumps.

3.3.2 PITCHING

- 3.3.2.1 The interpretation of “pitches in line between wicket and wicket” in Law 36.1.2 shall refer to the position of the centre of the ball at the point of pitching, in relation to the Pitching Zone.
- 3.3.2.2 The Pitching Zone is defined as a two-dimensional area on the pitch between both sets of stumps with its boundaries consisting of the base of both sets of stumps and a line between the outside of the outer stumps at each end.
- 3.3.2.3 Where applicable, the ball-tracking technology shall report that the ball pitched in one of the following three areas in relation to the Pitching Zone:

In Line	The centre of the ball was inside the Pitching Zone
Outside Off	The centre of the ball was outside, and to the off side of, the Pitching Zone
Outside Leg	The centre of the ball was outside, and to the leg side of, the Pitching Zone

3.3.2.4 Subject to the satisfaction of the other elements of Law 36, the batsman can be Out if the ball-tracking technology reports that the ball pitched Outside Off or In Line, but the batsman shall be Not Out if the ball pitched Outside Leg.

3.3.3 IMPACT

3.3.3.1 The interpretation of “the (first) point of impact, even if in above the level of the bails, is between wicket and wicket” in Law 36.1.4 shall refer to the position of the ball at the point of first interception, in relation to the Impact Zone.

3.3.3.2 The Impact Zone is defined as a three-dimensional space extending between both wickets to an indefinite height and with its boundaries consisting of a line between the outside of the outer stumps at each end.

3.3.3.3 The ball-tracking technology shall report that the point of first interception was in one of the following categories in relation to the Impact Zone:

In Line	The centre of the ball was inside the Impact Zone
Umpire’s Call	Some part of the ball was inside the Impact Zone, but the centre of the ball was outside the Impact Zone, with the further sub-category of ‘Umpire’s Call (off side)’ where the centre of the ball was to the off side of the Impact Zone and the bowler’s end umpire communicates to the Third Umpire that no genuine attempt to play the ball was made by the batsman.
Outside	No part of the ball was inside the Impact Zone, with the further sub-categories of ‘Outside (off)’ and ‘Outside (leg)’ to indicate the location of the point of first interception in relation to the Impact Zone when the bowler’s end umpire communicates to the Third Umpire that no genuine attempt to play the ball was made by the batsman.

3.3.3.4 Where a Not Out decision is being reviewed, and it is judged that the batsman has made a genuine attempt to play the ball, the ball-tracking technology must report that the point of first interception was In Line for the batsman to be eligible to be given Out, otherwise the batsman shall remain Not Out.

3.3.3.5 Where a Not Out decision is being reviewed, and it is judged that the batsman has made no genuine attempt to play the ball, the ball-tracking technology must report that the point of impact was In Line, or Umpire’s Call (off side), or Outside (off) for the batsman to be eligible to be given Out, otherwise the batsman shall remain Not Out.

3.3.3.6 Where an Out decision is being reviewed, and it is judged that the batsman has made a genuine attempt to play the ball, the ball-tracking technology must report that the point of first interception was Outside for the decision to be reversed to Not Out, otherwise the batsman shall remain eligible to be given Out.

3.3.3.7 Where an Out decision is being reviewed, and it is judged that the batsman has made no genuine attempt to play the ball, the ball-tracking technology must report that the point of first interception was Outside (leg) for the decision to be reversed to Not Out, otherwise the batsman shall remain eligible to be given Out.

3.3.4 WICKET

3.3.4.1 The interpretation of whether “the ball would have hit the wicket” in Law 36.1.5 shall refer to position of the ball as it either hits or passes the wicket, in relation to the Wicket Zone.

3.3.4.2 The Wicket Zone is defined as a two-dimensional area whose boundaries are the outside of the outer stumps, the base of the stumps and the top of the stumps.

3.3.4.3 The ball-tracking technology shall report whether the ball would have hit the wicket with reference to the following three categories:

Hitting	The ball was hitting the wicket, and the centre of the ball was inside the Wicket Zone
Umpire’s Call	The ball was hitting the wicket, but the centre of the ball was not inside the Wicket Zone
Missing	The ball was missing the wicket

3.3.4.4 Where a Not Out decision is being reviewed, the ball-tracking technology must report that the ball was Hitting for the batsman to be eligible to be given Out, otherwise the batsman shall remain Not Out.

3.3.4.5 However, where the evidence shows that the ball was Hitting, the point of first interception was In Line, and the ball pitched In Line or Outside Off, but that:

3.3.4.5.1 The point of first interception was 300cm or more from the stumps;

or

3.3.4.5.2 The point of first interception was more than 250cm but less than 300cm from the stumps and the distance between the point of pitching and the point of first interception was less than 40cm, the on-field decision shall stand (that is, Not Out).

3.3.4.6 Where an Out decision is being reviewed, the ball-tracking technology must report that the ball was Missing for the on-field decision to be reversed to Not Out, otherwise the batsman shall remain eligible to be given Out.

3.3.4.7 When the ball strikes the batsman on the full, and the evidence provided by the ball-tracking technology indicates that the ball would have pitched before striking or passing the wicket, there will be no information available from that delivery that will allow the ball-tracking technology to accurately predict the height of the ball after pitching.

3.3.4.8 With regard to determining whether the ball would have hit the wicket under these circumstances, the ball-tracking technology shall project the line of the ball in accordance with Law 36.2.3 (it is to be assumed that the path of the ball before interception would have continued after interception, irrespective of whether the ball might have pitched subsequently or not) and display the simulated path of the ball from directly above the wicket.

3.3.4.9 The Third Umpire shall advise the bowler’s end umpire only on the point of first interception and whether the ball would have hit the stumps (in line with the process set out above) but shall make no comment on the predicted height of the ball after pitching, which shall remain a judgment of the bowler’s end umpire.

3.4 The process for communicating the final decision

- 3.4.1 For Player Reviews concerning potential dismissals, the relevant on-field umpire shall indicate Out by raising his/her finger above his/her head in a normal yet prominent manner or indicate Not Out by the call of 'Not Out' and by crossing his/her hands in a horizontal position side to side in front and above his/her waist three times. Where the decision is a reversal of the on-field umpire's previous decision, he/she shall make the 'revoke last signal' indication immediately prior to the above.
- 3.4.2 If the mode of dismissal is not obvious or not the same as that on which the original decision was based, then the umpire shall advise the scorers via the Third Umpire.

3.5 Number of Player Review requests permitted

- 3.5.1 In each innings, each team shall be allowed to make a maximum of three Player Review requests that are categorised as 'Unsuccessful' (as set out in paragraph 3.5.3 below).
- 3.5.2 Where a request for a Player Review results in the original on-field decision being reversed, then the Player Review shall be categorised as 'Successful' and shall not count towards the innings limit.
- 3.5.3 Where a request for a Player Review results in the original on-field decision remaining unchanged, the Player Review shall be categorised as 'Unsuccessful'.
- 3.5.4 Where a request for a Player Review of an LBW decision results in the on-field decision remaining unchanged solely on the basis of an Umpire's Call, the Player Review shall be categorised as 'Unchanged – Umpire's Call'. A Player Review categorised as 'Unchanged – Umpire's Call' shall not count towards the innings limit.
- 3.5.5 Where, following a request for a Player Review, the original on-field decision of Out is unchanged, but for a different mode of dismissal from the original on-field decision, then the Player Review shall still be categorised as 'Unsuccessful'.
- 3.5.6 Where, following a request for a Player Review, the original on-field decision of Not Out is unchanged on account of the delivery being a No Ball (for any reason), thereby not requiring any further evaluation, the Player Review shall not be counted as 'Unsuccessful' and accordingly shall not count towards the innings limit.
- 3.5.7 Where a Player Review and an Umpire Review are requested from the same delivery and the decision of the Third Umpire from the Umpire Review renders the Player Review unnecessary, the Player Review request shall be disregarded and accordingly shall not count towards the innings limit.
- 3.5.8 A Player Review categorised as 'Unsuccessful' may be reinstated by the ICC Match Referee at his/her sole discretion (if appropriate after consultation with the ICC Technical Official and/or the television broadcast director) if the Player Review could not properly be concluded due to a failure of the technology. Any such decision shall be final and shall be taken as soon as possible, being communicated to both teams once all the relevant facts have been ascertained by the ICC Match Referee. A Player Review categorised as 'Unsuccessful' shall not be reinstated if, despite any technical failures, the correct decision could still have been made using the other available technology. Similarly, a Player Review categorised as 'Unsuccessful' shall not be reinstated where the technology worked as intended, but the evidence gleaned from its use was inconclusive.

3.5.9 The Third Umpire shall be responsible for counting the number Player Reviews categorised as ‘Unsuccessful’ and shall advise the on-field umpires once either team has exhausted their allowance for the innings.

3.5.10 The scoreboard shall display, for the innings in progress, the number of Player Reviews remaining available to each team.

Category of Player Review	Outcome of Player Review	Consequence of Player Review
Successful	On-field decision reversed	Does not count towards innings limit
Unsuccessful	On-field decision unchanged	Counts towards innings limit
Unchanged – Umpire’s Call	On-field decision unchanged	Does not count towards innings limit
No Ball – no evaluation required	On-field decision unchanged	Does not count towards innings limit
Failure of technology	On-field decision unchanged	Does not count towards innings limit

3.6 Dead ball

3.6.1 If following a Player Review request, an original decision of Out is changed to Not Out, then the ball is still deemed to have become dead when the original decision was made (as per Law 20.1.1.3). The batting side, while benefiting from the reversal of the dismissal, shall not benefit from any runs that may subsequently have accrued from the delivery had the on-field umpire originally made a Not Out decision, other than any No Ball penalty that could arise.

3.6.2 If an original decision of Not Out is changed to Out, the ball shall retrospectively be deemed to have become dead from the moment of the dismissal event. All subsequent events, including any runs scored, shall be ignored.

3.7 Use of technology

3.7.1 The following technology may be used by the Third Umpire during a Player Review:

3.7.1.1 Replays, at any speed, from any available broadcast camera

3.7.1.2 Sound from the stump microphones with the replays at normal speed and slow motion

3.7.1.3 Approved ball-tracking technology:

3.7.1.3.1 Hawk-Eye (Hawk-Eye Innovations), or;

3.7.1.3.2 VirtualEye (ARL)

3.7.1.4 Approved sound-based edge detection technology:

3.7.1.4.1 Real-Time Snickometer (BBG Sports), or;

3.7.1.4.2 Ultra-Edge (Hawk-Eye Innovations)

3.7.1.5 Approved heat-based edge detection technology:

3.7.1.5.1 Hot Spot cameras (BBG Sports)

3.7.1.6 LED Wickets (using the lights to indicate if the wicket is broken, as set out in paragraph 4.2):

3.7.1.6.1 Zing Bails and Stumps

3.7.2 In addition, other forms of technology may be used subject to the ICC being satisfied that the required standards of accuracy and time efficiency can be met.

3.7.3 Where practical usage or further testing indicates that any of the above forms of technology cannot reliably provide accurate and timely information, then it may be

removed prior to or during the match. The final decision regarding the technology to be used shall be taken by the ICC Match Referee in consultation with the ICC Technical Official, ICC management and the competing teams' governing bodies.

3.8 Combining Umpire Review with Player Review

- 3.8.1 If an Umpire Review and a request for a Player Review (under paragraph 3) are made following the same delivery but relating to separate modes of dismissal, the following process shall apply.
- 3.8.2 The Umpire Review shall be carried out prior to the Player Review if all of the following conditions apply:
 - 3.8.2.1 The Player Review has been requested by the fielding side
 - 3.8.2.2 The Umpire Review and the Player Review both relate to the dismissal of the same batsman
 - 3.8.2.3 If the batsman is Out, the number of runs scored from the delivery would be the same for both modes of dismissal
 - 3.8.2.4 If the batsman is Out, the batsman on strike for the next delivery would be the same for both modes of dismissal.
- 3.8.3 If the Umpire Review leads the Third Umpire to make a decision of Out, then this shall be displayed in the usual manner and the Player Review shall not be undertaken. If the Umpire Review results in a Not Out decision, then the Third Umpire shall make no public decision but shall proceed to address the request for a Player Review.
- 3.8.4 For illustration, following an LBW appeal which is given Not Out by the bowler's end umpire, the striker sets off for a run, is sent back and there is an appeal for his Run Out. The players request that the LBW decision is reviewed and the umpires request that the Run Out be reviewed. The four criteria above are satisfied, so the Run Out referral is determined first. Should the appeal for Run Out be Out, then there is no requirement for the LBW review to take place.
- 3.8.5 In all other circumstances, the incidents shall be addressed in chronological order. If the conclusion from the first incident is that a batsman is dismissed, then the ball would be deemed to have become dead at that point, rendering investigation of the second incident unnecessary.

4. Interpretation of Playing Conditions

- 4.1 When using a replay to determine the moment at which the wicket has been put down (as per Law 29.1), the Third Umpire shall deem this to be the first frame in which one of the bails is shown (or can be deduced) to have lost all contact with the top of the stumps and subsequent frames show the bail permanently removed from the top of the stumps.
- 4.2 Where LED Wickets are used the moment at which the wicket has been put down (as per Law 29.1) shall be deemed to be the first frame in which the LED lights are illuminated, and subsequent frames show the bail permanently removed from the top of the stumps.

Appendix C – Ethics Approval

This approval constitutes ethical clearance by the Tasmania Social Sciences HREC. The decision and authority to commence the associated research may be dependent on factors beyond the remit of the ethics review process. For example, your research may need ethics clearance from other organisations or review by your research governance coordinator or Head of Department. It is your responsibility to find out if the approvals of other bodies or authorities are required. It is recommended that the proposed research should not commence until you have satisfied these requirements.

In accordance with the National Statement on Ethical Conduct in Human Research, it is the responsibility of institutions and researchers to be aware of both general and specific legal requirements, wherever relevant. If researchers are uncertain they should seek legal advice to confirm that their proposed research is in compliant with the relevant laws. University of Tasmania researchers may seek legal advice from Legal Services at the University.



24 September 2019

Dr Robert Ollington
C/- University of Tasmania

Sent via email

Dear Dr Ollington

REF NO: H0018397
TITLE: Adapting the Technology Adoption Model to account for Multi-Stakeholder Perspective in presence of Decision Review System

We are pleased to advise that acting on a mandate from the Tasmania Social Sciences HREC, the Chair of the committee considered and approved the above project on 24 September 2019.

Please ensure that all investigators involved with this project have cited the approved versions of the documents listed within this letter and use only these versions in conducting this research project.

All committees operating under the Human Research Ethics Committee (Tasmania) Network are registered and required to comply with the *National Statement on the Ethical Conduct in Human Research* (NHMRC 2007 updated 2018).

Therefore, the Chief Investigator's responsibility is to ensure that:

- (1) All investigators are aware of the terms of approval, and that the research is conducted in compliance with the HREC approved protocol or project description.
- (2) Modifications to the protocol do not proceed until **approval** is obtained in writing from the HREC. This includes, but is not limited to, amendments that:
 - (i) are proposed or undertaken in order to eliminate immediate risks to participants;
 - (ii) may increase the risks to participants;
 - (iii) significantly affect the conduct of the research; or
 - (iv) involve changes to investigator involvement with the project.

Please note that all requests for changes to approved documents must include a version number and date when submitted for review by the HREC.

- (3) Reports are provided to the HREC on the progress of the research and any safety reports or monitoring requirements as indicated in NHMRC guidance. Researchers should notify the HREC immediately of any serious or unexpected adverse effects on participants.
- (4) The HREC is informed as soon as possible of any new safety information, from other published or unpublished research, that may have an impact on the continued ethical acceptability of the research or that may indicate the need for modification of the project.
- (5) All research participants must be provided with the current Participant Information Sheet and Consent Form, unless otherwise approved by the Committee.
- (6) This study has approval for four years contingent upon annual review. A *Progress Report* is to be provided on the anniversary date of your approval. Your first report is due 24/9/2019 and you will be sent a courtesy reminder closer to this due date. Ethical approval for this project will lapse if a Progress Report is not submitted in the time frame provided
- (7) A *Final Report* and a copy of the published material, either in full or abstract, must be provided at the end of the project.
- (8) The HREC is advised of any complaints received or ethical issues that arise during the course of the project.
- (9) The HREC is advised promptly of the emergence of circumstances where a court, law enforcement agency or regulator seeks to compel the release of findings or results. Researchers must develop a strategy for addressing this and seek advice from the HREC.

Should you have any queries please do not hesitate to contact me on (03) 6226 2975 or via email ss.ethics@utas.edu.au.

Yours sincerely

Jude Vienna-Hallam
Executive Officer I Social Sciences

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Appendix D – Information Sheet

Enhancing TAM 3 for DRS technologies and related stakeholder management practices: a case study of international cricket

PARTICIPANT INFORMATION SHEET

This information sheet is for the cricketing stakeholder groups who have in some capacity experienced the implementation of the Decision Review System - DRS in the game and can comment on the issues and the benefits the sport has perceived in the process.

1. Invitation

You are invited to participate in this research study to share your experience on the historic implementation of the DRS technology by the International Cricket Council - ICC. This information sheet will explain the project so that you can decide whether or not to take part. This study is a partial fulfilment of the PhD of Muhammad Qureshi under the supervision of Dr Robert Ollington, Dr Soonja Yeom and Dr Mark Wickham.

2. What is the purpose of this study?

The ICC's implementation of the DRS represents a valid exploratory case for the investigation of the stakeholder issues apparent since its introduction in the game. This study aims to find out how DRS technologies can more effectively and efficiently be adopted in the future.

3. Why have I been invited to participate?

You have been identified as a salient stakeholder in cricket who has in some capacity experienced the implementation of DRS in the game and can comment on the issues and the benefits the sport has perceived in the process. It is essential to understand that your involvement in this study is voluntary. While we would be pleased to have you participate, we respect your right to decline.

4. What will I be asked to do?

Participation in this study will involve a semi-structured interview with questions to ascertain your experience with the implementation of the DRS in cricket, providing information about the issues and the benefits the game has perceived in the process. All the questions will require a verbal explanation of individual experiences. The interview would take approximately 60 minutes to complete.

5. Are there any possible benefits from participation in this study?

There are no personal benefits, but the data collected through this study will help to provide information about what constitutes an ideal stakeholder consultation in future technological implementations in the sport. This research allows stakeholders in projecting their voice on what they feel about the technological indulgence in their beloved game.

6. Are there any possible risks from participation in this study?

There are no foreseeable risks anticipated with participation in this study. The questions asked will specifically be related to stakeholders' profession and pose no threat to their professional or social life.

7. What if I change my mind during or after the study?

If you decide to discontinue participation at any time before or after the semi-structured interview, you can do so without providing any explanation. After the interview, you will be provided with the interview transcript copy and will have 8 weeks to withdraw before the data analysis phase of the study.

8. What will happen to the data when this study is over?

The University of Tasmania has a secure virtual private network; the research documents and data for this research will only be accessible by the research team of this project. The data will be destroyed after five years, through secure shredding and electronic wiping.

9. How will the results of the study be published?

The results will be published as part of Muhammad Qureshi's PhD thesis; no individual stakeholder will be directly identifiable through the results. It is expected that parts of this study will be published in academic journals and conferences.

10. What if I have questions about this study?

If you have any queries, concerns or issues with this study, please feel free to contact any of the below-mentioned team members of this project to discuss any related matters:

Dr Robert Ollington (Chief Investigator),
Discipline of ICT | School of Technology, Environments and Design
Ph + 61 3 6226 2991 Email Robert.Ollington@utas.edu.au

Dr Soonja Yeom (Associate Investigator),
Discipline of ICT | School of Technology, Environments and Design
Ph + 61 3 6226 2963 Email Soonja.Yeom@utas.edu.au

Dr Mark Wickham (Associate Investigator),
Tasmanian School of Business and Economics
Ph + 61 3 6226 2159 Email Mark.Wickham@utas.edu.au

Mr Muhammad Qureshi (PhD student)
Discipline of ICT | School of Technology, Environments and Design
Ph + 61 3 6226 2964 Email Muhhammad.Qureshi@utas.edu.au

This study has been approved by the Tasmania Social Sciences Human Research Ethics Committee. If you have concerns or complaints about the conduct of this study, you can contact the Executive Officer of the HREC (Tasmania) Network on (03) 6226 2975 or email ss.ethics@utas.edu.au. The Executive Officer is the person nominate to receive complaints from research participants. You will need to quote H0018397.

If you wish to take part in it, please sign the attached consent form. This information sheet is for you to keep.

Thank you for your time to consider this study.

Appendix E – Consent Form

Enhancing TAM 3 for DRS technologies and related stakeholder management practices: a case study of international cricket

PARTICIPANT CONSENT FORM

By signing below, I confirm that I have read and understood the information sheet and in particular:

- I understand that the study undertaken is in partial fulfilment of the PhD degree of Muhammad Qureshi under the supervision of Dr Robert Ollington, Dr Soonja Yeom and Dr Mark Wickham.
- I understand that the study involves participating in face-to-face semi-structured interview and will take approximately 60 minutes to complete.
- I understand that participation involves no foreseeable risks. The questions asked will specifically be related to my profession and pose no threat to my professional or social life.
- I understand that all research data will be securely stored in the discipline of ICT's research server which can only be accessed through University of Tasmania's Virtual Private Network – VPN accessed only by the research team of this project.
- I understand that the data collected through the interview will be kept for five years from the publication of the study results, and will then be destroyed. No data will be placed in any cloud accounts to reduce the risk of tampering or unauthorised access.
- I understand that the researchers will maintain confidentiality and that any information I supply for this study will only be used for research purposes.
- I understand that my participation is voluntary and that I will be able to withdraw my data at any time after completing the semi-structured interview. I will also be able to pull out after reviewing a copy of the interview transcript and before the start of the data analysis phase of this research.
- I understand that no individual stakeholder will be directly identifiable through the results of the research and will be published in academic journals and conferences.
- I understand that this consent form requires submission within the semi-structured interview to the research team of this project, and I agree to participate in this study:

Name	
Signature	
Date	

Statement by Researcher

☐ I have explained the project, and the implications of participation in it to this volunteer and I believe that the consent is informed and that he/she understands the impact of involvement.

If the researcher has not had an opportunity to talk to participants before them participating, the following must be ticked.

☐ The participant has received the Information Sheet where my details have been provided, so participants have had the opportunity to contact me before consenting to participate in this project.

Name	
Signature	
Date	

Appendix F – Interview Questions

Enhancing TAM 3 for DRS technologies and related stakeholder management practices: a case study of international cricket

Role of the Stakeholder	
Participant or Non-participant Stakeholder	

Set of Semi-Structured Questions

1. **What are your views about the benefits of Decision Review System technology in cricket? (This is based on Chapter Two, sections 2.2.1, 2.2.2 and 2.2.3)**
 - a. For yourself as a stakeholder (based on Chapter Two, section 2.5)
 - b. For any of your colleagues (based on Chapter Two, section 2.5)
 - c. For other stakeholders (based on Chapter Two, section 2.5)
2. **What are your views about the problems/issues of Decision Review System technology in cricket? (This is based on information from Chapter Four, section 4.1)**
 - a. For yourself as a stakeholder (based on Chapter Two, section 2.5)
 - b. For any of your colleagues (based on Chapter Two, section 2.5)
 - c. For other stakeholders (based on Chapter Two, section 2.5)
3. **What were the issues underpinning the drawbacks from your perspective (This is based on Chapter Two, section 2.5)?**
4. **How can the issues be minimised or eradicated from future Decision Review technology implementation in cricket (This is based on Chapter Two, section 2.5)?**

Thank the participant for their time.

Advise them that their feedback is vital to us.

Appendix G – Interview Invitation to Cricket Australia

REQUEST FOR SUPPORT TO CONDUCT SEMI-STRUCTURED INTERVIEWS

Dear ABC,

We are a team of researchers from the University of Tasmania and have obtained Ethics approval from our university to conduct semi-structured interview with the salient stakeholder groups in the game of cricket who have in some capacity experienced the implementation of the Decision Review System (DRS) and are able to comment on the issues and the benefits the sport has perceived in the process.

This study seeks to explore the International Cricket Council's implementation of the DRS through the perceptions of both the playing and non-playing stakeholder groups. This research will allow participants in projecting their voice on what they feel about the technology implementation in the game.

The data collected through the interviews will help to provide information about what constitutes an ideal stakeholder consultation in future technological implementations for DRS in the sport.

We hereby request for your support to conduct the interviews and have attached a list of possible stakeholders in your organisation but welcome you to alter and suggest any further names relevant to participate in this research. We have also attached a copy of the University of Tasmania Ethics approval along with the Information & Consent forms for the relevant stakeholders.

If there are any questions or require any further information, please do not hesitate to contact me on 03 6226 2991.

Thank you for your time and consideration in this matter.

Yours sincerely,

Dr Robert Ollington
Lecturer in Information Systems

Appendix H – The Preamble to the Laws of Cricket:

The Spirit of Cricket

Cricket owes much of its appeal and enjoyment to the fact that it should be played not only according to the Laws, but also within the Spirit of Cricket.

The major responsibility for ensuring fair play rests with the captains, but extends to all players, umpires and, especially in junior cricket, teachers, coaches and parents.

Respect is central to the Spirit of Cricket.

Respect your captain, team-mates, opponents and the authority of the umpires.

Play hard and play fair.

Accept the umpire's decision.

Create a positive atmosphere by your own conduct, and encourage others to do likewise.

Show self-discipline, even when things go against you.

Congratulate the opposition on their successes, and enjoy those of your own team.

Thank the officials and your opposition at the end of the match, whatever the result.

Cricket is an exciting game that encourages leadership, friendship and teamwork, which brings together people from different nationalities, cultures and religions, especially when played within the Spirit of Cricket.

(Source: Marylebone Cricket Club, cited in MCC Lord's Website 2021)